خدمتكم طريق خضناه لرضى الله

2021

# ملخص
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## تفاضل وتكامل 101

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Calculus 101

# *Chapter Zero :- Functions

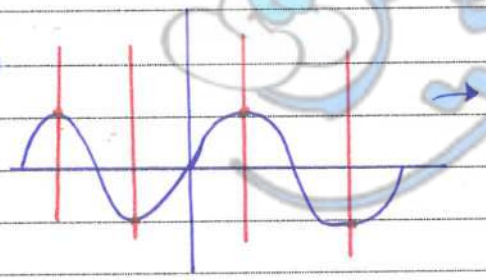
*Def:- A function is a rule that assigns every element in the domain ( المجال ) with only one element in the range ( المدى ).

* تعريف :- الإقتران هو علاقة بين مجموعتين بحيث انه يكون لكل عنصر داخل المجال صورة واحدة فقط في المدى.

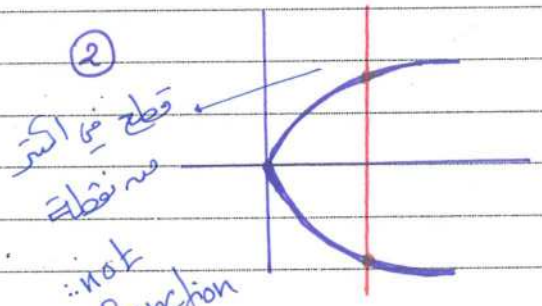
## * Vertical line test. (اختبار الخط العمودي)

↳ If the vertical line cuts the curve in one point then it's a function, otherwise it's not.

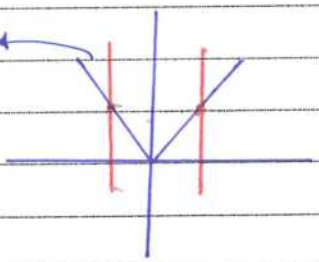
Ex:- Which of the following is a function ?

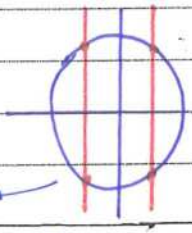
① 

حسب اختبار الخط العمودي فإنه قطع الإقتران في نقطة واحدة فقط اذاً هذا اقتران
∴ yes it's a function.

② 

قطع في اكثر من نقطة
∴ not function

قطع في نقطة
∴ function ③ 

④ 

قطع في اكثر من نقطة
not function

Ex $f(x) = 2x^2 - 7$ , find:-

① $f(2) = \ldots$ ?

② $f(t^2) = \ldots$ ?

↳ ① $f(2) = 2(4) - 7 = 8 - 7 = 1$

② $f(t^2) = 2t^4 - 7$ .

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* Domain ( المجال ) :- هو القيم التي نستطيع تعويضها داخل الإقتران

* Range ( المدى ) :- هي صور عناصر المجال .

Ex find the domain and the range:-

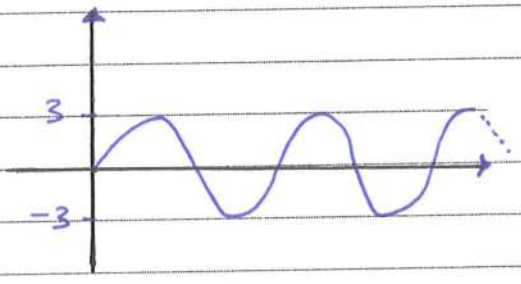
①

• لاحظ أن :- $x \geq 0$

domain : $[0, 4]$

range : $[-1, 1]$ .

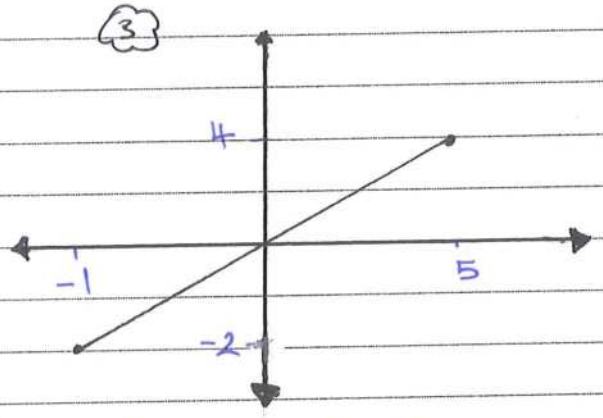
② 




domain : $[0, \infty)$

range : $[-3, 3]$

③ 




domain : $[-1, 5]$

range : $[-2, 4]$

## * Rule :-

① The domain of polynomial (كثير الحدود) is $\mathbb{R}: (-\infty, \infty)$

② The domain of fraction $\frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials is: $\mathbb{R} - \{$ أصفار المقام $\}$

↳ مجال الإقتران النسبي (الكسر) : هو كل الأعداد الحقيقية ما عدا اصفار المقام لأنها تجعل الإقتران غير معرف.

③ The domain of square root $\sqrt{f(x)}$ is all values make $f(x) \geq 0$.

↳ مجال الجذر التربيعي :- هو القيم التي تجعل ما داخل الجذر $\geq 0$.

* To find range, there are two ways :-
① عن طريق الرسم.
② بطرق المجال.

## * Examples :-

* Find the domain ?

① $f(x) = \frac{3}{x^2+2x}$ ?

↳ Sol أصفار المقام :- $x^2+2x = 0$
$x(x+2) = 0$
$x = 0, x = -2$

∴ domain $f(x) : (-\infty, \infty) - \{-2, 0\}$

② $f(x) = \frac{7}{5x^2+3}$ ?

↳ Sol أصفار المقام

$5x^2 + 3 = 0$
$5x^2 = -3$ !!

∴ لا يوجد اصفار للمقام.

∴ domain $f(x) : (-\infty, \infty)$.

Note: $\mathbb{R} = (-\infty, \infty)$

③ $f(x)= 3x^7 - 3x^2 + 5$ ?

↳ domain : $(-\infty, \infty)$

④ $f(x)= x^{-2}$ ?

↳ $f(x)= \frac{1}{x^2}$

∴ domain : $(-\infty, \infty) - \{0\}$

⑤ $g(x)= \frac{1}{x^3-8}$ ?

↳ أصفار المقام $x^3-8=0$ — طريقة a ، b

$(x-2)(x^2+2x+4)=0$

∴ $x=2$

∴ dom. $\mathbb{R} - \{2\}$

⑥ $f(x)= \sqrt{x}$ ?

↳ $x \geq 0$ , $[0, \infty)$

⑦ $f(x)= \sqrt{x-3}$ ?

↳ $x-3 \geq 0$

$x \geq 3 \Rightarrow \therefore D_f : [3, \infty)$

⑧ $f(x)= \sqrt{2-x}$ ?

↳ $2-x \geq 0$

$2 \geq x \Rightarrow D_f : (-\infty, 2]$

**Rule:**

① $x^2-a^2 = (x-a)(x+a)$

② $x^3-a^3 = (x-a)(x^2+ax+a^2)$

③ $x^3+a^3 = (x+a)(x^2-ax+a^2)$

⑨ $f(x)= \sqrt[3]{x-7}$ ?

* الجذور الزوجية نعاملها كأنها جذر تربيعي ...

$x-7 \geq 0$

$x \geq 7 \Rightarrow D_f : [7, \infty)$

⑩ $f(x)= \sqrt{x^2-5x+6}$ ?

↳ $\sqrt{(x-2)(x-3)}$

| | ++ | | -- | | ++ |
|---|---|---|---|---|---|
| | | 2 | | 3 | |

$\therefore D_f : (-\infty, 2], [3, \infty)$

⑪ $f(x)= \sqrt{\frac{x-1}{x^2-1}}$ ?

* يجب التركيز هنا لأنه لدينا :- أصفار مقام ، ما داخل جذر $\geq 0$

$\Rightarrow$ أصفار مقام :- $x^2-1=0 \rightarrow x = \pm 1$

$\Rightarrow (x-1)$ — — — + — — + + +

$x^2-1=(x-1)(x+1)$ ++ — — — + + +

$\frac{x-1}{x^2-1}$ — — — + + + + + +

(-1 , 1)

$\therefore D_f : (-1, 1), (1, \infty)$

OR $(-1, \infty) - \{1\}$

* مهم جداً :- (علينا انه نميز بين ما الذي يؤثر في المجال وما الذي يؤثر في المدى)

⇐ اليكم التوضيح الآتي :-

$a f(X-b) + c$

هذا ما يؤثر في المجال — هذا ما يؤثر على المدى

ex) If domain $f(x)$ is $[3,\infty)$, find domain $2f(5-2X)+1$ ?

↳ Sol: $2f(5-2X)+1$ (مجال)

$3 \le 5-2X < \infty$

$-5 \quad -5 \quad -5$

$-2 \le -2X < \infty$

$\div -2 \quad \div -2 \quad \div -2$ ← عند القسمة على سالب فاننا نعكس العلامات

$1 \ge X > -\infty \Rightarrow \therefore (-\infty, 1]$ .

ex) If range $f(x)$ is $[1,\infty)$ find range $2f(3+2X)+5$ ?

↳ $2f(3+2X)+5$

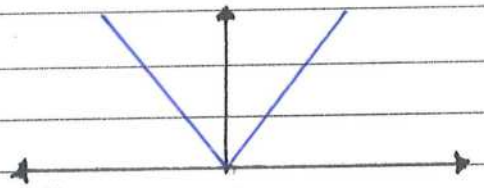
$2 * [1,\infty) + 5$

$[2,\infty) + 5$

$[7,\infty)$

* Absolute Value: القيمة المطلقة.

$$|x| = \begin{cases} x & , x \geq 0 \\ -x & , x < 0 \end{cases}$$



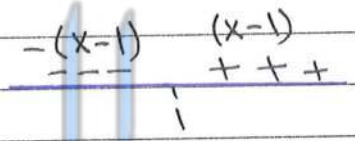
Domain : $(-\infty, \infty)$ ⇒ x-axis.

range : $[0, \infty)$ ⇒ y-axis.

* كيف نعبر تعريف القيمة المطلقة

↳ i) $|x-1|$.

↳ $|x-1| = \begin{cases} x-1 & , x \geq 1 \\ 1-x & , x < 1 \end{cases}$




ii) $|x+3|$. (homework)

* Properties of absolute value :- (الخواص)

1. $|xy| = |x| \cdot |y|$
2. $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$.
3. $|x| = |-x|$
4. $\sqrt{x^2} = |x|$ ← البرهان
5. $|x| = a \Rightarrow x = \pm a$ ← البرهان
6. $|x| \leq a \Rightarrow -a \leq x \leq a$ ← البرهان
   $(-a, a)$
7. $|x| \geq a \Rightarrow x \geq a$
   $x \leq -a$
   $[a, \infty)$ , $(-\infty, -a]$

ex) Solve:-

[1] $|x| = 9 \Rightarrow x = \pm 9$

[2] $|x| \leq 3 \Rightarrow -3 \leq x \leq 3 , [-3,3]$

[3] $|x| < 4 \Rightarrow -4 < x < 4 , (-4,4)$

[4] $|2x-1| = 3$

↳ لدينا حالة :- $2x-1 = 3 \Rightarrow x = 2$

OR $2x-1 = -3 \Rightarrow x = -1$

[5] $|x| \geq 5$

↳ $x \geq 5 , x \leq -5 \Rightarrow [5,\infty) , (-\infty,-5]$

[6] $x^2 \geq 9$.

↳ $\sqrt{x^2} \geq \sqrt{9}$

$|x| \geq 3$

$x \geq 3 , x \leq -3$

$\therefore [3,\infty) , (-\infty,-3]$

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* Integer function: $[x]$

↳ $[x] = n$ , then $n \leq x < n+1$

*اقتران اكبر عدد صحيح : يعطي العدد الصحيح السابق اذا كان العدد عشرياً .

ex) i) $[1.99] = 1$

ii) $[0.91] = 0$

iii) $[-2.3] = -3$

iv) $[-1/2] = -1$

v) $[x] = 3 \Rightarrow 3 \leq x < 4$

$[3,4)$

* Piecewise function: الاقتران المتشعب

↳ وهو اقتران معرف على أكثر من قاعدة خلال الفترة الواحدة.

ex $f(x) = \begin{cases} 7 , & x \le 0 \\ \sqrt{x} , & 0 < x < 1 \\ 2x , & x \ge 1 \end{cases}$

Q Find Range:-

① $f(x) = \sqrt{x-5}$ ?

↳ $D_f : [5, \infty)$ ← أصغر نقطة في المجال.

| x | 5 | 9 | 21 |
|---|---|---|---|
| f(x) | 0 | 2 | 4 |

$\Rightarrow \therefore R_f : [0, \infty)$

② $f(x) = 3 - \sqrt{x-1}$

↳ $D_f : [1, \infty)$

| x | 1 | 5 | 10 |
|---|---|---|---|
| f(x) | 3 | 1 | 0 |

$\Rightarrow \therefore R_f : (-\infty, 3]$.

* Even and Odd functions:- Even: زوجي ، Odd: فردي

① A function f is said to be even if $f(-x) = f(x)$.

↳ لو وضعنا محل كل x ← $-x$ ثم عاد الاقتران كما هو يكون زوجي.

Ex Which of the following is even, odd or neither?

1) $f(x) = |x|$ ?

↳ $f(-x) = |-x| = |x| \Rightarrow f(x)$ نفسها

$\therefore f(x)$ is even.

2) $f(x) = x^6$ ?

↳ $f(-x) = (-x)^6 = x^6 = f(x)$

$\therefore f$ is even.

3) $f(x) = \frac{x^2}{x^4+1}$ ؟

↳ $f(-x) = \frac{(-x)^2}{(-x)^4+1} = \frac{x^2}{x^4+1} \Rightarrow f(x)$ نفسها $\Rightarrow \therefore$ even.

4) $f(x) = \frac{x^4|x|}{x^2+3}$ (homework).

∴ $x^2, x^4, x^6, x^8, \ldots$ even

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(2) A function $f$ is said to be odd if $f(-x) = -f(x)$.

ex ① $f(x) = x^3$

↳ $f(-x) = (-x)^3 = -x^3 = -f(x)$ ∴ $f$ is odd.

② $f(x) = x|x|$

↳ $f(-x) = -x|-x| = -x|x| = -f(x)$ ∴ $f$ is odd.

∴ $x, x^3, x^5, x^7, \ldots$ odd

③ $f(x) = 2x - x^2$

↳ $f(-x) = 2(-x) - (-x)^2 = -2x - x^2 \Rightarrow$ not even $(f(-x) \neq f(x))$

$-f(x) = -(2x - x^2) = -2x + x^2 \Rightarrow$ not odd $(f(-x) \neq -f(x))$

∴ neither even nor odd.

④ $f(x) = x^3 + 1$

↳ $f(-x) = (-x)^3 + 1 = -x^3 + 1 \Rightarrow$ not even $(f(-x) \neq f(x))$

$-f(x) = -(x^3+1) = -x^3 - 1 \Rightarrow$ not odd $(f(-x) \neq -f(x))$

∴ neither even nor odd.

---

* even (+) , odd (−) ...

* even · even = even

* even · odd = odd

* odd · odd = even

* $\frac{\text{even}}{\text{odd}}$ = odd

* $\frac{\text{odd}}{\text{odd}}$ = even

* $\frac{\text{even}}{\text{even}}$ = even

* Quadratic equation : المعادلة التربيعية

* $ax^2 + bx + c \Rightarrow$ الصورة العامة لها .

* Discriminant : المميز

↳ $\Delta = b^2 - 4ac$ قانونه

* If : ① $\Delta > 0$, then the eqn. $(ax^2 + bx + c) = 0$ has two roots (solutions)
② $\Delta = 0$, then it has one solution (root).
③ $\Delta < 0$, then it has No solution (No roots).

* Quadratic formula for solving $ax^2 + bx + c = 0$ :-

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$ القانون العام المستخدم لحل العبارة التربيعية .

* Ex Solve :- $x^2 - x - 20 = 0$

↳ $x^2 - x - 20 = 0$

$(x-5)(x+4) = 0$

$\boxed{x=5}$ $\boxed{x=-4}$

OR $X = \frac{-(-1) \pm \sqrt{1+4(20)}}{2(1)}$

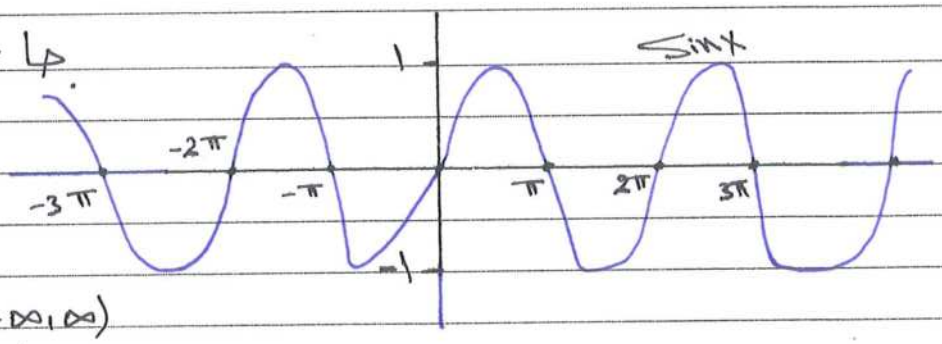
$= \frac{1 \pm \sqrt{81}}{2}$

$\frac{1+9}{2}$ $\frac{1-9}{2}$

$\therefore x = 5$ $\therefore x = -4$

## * Trigonometric Functions :- الإقترانات المثلثية (الدائرية)

### 1] $\sin x$ ≡ ب ـ ل



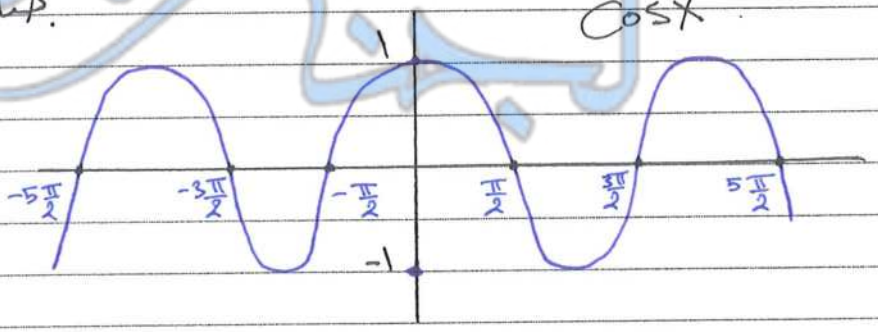

⇒ Domain $\sin x$ : $(-\infty, \infty)$
Range $\sin x$ : $[-1, 1]$

* $\sin x$ is periodic ( دوري ) with period = $2\pi$
* $\sin(x + 2\pi) = \sin x$ . دورة $2\pi$ كل ب ـ ل يكرر نفسه

* $\forall\ -1 \le \sin x \le 1$ ⇒ ب ـ ل محصورة قيمتها بين $-1, 1$
$\forall\ 0 \le \sin^2 x \le 1$
$\forall\ 0 \le |\sin x| \le 1$

* $\forall\ \sin x = 0$ at $x = 0, \pm\pi, \pm 2\pi, \pm 3\pi \dots$ اصفار $\sin x$
$\therefore x = n\pi$ , $n = 0, \pm 1, \pm 2 \dots$

### 2] $\cos x$ ≡ ب ـ ح



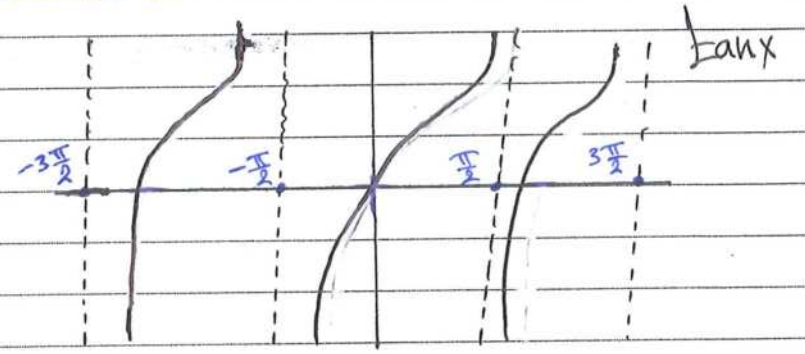

* D $\cos x$ = $(-\infty, \infty)$
R $\cos x$ = $[-1, 1]$

* $\cos x$ is periodic with period = $2\pi$
* $\cos(x + 2\pi) = \cos x$
* $-1 \le \cos x \le 1$
$0 \le \cos^2 x \le 1$
$0 \le |\cos x| \le 1$

* $\cos x = 0$ where $x = \pm \frac{\pi}{2}$
$\therefore x = \pm(2n+1)\frac{\pi}{2}$ $\pm 3\frac{\pi}{2}$
$\pm 5\frac{\pi}{2} \dots$
اصفار الجتا

* $\sin(2x)$ is Periodic with period $= \frac{2\pi}{2} = \pi$
  $\cos(5x)$ is Periodic with period $= \frac{2\pi}{5}$ ← 2 معامل الزاوية x

[3] $\tan x \equiv$ ظا




* $\tan x$ is periodic with period $= \pi$
* $\tan(x+\pi) = \tan x$
* $\tan x = \frac{\sin x}{\cos x}$

domain $\tan x$ $\quad \mathbb{R} - \{\pm(2n+1)\frac{\pi}{2}\}$ ← الظل هو اقترانه جاس / جتاس فنستبعد اصفار المقام

Range $\tan x$ $(-\infty, \infty) = \mathbb{R}$

* other Trigonometric fxn's :-

① $\sec x = \frac{1}{\cos x}$ قاس

② $\csc x = \frac{1}{\sin x}$ قتاس

③ $\cot x = \frac{\cos x}{\sin x} = \frac{1}{\tan x}$ ظتا

* $\sin x \Rightarrow$ odd fxn.
  $\sin(-x) = -\sin x$

* $\csc x = \frac{1}{\sin x}$ = odd fxn

* $\cos x \Rightarrow$ even fxn
  $\cos(-x) = \cos x$

* $\sec x = \frac{1}{\cos x}$ = even fxn

* $\tan x = \frac{\sin x}{\cos x}$ → odd / → even

* $\cot x = \frac{1}{\tan x}$ = odd fxn.

$\Rightarrow \therefore$ odd fxn.

ex3 which of the following is even, odd or neither ?

① $f(x) = x^2 \sin x$ ?

↳ $f(-x) = (-x)^2 \cdot \sin(-x)$

$= -x^2 \sin x$ ∴ odd

OR $x^2$: even

$\sin x$: odd

even · odd = odd fxn.

② $f(x) = \tan^2 x$ ?

↳ $\tan x$: odd ⇒ $(\tan x)^2 = (odd)^2 = (-)^2 = +$ ∴ even fxn.

③ $f(x) = \dfrac{\sin x}{x^4+1}$ ? (odd / even) ⇒ $\dfrac{odd}{even}$ = odd fxn.

④ homework $f(x) = \dfrac{\sec x}{\tan x + x^3}$ ?

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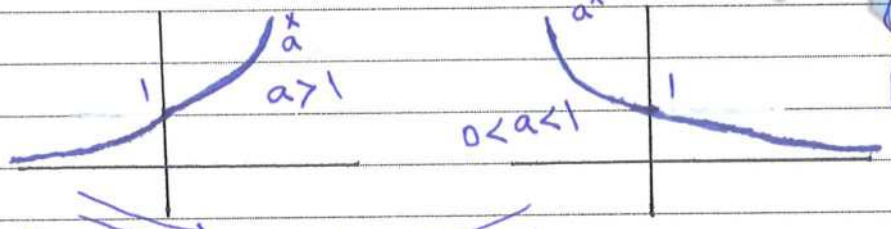
* Exponential function : الأسية اقترانات

↳ is of the form $a^x$, $a>0$.

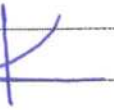
* $a^x > 0$ لا يمكن أن يساوي صفر أبداً




* Domaine $(-\infty, \infty)$

Range $(0, \infty)$

أشهر اقترانات أسية هو $e^x$ حيث e هو العدد النيبيري و يساوي $e \approx 2.718$




* Properties :

① $a^n = a \cdot a \cdot a \cdots a$ (n-times)

② $a^0 = 1$

③ $a^{-n} = \dfrac{1}{a^n}$

④ $(a^n)^m = a^{n \cdot m}$

⑤ $a^{\frac{m}{n}} = \sqrt[n]{a^m}$

⑥ $a^n \cdot a^m = a^{n+m}$

⑦ $\dfrac{a^n}{a^m} = a^{n-m}$

⑧ $(a \cdot b)^n = a^n \cdot b^n$

* Evaluate: (ب ـ ١)

1) $2^3$ ? $\Rightarrow (2)(2)(2) = 8$

2) $3^{-2}$ ? $\Rightarrow \frac{1}{3^2} = \frac{1}{9}$.

3) $9^0$ ? $\Rightarrow 9^0 = 1$

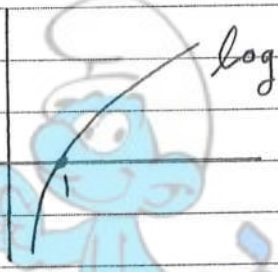
4) $4^{\frac{1}{2}}$ ? $\Rightarrow \sqrt[2]{4^1} = 2$

5) $5^{\frac{3}{2}}$ ? $\Rightarrow \sqrt[2]{5^3} = \sqrt{125}$

6) $(x y^3)^2$ ? $\Rightarrow x^2(y^3)^2 = x^2 y^6$.

* Logarithmic function :- اقتران اللوغاريتمي

↳ is the farm $\log_a^x$ = Logarithm x to the base a.




$\Rightarrow$ D $\log^x$ = $(0,\infty)$
R $\log^x$ = $(-\infty,\infty)$

* Notes ① Logarithmic function is the inverse (معكوس) of exponential function

② $0 <$ (اللوغاريتم) ما داخل ∴ مجال اللوغاريتم

* Logarithms :- $\log_b^a = z \Longrightarrow$ means that :- $b^z = a$

* Proper ties :-

① $\log_b^{(ac)} = \log_b^a + \log_b^c$

② $\log_b^{(\frac{a}{c})} = \log_b^a - \log_b^c$

③ $\log_b^{a^n} = n \log_b^a$

④ $\log_b^{(\frac{1}{c})} = \log_b^{c^{-1}} = -\log_b^c$

⑤ $\log^a = \log_{10}^a$

⑥ $\log_e^a = \ln a$

⑦ $\log_b^{(a \pm c)} \neq \log_b^a \pm \log_b^c$

⑧ $\log_b^a = \frac{\ln a}{\ln b}$.

* Evaluate :-

① $\log_2^{16} = ?$ نفرض أن $\log_2^{16} = x$ ثم نحوّل للصيغة الأسية

↳ $2^x = 16$

$2^x = 2^4 \Rightarrow (x = 4)$ * اذا تساوت الأساسات تتساوى الأسس

② $\log_{10}^{100} = ?$

↳ $10^x = 100 \Rightarrow 10^x = 10^2 \Rightarrow (x = 2)$

③ $\log_{10}^{\frac{1}{1000}} = ? \rightarrow \log_{10}^{10^{-3}} = -3$.

④ $\log_{\frac{1}{4}}^{1} = ? \rightarrow$ Zero (0).

⑤ $\ln 1 = ? \rightarrow \log_e^1 =$ Zero.

⑥ $\ln e = ?$

↳ $\log_e^e = 1$

⑦ $\log_8^8 = 1$

⑧ $\log_2^4 = \frac{\ln 4}{\ln 2}$.

* ex3 expand $\log\left(\frac{xy^5}{\sqrt{z}}\right)$ ?

↳ $= \log(xy^5) - \log(\sqrt{z}) = \log x + \log y^5 - \log(z)^{\frac{1}{2}}$

$= \log x + 5\log y - \frac{1}{2}\log z$.

* ex3 Write as Single Logarithm :-

[1] $5\log 2 + \log 3 - \log 8$ ?

↳ $\log 2^5 + \log 3 - \log 8 = \log^{32} + \log^3 - \log 8 = \log(32*3) - \log 8$

$= \log\left(\frac{32*3}{8}\right) = \log(12)$.

[2] $\frac{1}{3}\ln x - \ln(x^2-1) + 2\ln(x+3)$ ?

$\ln x^{\frac{1}{3}} - \ln(x^2-1) + \ln(x+3)^2$

$= \ln\left(\frac{\sqrt[3]{x}}{x^2-1}\right) + \ln(x+3)^2 = \ln\left(\frac{\sqrt[3]{x}\,(x+3)^2}{x^2-1}\right)$.

---

* find domain ?

* القاعدة : ما داخل ln, log يجب ان يكون اكبر من صفر

[1] $f(x) = \ln(x-3)$.

$x-3>0 \therefore (3,\infty)$

$x>3$

[3] $f(x) = \log_7(x-1)$.

$(x-1)>0$

$x>1 \therefore (1,\infty)$

[2] $f(x) = \ln(8-x)$.

$8-x>0 \therefore (-\infty,8)$

$8>x$

[4] $f(x) = \ln(2x-3)$.

$2x-3>0$

$x>\frac{3}{2} \therefore (\frac{3}{2},\infty)$.

---

* Rule :- ① $\ln e^{f(x)} = f(x)$ ② $\log_b b^{f(x)} = f(x)$ ③ $e^{\ln f(x)} = f(x)$ ④ $b^{\log_b f(x)} = f(x)$

---

* Evaluate :-

① $e^{\ln 7} = 7$

② $\ln(e^9) = 9$

③ $e^{\ln(\ln 3)} = \ln 3$

④ $\ln(\ln e^{e^{10}}) = \ln(e^{10}) = 10$

⑤ $e^{\ln \pi} = \pi$

⑥ $e^{2\ln\pi} = e^{\ln \pi^2} = \pi^2$

⑦ $\ln(2e^7) = \ln 2 + \ln e^7$

$= \ln 2 + 7$.

⑧ $\ln(2e^{\pi+\ln 2}) =$

$\ln 2 + \ln e^{\pi+\ln 2}$

$\ln 2 + \pi + \ln 2 = 2\ln 2 + \pi$

$= \ln 4 + \pi$.

* Find domaine:-

① $f(x) = \frac{7}{\ln(5-x)}$ ?

↳ $5-x > 0$

$5 > x \Rightarrow (-\infty, 5)$

أصفار المقام $\Rightarrow \ln(5-x) = 0 \Rightarrow$ (e) * نرفع القوة

$5-x = 1$

$x = 4$ ∴ $D_f : (-\infty, 5) - \{4\}$

OR : $(-\infty, 4) \cup (4, 5)$.

② $f(x) = \sqrt{3 - e^{2x}}$ ?

↳ $3 - e^{2x} \geq 0$

$3 \geq e^{2x}$

$\ln 3 \geq \ln e^{2x}$

$\ln 3 \geq 2x \Rightarrow x \leq \frac{1}{2}\ln 3$

∴ $D_f : (-\infty, \frac{1}{2}\ln 3]$.

* Solve (find x) ?

① $\ln x > -1$ ?

↳ $e^{\ln x} > e^{-1}$

$x > e^{-1} \Rightarrow (\frac{1}{e}, \infty)$

② $\log_{10}(1+x) = 3$ ?

↳ $1+x = 10^3$ → نحول للصيغة الأسية

$1+x = 1000$

$x = 1000 - 1$

∴ $x = 999$

(3) $\log^{x} = \sqrt{2}$ ?

↳ $10^{\sqrt{2}} = x$ .

(4) $\ln(x+1) = 5$ ?

↳ $e^{\ln(x+1)} = e^{5} \Rightarrow x+1 = e^5 \therefore x = e^5 - 1$ .

OR $\ln(x+1) = 5$

$\log_e^{(x+1)} = 5$

$e^5 = x+1 \Rightarrow \therefore x = e^5 - 1$ .

(5) $5^{2x} = 8$ ? $\log_5 5^{2x} = \log_5 8$

↳ $2x \log_5^5 = 8$

$2x = 8 \Rightarrow \boxed{x = 4}$

(6) $5^{x} = 7$ ?

↳ $\log_5^{5^x} = \log_5^7$

$x = \log_5^7$

(7) $3^{-2x} = 2$ ?

↳ $\log_3^{3^{-2x}} = \log_3^2$

$-2x = \log_3^2$

$\therefore x = -\frac{1}{2}\log_3^2$

(8) $\log_{10}^{x^2} + \log_{10}^{x} = 30$ ?

↳ $\log_{10}(x^2 \cdot x) = 30$

$\log_{10}^{x^3} = 30$

$x^3 = 10^{30}$ (نأخذ الجذر التكعيبي)

$(x^3)^{1/3} = (10^{30})^{1/3}$

$\therefore x = 10^{10}$

(9) $2e^{3x} = 7$ ?

↳ $\ln(2e^{3x}) = \ln 7$

$\ln 2 + \ln e^{3x} = \ln 7$

$\ln 2 + 3x = \ln 7$

$3x = \ln 7 - \ln 2 \rightarrow \ln(\frac{7}{2})$

$\therefore x = \frac{1}{3}\ln(\frac{7}{2})$ .

(10) $\ln\left(\frac{1}{x}\right) + \ln(2x^3) = \ln 3$ ?

$\ln\left(\frac{1}{x} \cdot 2x^{\frac{2}{3}}\right) = \ln 3$

$\ln(2x^2) = \ln 3$

$e^{\ln(2x^2)} = e^{\ln 3}$

$2x^2 = 3 \Rightarrow x^2 = \frac{3}{2} \Rightarrow x = \pm\sqrt{\frac{3}{2}}$

* لكنه نهمل السالب لأنه ماداخل (ln) يجب ان يكون موجب .

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(11) $(25)^x + 5^x = 6$ ?

suppose that $5^x = z$ نفرض ان ، $25^x = (5^x)^2$

$\Rightarrow (5^x)^2 + 5^x = 6$

$z^2 + z = 6$

$z^2 + z - 6 = 0 \Rightarrow (z+3)(z-2) = 0$

$z = -3$ ، $z = 2$

$5^x = -3$ وهذا لا يجوز لأنه مدى الأس يكون اكبر من صفر $(0,\infty)$

$5^x = 2$

$\log_5^{5^x} = \log_5^2$

$x = \log_5^2$

(12) $\frac{e^{x}-e^{-x}}{2}=1$ ؟

$\hookrightarrow (e^{x}-e^{-x}=2)$ نضرب كل المعادلة بـ $e^{x}$

$e^{2x}-1=2e^{x}$

$e^{2x}-2e^{x}-1=0 \Rightarrow$ نفرض $e^{x}=y$ لتسهيل الحل .

$y^{2}-2y-1=0$

$(y-?)(y+?)=0 \Rightarrow \Delta=b^{2}-4ac=4-4(1)(-1)=8$

$\Rightarrow$ نلجأ للقانون العام $\Rightarrow y=\frac{-b\pm\sqrt{\Delta}}{2a}=\frac{2\pm\sqrt{8}}{2}$

$y=1\pm\sqrt{2}$

$e^{x}=1\pm\sqrt{2}$

$e^{x}=1+\sqrt{2}$ ←

$\ln e^{x}=\ln(1+\sqrt{2})$

$x=\ln(1+\sqrt{2})$

$e^{x}=1-\sqrt{2}$ X

⇓

نهملها لأنه جواب $e^{x}$ يجب انه يكون اكبر من صفر

Range $e^{x}=(0,\infty)$

ex3 find domain: $\frac{13}{1-e^x}$ ?

↳ نبحث عن : $1-e^x=0$

المقام

$e^x=1$

$\ln e^x=\ln 1$

$\therefore x=0$

$\therefore D_f=\mathbb{R}-\{0\}$

---

ex3 Evaluate

$\log_2^3 \cdot \log_3^4 \cdot \log_4^5 \cdots \log_{14}^{15} \cdot \log_{15}^{16}$ ?

↳ $\frac{\ln 3}{\ln 2}\cdot\frac{\ln 4}{\ln 3}\cdot\frac{\ln 5}{\ln 4}\cdots\frac{\ln 15}{\ln 14}\cdot\frac{\ln 16}{\ln 15}$

$=\frac{\ln 16}{\ln 2}=\frac{\ln 2^4}{\ln 2}=\frac{4\ln 2}{\ln 2}=4$

OR $\log_2^3 \cdot \log_3^4 \cdot \log_4^5 \cdots \log_{14}^{15} \log_{15}^{16}$

$=\log_2^{16}=x$

$\rightarrow 2^x=16 \rightarrow 2^x=2^4$

$\therefore x=4$

---

## * New functions from old.

If $f(x)$ and $g(x)$ two functions, then:

1) $(f\pm g)(x)=f(x)\pm g(x)$

↳ $D_{f\pm g}=D_f\cap D_g$

2) $(f\cdot g)(x)=f(x)\cdot g(x)$

↳ $D_{f\cdot g}=D_f\cap D_g$

3) $\left(\frac{f}{g}\right)(x)=\frac{f(x)}{g(x)}$ , $g(x)\neq 0$

↳ $D_{\frac{f}{g}}=D_f\cap D_g-\{\text{Zero's of } g(x)\}$

---

ex3 find $f+g$, $f-g$, $f\cdot g$ and $\frac{f}{g}$ and state the domains :-

$f(x)=2\sqrt{x-1}$ , $g(x)=\sqrt{x-1}$ ?

↳ ① $f+g=f(x)+g(x)$

$=2\sqrt{x-1}+\sqrt{x-1}=3\sqrt{x-1}$

$\Rightarrow D_{f+g}=D_f\cap D_g=[1,\infty)$.

② $f-g=f(x)-g(x)$

$=2\sqrt{x-1}-\sqrt{x-1}=\sqrt{x-1}$

$\Rightarrow D_{f-g}=D_f\cap D_g=[1,\infty)$

③ $f \cdot g = f(x) \cdot g(x)$
$= 2\sqrt{x-1} \cdot \sqrt{x-1}$
$= 2(x-1) \cdot = 2x-2$

$D_{f \cdot g} = D_f \cap D_g = [1,\infty)$

④ $\frac{f}{g} = \frac{f(x)}{g(x)} = \frac{2\sqrt{x-1}}{\sqrt{x-1}} = 2$

$D_{\frac{f}{g}} = D_f \cap D_g - \{ \text{اصفار } g \}$
$= [1,\infty) - \{1\} = (1,\infty)$.

---

* $f(x) = \frac{x}{1+x^2}$ , $g(x) = \frac{1}{x}$

↳ ① $f+g = f(x) + g(x) = \frac{2x^2+1}{x(x^2+1)}$.

$D_{f+g} = D_f \cap D_g$
$= \mathbb{R} \cap (\mathbb{R} - \{0\}) = \mathbb{R} - \{0\}$

② $(f-g)(x) = \frac{-1}{x(x^2+1)}$ , $D_{f-g} = \mathbb{R} - \{0\}$

③ $(f \cdot g)(x) = \frac{1}{(x^2+1)}$ , $D_{f \cdot g} = \mathbb{R} - \{0\}$

Note :- $D_{f+g} = D_{f-g} = D_{f \cdot g} = D_f \cap D_g$

④ $\left(\frac{f}{g}\right)(x) = \frac{x^2}{x^2+1}$

↳ $D_{\frac{f}{g}} = D_f \cap D_g - \{ \text{اصفار } g \}$

$\therefore = (\mathbb{R} - \{0\}) - \{0\}$
$= \mathbb{R} - \{0\}$.

---

* Find the domain of the function

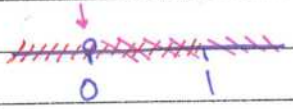
① $f(x) = \sqrt{1-x} \cdot \ln x$ ?

a) $(0,\infty)$ b) $(-\infty,1)$

c) $(0,1]$ d) $[0,1]$

↳ $D_{\sqrt{1-x}}$ : $1-x \geq 0$
$1 \geq x \Rightarrow (-\infty,1]$

→ $\ln x \Rightarrow D_{\ln x} (0,\infty)$ , $x > 0$.

→ $D_{\sqrt{1-x}} \cap D_{\ln x}$ 


$\therefore = (0,1]$ ⓒ نعم

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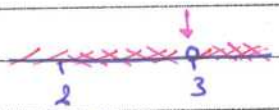
② $f(x) = \frac{\sqrt{x-2}}{x-3}$ ?

↳ $D_{\sqrt{x-2}} = [2,\infty)$ , $x-2 \geq 0$

$D_{(x-3)} = \mathbb{R}$ ، لا تنتهي المقام !

أصفار المقام :- $x-3=0 \Rightarrow x=3$

$\therefore D_{\sqrt{x-2}} \cap D_{(x-3)}$ 


$\therefore D_f = [2,3) ; (3,\infty)$

---

③ $f(x) = \ln\left(\frac{x-3}{x^2+x+1}\right)$ ?

$\frac{x-3}{x^2+x+1} > 0$

$x-3 > 0 \rightarrow x > 3$

$\therefore D_f = (3,\infty)$.

* المقام لا يحل وليس له أصفار

④ $f(x) = \frac{3x^2+1}{\ln(3-x)}$ ?

↳ $D_{(3x^2+1)} = \mathbb{R}$

$D_{\ln(3-x)} \rightarrow 3-x > 0$

$3 > x \Rightarrow (-\infty, 3)$.

أيضاً مقام :- $\ln(3-x) = 0$

$\ln(3-x) \quad 0$

$\not{e} \quad = e$

$3-x = 1 \Rightarrow \therefore x = 2$

$\therefore$


$\therefore D_{f(x)} = (-\infty, 3) - \{2\}$

OR $(-\infty, 2), (2, 3)$.

---

homework $f(x) = x^2$ , $g(x) = x-2$

find $D_{\frac{f}{g}}$ ?

↳ Sol $\mathbb{R} - \{2\}$.

---

* $f(x) = x$ , $g(x) = \frac{-1}{x}$ , find :-

$D_{f+g}$ , $D_{f-g}$ , $D_{f \cdot g}$

↳ * السؤال عادة يطلب فقط واحدة

$\rightarrow D_f = \mathbb{R}$.

$D_g = \mathbb{R} - \{0\}$.

$D_{f+g} = D_{f-g} = D_{f \cdot g} = D_f \cap D_g$

$= (\mathbb{R} \cap \mathbb{R} - \{0\}) = \mathbb{R} - \{0\}$.

---

homework $f(x) = 1 + \sqrt{x+2}$ , $g(x) = x+3$

find $D_{\frac{f}{g}}$ ?

↳ Sol $[-2, \infty) - \{3\}$.

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* Composition Functions :-

تركيب الإقترانات

* $(f \circ g)(x) = f(g(x))$ { تركيب $g(x)$ داخل $f(x)$

ex $f(x) = x^2$ , $g(x) = x+1$ , find

① $(f \circ g)(x)$ , ② $(g \circ f)(x)$ ?

↳ ① $(f \circ g)(x) = f(g(x)) = f(x+1)$

$= (x+1)^2$.

→ ② $(g \circ f)(x) = g(f(x)) = g(x^2)$

$= x^2 + 1$

* نستنتج من المثال السابق أن $f \circ g \neq g \circ f$

* $D_{f \circ g} = D_{f(g(x))} \cap D_{g(x)}$

(الناتج من التركيب) (الداخلي)

* $D_{g \circ f} = D_{g(f(x))} \cap D_{f(x)}$

(الناتج) (الداخلي)

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ex $f(x) = x^2 + 3$ , $g(x) = \sqrt{x}$ , find

$D_{f \circ g}$ , $D_{g \circ f}$ ?

↳ $f \circ g(x) = f(g(x)) = f(\sqrt{x})$

$= (\sqrt{x})^2 + 3 = x + 3$.

* $f(x)=\frac{1+x}{1-x}$ , $g(x)=\frac{x}{1-x}$ find

$D_{fog}$ ?

$\hookrightarrow f(g(x)) = f\left(\frac{x}{1-x}\right) = \frac{1+\frac{x}{1-x}}{1-\frac{x}{1-x}}$

* انتبه : لا توجد مقامات

$\rightarrow D_{f(g(x))} = D_{\text{بسط}} \cap D_{\text{مقام} - \{\text{أصفار مقام}\}}$

$= (\mathbb{R}-\{1\} \cap \mathbb{R}-\{1\}) - \{\frac{1}{2}\}$

$\therefore = \mathbb{R} - \{1, \frac{1}{2}\}$.

$\Rightarrow D_{fog} = D_{f(g)} \cap D_g$

$= (\mathbb{R}-\{1,\frac{1}{2}\}) \cap (\mathbb{R}-\{1\})$

$= \mathbb{R} - \{1, \frac{1}{2}\}$.

* homework find $D_{gof}$ ?

→ sol : $\mathbb{R} - \{0,1\}$

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* $(fogoh)(x) = f(g(h(x)))$ ...

↳ ex) $f(x)=\sqrt{x}$ , $g(x)=\frac{1}{x}$ , $h(x)=x^2$

find $(fogoh)(x)$ ?

↳ $(fogoh) = f(g(h(x)))$.

$= f(g(x^2)) = f\left(\frac{1}{x^2}\right)$

$= \sqrt{\frac{1}{x^2}} = \frac{1}{\sqrt{x^2}} = \frac{1}{|x|}$.

---

$D_{fog} = D_{f(g(x))} \cap D_{g(x)}$.

$= \mathbb{R} \cap [0,\infty)$

$= [0,\infty)$.

↳ $(gof)(x) = g(f(x)) = \sqrt{x^2+3}$

$D_{gof} = D_{g(f(x))} \cap D_{f(x)}$

$= \mathbb{R} \cap \mathbb{R}$

$= \mathbb{R}$

---

* $f(x)=\sqrt{x}$ , $g(x)=\sqrt{2-x}$ , find :

$D_{fog}$ , $D_{gog}$ ?

↳ $fog = f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}} = \sqrt[4]{2-x}$

$D_{fog} = D_{f(g(x))} \cap D_{g(x)}$

$= (-\infty,2] \cap (-\infty,2]$

$= (-\infty,2]$.

↳ $(gog)(x) = g(\sqrt{2-x}) = \sqrt{2-\sqrt{2-x}}$.

↳ $D_{g(g(x))}$ : $2-\sqrt{2-x} \geq 0$

$2 \geq \sqrt{2-x}$

$4 \geq 2-x$

$\therefore x \geq -2 \Rightarrow [-2,\infty)$

$\Rightarrow D_{gog} = D_{g(g(x))} \cap D_{g(x)}$.

$= [-2,\infty) \cap (-\infty,2]$

$= [-2,2]$

*homework If $f(x)=\sqrt{x-1}$, $g(x)=\sqrt{x-3}$, then the domain of gof is..... ?

↳ sol $[10,\infty)$

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* If $f(x)=\frac{x+1}{x^2+1}$, $g(x)=\sqrt{x^2+4x-1}$ then $(fog)(1)$ ?

↳ $f(g(1)) = f(2) = \frac{3}{5}$

$\Rightarrow g(1) = \sqrt{1+4-1} = \sqrt{4} = 2$

---

* If the domain of $f(x)$ is $[-4,3)$ and $g(x)=2x+1$, then the domain of $(fog)(x)$ ?

↳ $fog = f(g(x)) = f(2x+1)$

$-4 \le 2x+1 < 3$

$-1 \quad -1 \quad -1$

$\frac{-5}{2} \le \frac{2x}{2} < \frac{2}{2}$

$-\frac{5}{2} \le x < 1 \Rightarrow [-\frac{5}{2}, 1)$

---

*homework Given that the function $f(x)=\sqrt{x-1}$ and, $g(x)=\frac{1}{x^2+1}$, the domain of $(gof)(x)$ is ...

a) $[1,\infty)$ b) $(-\infty,1),(1,\infty)$

c) $(0,\infty)$ d) $(-\infty,0),(0,\infty)$

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*homework The value of $gof(-1)$ when $f(x)=\frac{x+4}{x}$, $g(x)=x^2-2$

a) 5 b) 6 c) 8

d) 9 e) 7

---

*homework $f(x)=\sqrt{x}$. Find $(fof)(x)$

a) $\sqrt{x}$, $x\ge 0$ b) $\sqrt[4]{x}$, $x\ge 0$

c) $x$ d) None of these.

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*homework $f(x)=\frac{7}{x+4}$, $g(x)=\frac{4}{5x}$, find $(fog)(x)$ :-

a) $\frac{35x}{4-20x}$ b) $\frac{35x}{4+20x}$

c) $\frac{4x+16}{35x}$ d) $\frac{7x}{4+20x}$ ?

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(أسئلة بوليتكنك) تاريخ 1-11-2016

* Given that $f(x)=\frac{x^2+6}{x^2-2x}$ and $g(x)=\frac{1}{x^2-1}$, then $D_{gof}$ is ? $x\neq 2x$

↳ $gof = \frac{1}{\left(\frac{x^2+6}{x^2-2x}\right)^2-1}$

اضفنا مقام

$D_{f(g(x))} = \mathbb{R} \cap (\mathbb{R}-\{0,2\}) - \{3\}$

$\therefore = \mathbb{R} - \{-3,0,2\}$

$\rightarrow D_{gof} = D_{g(f(x))} \cap D_f$

$= (\mathbb{R}-\{-3,0,2\}) \cap (\mathbb{R}-\{0,2\})$

$= \mathbb{R}-\{-3,0,2\}$.

## * Decomposition (فك التركيب)

ex) Express the following functions as composition of functions fog ?

↳ $(fog)(x) = f(g(x))$

← الـ g هو الاقتران المطبق على ... أول

① $h(x) = (x-4)^2$

↳ $f(x) = x^2$ , $g(x) = x-4$

② $h(x) = \sqrt{4-3x}$

↳ $f(x) = \sqrt{x}$ , $g(x) = 4-3x$

③ $q(x) = |x^2-3x+5|$

↳ $f(x) = |x|$ , $g(x) = x^2-3x+5$

④ $u(x) = [\sin x]$

↳ $f(x) = [x]$ , $g(x) = \sin x$

⑤ $h(x) = \frac{1}{x-3}$ ↳ $f(x) = \frac{1}{x}$ , $g(x) = x-3$

⑥ $r(x) = \sin(x^3)$

↳ $f(x) = \sin x$ , $g(x) = x^3$

⑦ $q(x) = \sin^3 x$

↳ $f(x) = x^3$ , $g(x) = \sin x$

## * Translations :-

$y = f(x)$

* التأثير الخارجي على الاقتران يحرك لأعلى أو لأسفل :-

1) $f(x)+c$ : translates $f(x)$ upward by $c$ units.

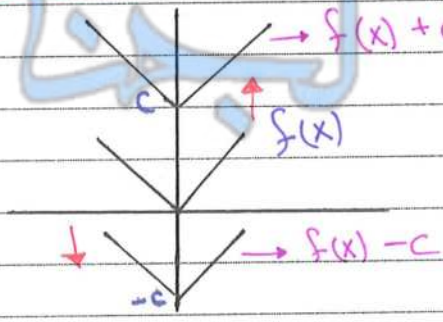
2) $f(x)-c$ : translates $f(x)$ downward by $c$ units.

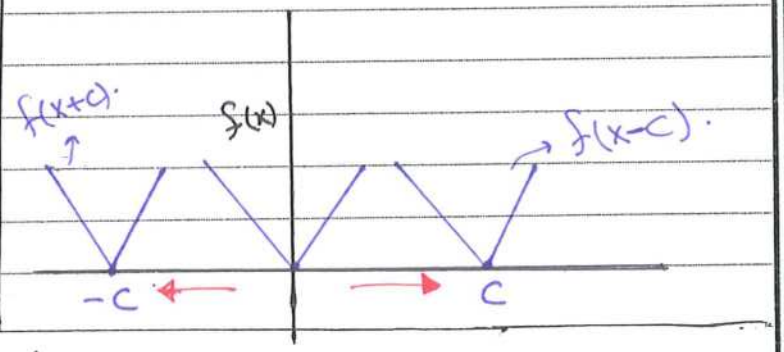
* التأثير الداخلي على الاقتران يحرك إلى اليمين أو اليسار :-

1) $f(x+c)$ : translates $f(x)$ to the left $c$ units.

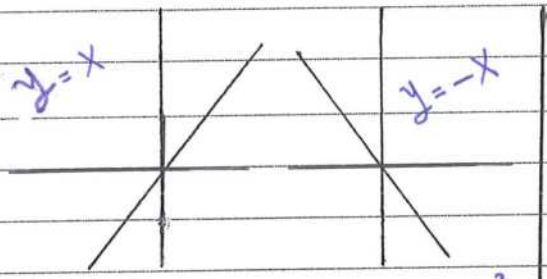
2) $f(x-c)$ : translates $f(x)$ to the right $c$ units.

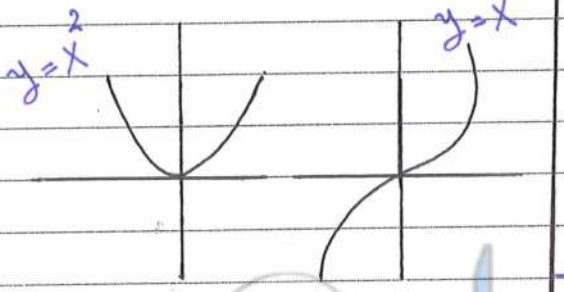
* $f(x \pm k) \pm c$ ← التأثير الخارجي . ← التأثير الداخلي .

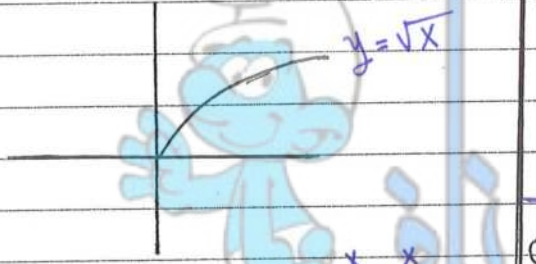


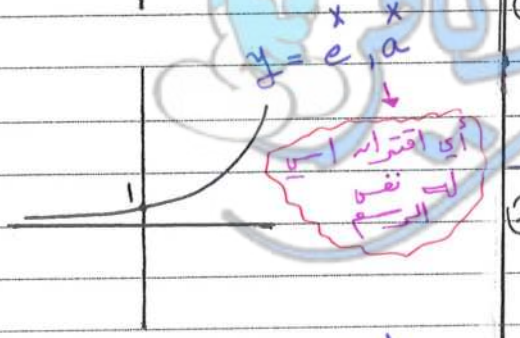



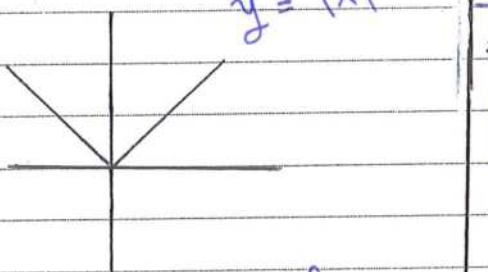
## بعض الرسمات الهامة

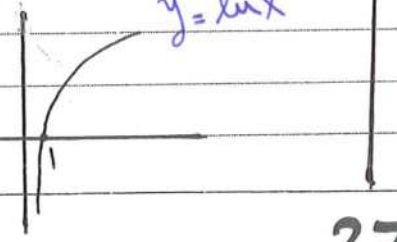


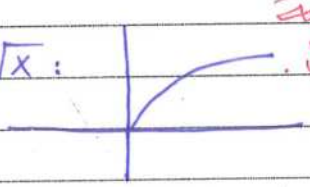



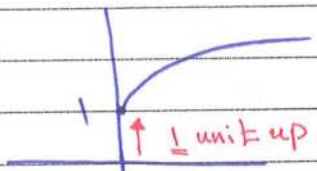

*ex) Sketch ? (ارسم)

① $\sqrt{x}+1$.

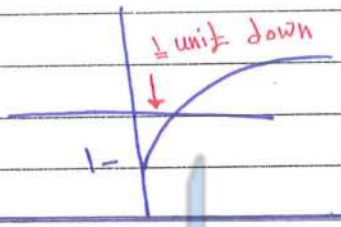
↳ $\sqrt{x}$ :



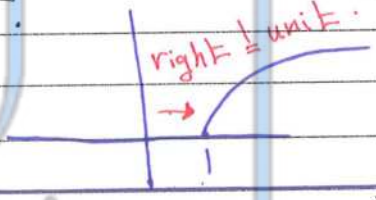

$\sqrt{x}+1$ :



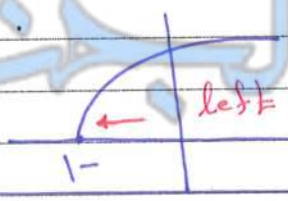

② $\sqrt{x}-1$.




③ $\sqrt{x-1}$.

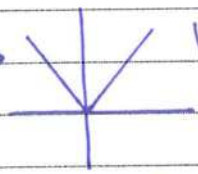



④ $\sqrt{x+1}$.

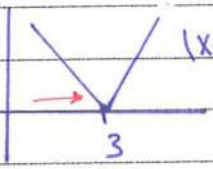



* Sketch $|x-3|+2$, and find range?

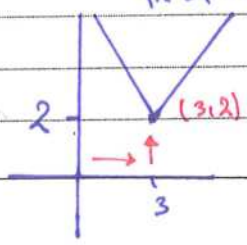
↳ $|x|$.



$|x-3|$




$|x-3|+2$.




⇒ Range : [2, 3)

العدى : من y-axis .

* يفيدنا الرسم في تحديد الـ Range الإقتراب وقد نستطيع إيجاده من دون الرسم ...

ex3 Find Range ?

① $f(x) = |x-3| + 2$ .

↳ $|x-3| \geq 0$ أصغر قيمة مطلقة أكبر أو يساوي صفر مهما كان داخلها ...

$|x-3|+2 \geq 0+2$

$|x-3|+2 \geq 2 \Rightarrow \therefore [2, \infty)$ .

② $f(x) = e^x + 3$ .

↳ Range $e^x = (0, \infty)$ دائماً موجب وأكبر من صفر.

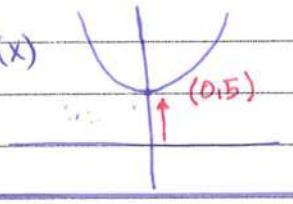
$e^x > 0 \Rightarrow e^x + 3 > 0 + 3$

$e^x + 3 > 3 \Rightarrow \therefore (3, \infty)$

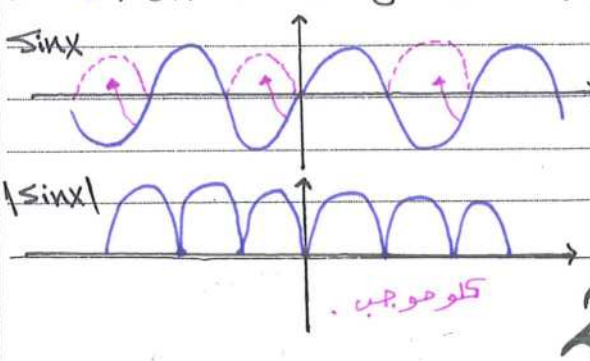
③ $f(x) = x^2 + 5$ .

↳ $x^2 \geq 0$

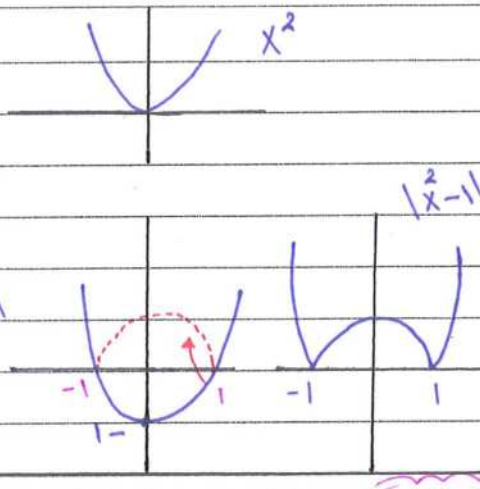
$x^2 + 5 \geq 0 + 5 \Rightarrow \therefore [5, \infty)$




* Sketch :- ① $y = |\sin x|$ ?




② $y = |x^2 - 1|$ ?




* إكمال المربع ⇐ نستخدمها إذا وجد اقتراب تربيعي على صورة $ax^2 + bx + c$ ... شروطها :

1) أن يكون معامل $x^2 = 1$

2) نصنف ونطرح $\left(\frac{\text{معامل } x}{2}\right)^2$ .

*ex $x^2 + 4x + 5$ ?

↳ $x^2 + 4x + 4 - 4 + 5$ ← $1 = x^2$ معامل

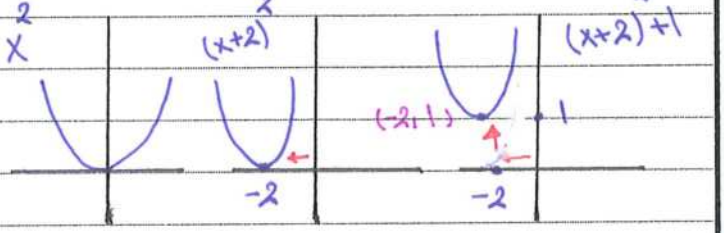
$4 = 2^2 = \left(\frac{4}{2}\right)^2 = \left(\frac{\text{معامل } x}{2}\right)^2$

$\rightarrow (x+2)(x+2) + 1 = (x+2)^2 + 1$ .

ex3 Sketch $x^2 + 4x + 5$ ?

↳ $x^2 + 4x + 5 = (x+2)^2 + 1$ من المثال السابق

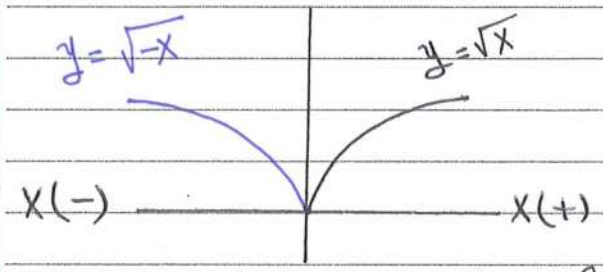
ظطوتا بـ لليسار . ظطوة لأعلى



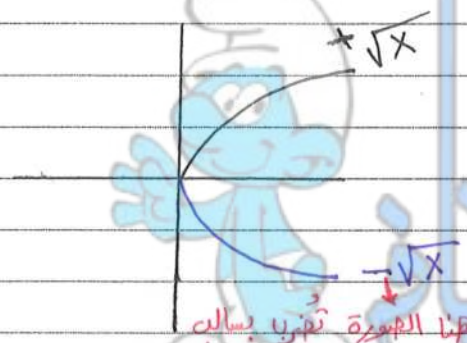

* Two units left and one unit up...

## * Reflection (الانعكاس)

→ $y = f(-x)$ is reflection for $f(x)$ about y-axis.

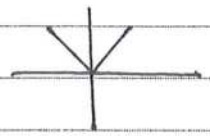



→ $y = -f(x)$ is reflection for $f(x)$ about X-axis.

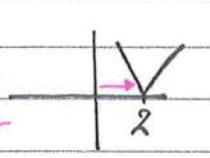



**ex3 Sketch $y = 4 - |x-2|$ and find the Range ?**

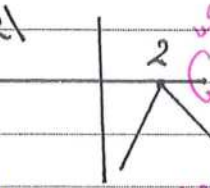
↳ $|x|$



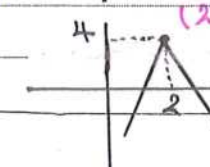
$|x-2|$ ظطويته بحيبه




$-|x-2|$ نعكس




$4-|x-2|$




∴ Range : $(-\infty, 4]$ .

**OR : Range $4-|x-2|$ :-**

↳ $|x-2| \geq 0$

$-|x-2| \leq 0$ عندما نضرب بسالب نعكس العتبا نيك

$4-|x-2| \leq 4$

∴ $(-\infty, 4]$

*** Find Range of $f(x) = 3 - 2^x$ ?**

↳ $2^x > 0$ x

$-2^x < 0 \Rightarrow 3-2 < 3$

∴ Range $(-\infty, 3)$

*** Find Range $-x^2 + 2x + 2$ ?**

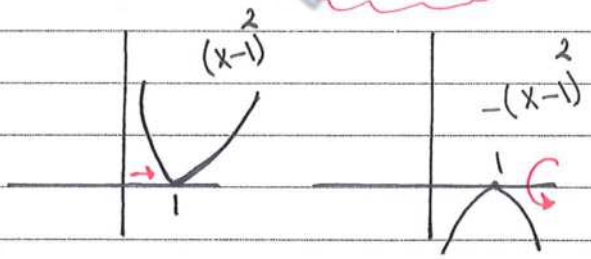
↳ هنا يجب انه نرسم ، نكمل مربع حتى تتضح الرسم ..

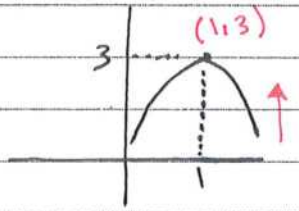
$\Rightarrow -(x^2 - 2x - 2) \Rightarrow$ ليصبح معامل $x^2$ يساوي 1

$-(x^2 - 2x + 1 - 1 - 2)$ → $(\frac{\text{معامل } x}{2})^2$ جمع وطرح

$-((x-1)^2 - 3)$

↳ $= 3 - (x-1)^2$




∴ Range $(-\infty, 3]$ .

سؤال بوليتكنك 2016

* Let $g(x)$ be the function obtained from $f(x) = |x| + x$ by shifting 3 units to the right, 2 units up and reflected about the x-axis, then $g(4) = \ldots$

(a) $-8$ (b) $-2$

(c) $-14$ (d) $-4$

$g(x) \Rightarrow$ 3 unit right (تأثير داخلي)

2 unit up (تأثير خارجي)

reflection x-axis (تأثير خارجي) $\rightarrow -f(x)$

$\therefore g(x) = -(|x-3| + (x-3) + 2)$

$g(4) = -(1 + 1 + 2)$

$\therefore g(4) = -4$ فرع (d)

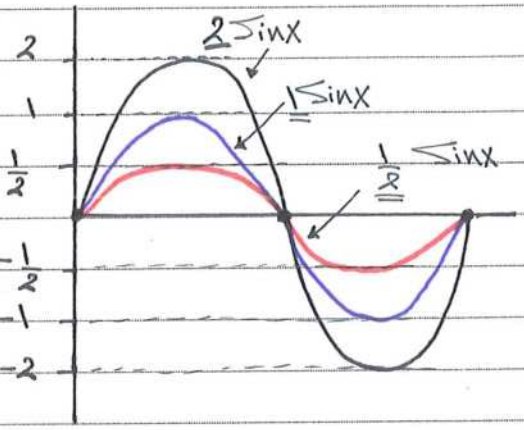
* Stretches and Compression :- التمدد و الانضغاط

* التأثير الخارجي *

* If $c > 1$, then $cf(x)$ stretches $f(x)$ vertically ($c$ units). تمدد (عامودياً)

* If $0 < c < 1$, then $cf(x)$ compresses $f(x)$ vertically $c$ units. بضغط

ex




* تمدد وانضغاط بشكل عامودي ...

* لاحظ ان التأثير الخارجي يؤثر على الـ Range ∴

$R_{\sin x}$ $[-1, 1]$

$R_{2\sin x}$ $[-2, 2]$

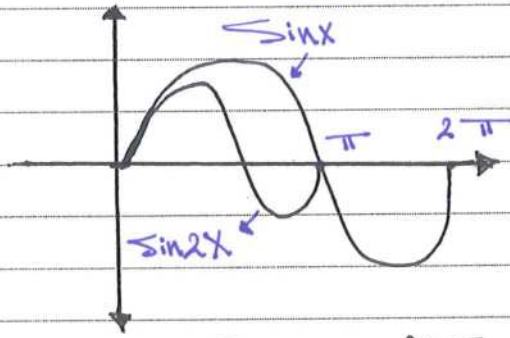
$R_{\frac{1}{2}\sin x}$ $[-\frac{1}{2}, \frac{1}{2}]$

* التأثير الداخلي *

* If $c > 1$, then $f(cx)$ Compresses $f(x)$ horizontally (أفقياً)

* If $0 < c < 1$, then $f(cx)$ Stretches $f(x)$ horizontally.

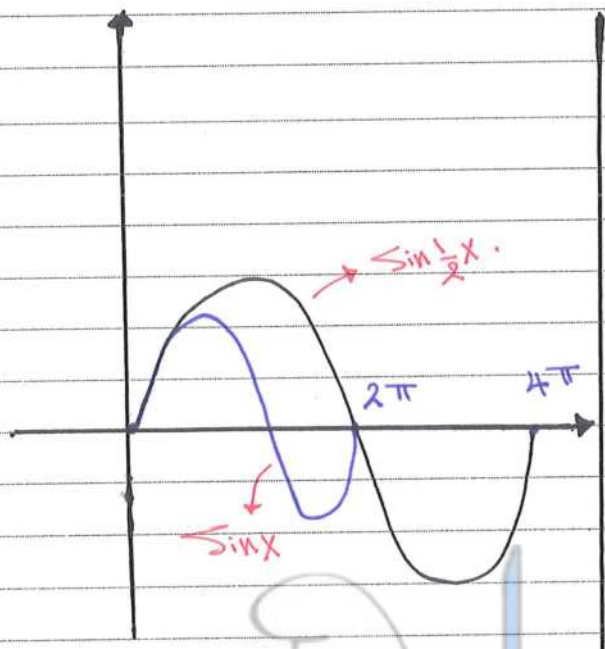
ex




Period

$\sin x : 2\pi$

$\sin 2x : \frac{2\pi}{2} = \pi$




Period

- $\sin x$ : $2\pi$
- $\sin \frac{x}{2} = \frac{2\pi}{\frac{1}{2}} = 4\pi$.

* Find the function g that is obtained from $f(x) = |4x+1|$ by the following steps:-

a) Reflection the graph of $y = f(x)$ about the y-axis?

↳ y-axis : $f(-x) = g$.

→ $g(x) = |4(-x)+1|$

$= |-4x+1|$.

b) Shifting 5 units down?

↳ $g(x) = f(x) - 5$

$= |4x+1| - 5$.

c) Stretching vertically by a factor of 2 units?

↳ $g(x) = 2f(x)$

$= 2|4x+1|$

d) Reflecting about x-axis?

↳ $g(x) = -f(x) = -|4x+1|$.

e) 2 units to the left, and 3 units up?

↳ $g(x) = f(x+2) + 3$

$= |4(x+2)+1| + 3$

$= |4x+9| + 3$.

## * Symmetry (تناظر)

a) Symmetric about the y-axis If:- replacing x by -x in its equation produces an equivalent equation.

b) Symmetric about the x-axis if replacing y by -y in its eqn. produces an equivalent equation

c) Symmetric about the origin if replacing x by -x and y by -y in its eqn. produces an equivalent eqn.

* يكون الاقتران متماثل حول y إذا وضعنا -x مكان كل x وعادت المعادلة كما هي.

* يكون متماثل حول x إذا وضعنا مكان كل y ← -y وعادت المعادلة كما هي.

* يكون متماثل حول نقطة الأصل إذا وضعنا مكان x ← -x ومكان y ← -y وعادت المعادلة كما هي.

ex Identify Symmetric of $x = y^2$ ?

↳ sym. about y-axis : $-x = y^2$

Symm. about x-axis : $x = (-y)^2$

$x = y^2$ ✓

∴ Symm. about x-axis.

---

* Determine wether the graph has Symm. about the x-axis, the y-axis or the origin ?

① $xy = 5$ ?

↳ Symm. about origin:-

$(-x)(-y) = 5 \Rightarrow xy = 5$ ✓

∴ Symm. about origin.

---

② $y^2 = |x| - 5$ ?

↳ Symm. about y-axis

$y^2 = |-x| - 5 \Rightarrow y^2 = |x| - 5$ ✓

↳ Symm. about x-axis

$(-y)^2 = |x| - 5 \Rightarrow y^2 = |x| - 5$ ✓

↳ symm. about origin.

$(-y)^2 = |-x| - 5 \Rightarrow y^2 = |x| - 5$ ✓

---

③ $x^2 - 2y^2 = 3$ ?

↳ y-axis :- $(-x)^2 - 2y^2 = 3 \rightarrow x^2 - 2y^2 = 3$

↳ x-axis :- $x^2 - 2(-y)^2 = 3 \rightarrow x^2 - 2y^2 = 3$

↳ origin:- $(-x)^2 - 2(-y)^2 = 3 \rightarrow x^2 - 2y^2 = 3$

* متماثل على x ، y ، نقطة الأصل.

home work ① $y = \frac{x}{3+x^2}$

② $x = 5y^2 + 9$

③ $x^4 = 2y^3 + y$

## * Chapter 0 أسئلة متنوعة على *

[1] The Solution of equation $e^{x^2} \cdot e^{4x+4} = 1$ is (are) $x =$

↳ عند الضرب تجمع الأسس.

$e^{x^2+4x+4} = 1$

$\ln e^{x^2+4x+4} = \ln 1$

$x^2 + 4x + 4 = 0$

$(x+2)(x+2) = 0 \Rightarrow \therefore x = -2$

② The value of $5^{\log_5^3 - 2\log_5^4}$ ?

↳ $5^{\log_5^3 + \log_5^{4^{-2}}}$

$= 5^{\log_5 (3 \cdot 4^{-2})} = (3 \cdot 4^{-2}) = \frac{3}{16}$.

③ If $g(x)$ is obtained from translating the graph of $f(x) = \frac{1}{1-x}$ 1-unit to the right and 2 unit upward, Then $g(x) = ?$

↳ $f(x-1) + 2 = \frac{1}{1-(x-1)} + 2$

$= \frac{1}{2-x} + 2 \Rightarrow g(x)$.

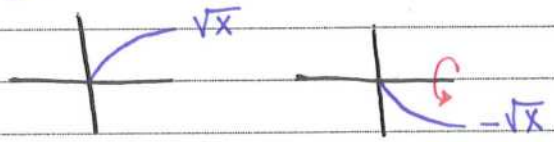
④ The range of $g(x) = 1 - \sqrt{x}$ is..

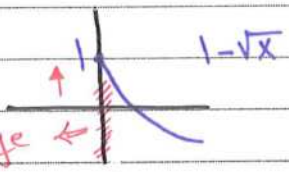
↳ Dom $g(x) = [0, \infty)$

| x | 0 | 4 | 9 |
|---|---|---|---|
| g(x) | 1 | -1 | -2 |

$\therefore R_{g(x)} = (-\infty, 1]$

OR




$(-\infty, 1]$ : Range ←

⑤ If $f(x) = \frac{2}{x}$ and $g(x) = \frac{x+1}{x+3}$, then the domain of $g \circ f$ is?

↳ $g(f(x)) = \frac{\frac{2}{x} + 1}{\frac{2}{x} + 3}$ → بسط / → مقام

→ $D_{g(f)} = D_{\text{بسط}} \cap D_{\text{مقام}} - \{\text{أصفار المقام}\}$

$= (\mathbb{R} - \{0\}) \cap (\mathbb{R} - \{0\}) - \{-\frac{2}{3}\}$

$\therefore = \mathbb{R} - \{0, -\frac{2}{3}\}$

→ $D_{f(x)} = \mathbb{R} - \{0\}$.

$D_{g \circ f} = (\mathbb{R} - \{0, -\frac{2}{3}\}) \cap (\mathbb{R} - \{0\})$

$\therefore = \mathbb{R} - \{0, -\frac{2}{3}\}$.

⑥ سؤال بوليتكنك 2016

If $f(x) = \frac{\cos x}{[x]-4}$, then the domain of $f(x)$ is ... ?

ⓐ $\mathbb{R} - \{4\}$ ⓑ $\mathbb{R} - (4,5]$
ⓒ $\mathbb{R} - [4,5)$ ⓓ $\mathbb{R} - [4,5]$

↳ أصفار المقام :- $[x]-4 = 0$

$[x] = 4$

$4+1 > x \geq 4$

$\therefore x \in [4,5)$

$\therefore = \mathbb{R} - [4,5)$ ⓒ

⑦ سؤال بوليتكنك 2016

$f(x) = \ln(1+\ln x)$, then the domain of $f$ is ... ?

↳ $1+\ln x > 0$

$\ln x > -1$

$e^{\ln x} > e^{-1}$

$\therefore x > \frac{1}{e}$

لكن $\ln x$ : $x > 0$ ، $\frac{1}{e} > 0$

$\therefore$ domain $f(x)$ $(\frac{1}{e}, \infty)$

⑧ If $\log_2^{(7x-1)} - \log_2^{x} = 1$, then $x = \ldots$ ?

↳ $\log_2\left(\frac{7x-1}{x}\right) = 1$

$\frac{7x-1}{x} = \frac{2}{1} \Rightarrow 7x-1 = 2x$

$5x = 1 \rightarrow \boxed{x = \frac{1}{5}}$

⑨ $\ln\left(\frac{1}{x}\right) + \ln(x^3) = \ln 3$ , Find $x$ ?

↳ $\ln\left(\frac{1}{x} \cdot x^{\frac{2}{3}}\right) = \ln 3$

$\ln x^{\frac{2}{3}} = \ln 3$

$e^{\ln x^{\frac{2}{3}}} = e^{\ln 3}$

$x^{\frac{2}{3}} = 3 \Rightarrow \therefore x = \pm\sqrt{3}$

لكن نهمل السالب لأنه ما داخل ln يجب أن يكون فقط موجب

$\therefore \boxed{x = \sqrt{3}}$

⑩ $\log_3(\log_3^{27}) = ?$

↳ $\log_3(3) = 1$

⑪ If $x = \frac{1}{2}\ln 9 + \ln 2$, then $e^{2x} = ?$

↳ $\ln 9^{\frac{1}{2}} + \ln 2 = x$

$\ln 3 + \ln 2 = x \Rightarrow \therefore \boxed{x = \ln 6}$

$\Rightarrow 2x = 2\ln 6 \Rightarrow \therefore \boxed{2x = \ln 6^2 = \ln 36}$

$\Rightarrow \therefore e^{2x} = e^{\ln 36} = 36$

⑫ H.W Solve $\log_2^x + \log_2^{(x-3)} = \log_3^9$

↳ Sol $x = 4$ only

* Remark

$-1 \leq \frac{\sin x}{\cos x} \leq 1$ ① وأيا قوة فردية

$0 \leq \frac{\sin x}{\cos^2 x} \leq 1$ ② وأيا قوة زوجية

(13) If $f(x)=\frac{2}{2+\cos x}$ then the Range of $f(x)$ is ... ?

$\hookrightarrow \quad 1 \geq \cos x \geq -1$

↓

ثم نحلها كما في السؤال العطلي.

$3 \geq 2+\cos x \geq 1$

$\frac{1}{3} \leq \frac{1}{2+\cos x} \leq 1$

$\frac{2}{3} \leq \frac{2}{2+\cos x} \leq 2$

$\therefore$ Range $f(x) = [\frac{2}{3}, 2]$

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(14) If $f(x)=2\sin^2 x - 1$, then the range of $f$ is?

$\hookrightarrow \quad 1 \geq \sin x \geq -1$

$1 \geq \sin^2 x \geq 0$

$2 \geq 2\sin^2 x \geq 0$

$1 \geq 2\sin^2 x - 1 \geq -1$

$\therefore$ Rang $f(x) = [-1, 1]$

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(15) If $g(x)=2\cos^2 x + 4$, then range $g$ is :-

$\hookrightarrow \quad 1 \geq \cos^2 x \geq 0$

$2 \geq 2\cos^2 x \geq 0$

$6 \geq 2\cos^2 x + 4 \geq 4$

$\therefore$ Range : $[4, 6]$.

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(16) $f(x)=\frac{3}{3+\cos x}$, then range of $f(x)$ is ?

$\hookrightarrow \quad 1 \geq \cos x \geq -1$

$4 \geq \cos x + 3 \geq 2$

$\frac{1}{4} \leq \frac{1}{\cos x + 3} \leq \frac{1}{2}$

$\frac{3}{4} \leq \frac{3}{\cos x + 3} \leq \frac{3}{2}$

$\therefore$ Range. $[\frac{3}{4}, \frac{3}{2}]$.

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(17) If $f(x)=\frac{4}{4+2\sin^3 x}$, then the range of $f$ is ... ?

سؤال بوليتكنك 2016

$\hookrightarrow \quad 1 \geq \sin^3 x \geq -1$

$2 \geq 2\sin^3 x \geq -2$

$6 \geq 4 + 2\sin^3 x \geq 2$

$\frac{1}{6} \leq \frac{1}{4+2\sin^3 x} \leq \frac{1}{2}$

$\frac{4}{6} \leq \frac{4}{4+2\sin^3 x} \leq \frac{4}{2}$

$\therefore$ Range : $[\frac{2}{3}, 2]$

---

(18) The range of the function $f(t) = 3 + \cos 2t$ is ... ?

لا يؤثر في المدى.

$\hookrightarrow \quad 1 \geq \cos 2t \geq -1$

$4 \geq \cos 2t + 3 \geq 2$

$\therefore R : [2, 4]$.

(19) Find domain for:-

A $f(x) = \cos(e^x)$ ?

$\hookrightarrow D_f = \mathbb{R}$

B $f(x) = \sin(\sqrt{x})$ ?

$\hookrightarrow D_f = [0, \infty)$

(20) أسئلة بوليتكنك 2016

If $f(x) = \sqrt{(\ln x)^2 - 3\ln x + 2}$, then

The domain of $f$ is .... ?

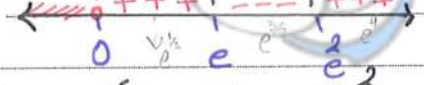
$\hookrightarrow (\ln x)^2 - 3\ln x + 2 \geq 0$

$(\ln x - 2)(\ln x - 1) \geq 0$

$\ln x = 2 \longrightarrow x = e^2$

$\ln x = 1 \longrightarrow x = e^1$

تنسى $\ln x \to x > 0$




$\therefore D_{f(x)} = (0, e], [e^2, \infty)$.

(21) Find domaine and Range:-

A $f(x) = \ln(x^2)$ ?

$\hookrightarrow D_f = \mathbb{R} - \{0\}$

هنا كل القيم داخل العمال لأنه ما داخل ln مربع لكن تبقى المشكلة عند صفر.

$\longrightarrow R_f = \mathbb{R} : (-\infty, \infty)$.

B $g(x) = 3\ln\sqrt[3]{x-1}$ ?

الجذر التكعيبي معرف على كل الأعداد فلا يوجد عنده مشاكل

$\sqrt[3]{x-1} > 0$

$x - 1 > 0$

$x > 1$

$\therefore D_g = (1, \infty)$

$R_g = \mathbb{R}$

C $f(x) = 1 - e^{-x+1}$ ?

$\hookrightarrow D_f = \mathbb{R}$.

$R_f$: $e^{-x+1} > 0$

$-e^{-x+1} < 0$

$1 - e^{-x+1} < 1$

$\therefore R_f = (-\infty, 1)$

D $g(x) = 3 + e^{x-2}$ homework

$\hookrightarrow$ Sol $D_g = \mathbb{R}$, $R_g = (3, \infty)$

E $f(x) = \left(\frac{1}{2}\right)^{x-1} - 1$ ?

$\hookrightarrow D_f = \mathbb{R}$

$\longrightarrow R_f$ : $\left(\frac{1}{2}\right)^{x-1} > 0$

$\left(\frac{1}{2}\right)^{x-1} - 1 > -1$

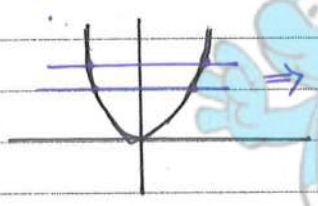
$\therefore R_f = (-1, \infty)$.

## * Inverse Functions :-

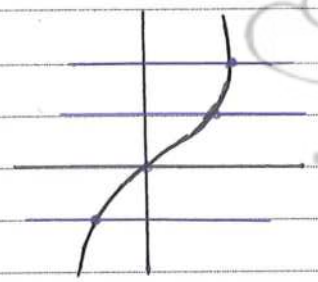
↳ A function $f$ is said to be one to one if for $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$

* Horizontal Line Test : if the horizontal line cuts the curve in one point only, then its One-One, otherwise it's not One-One

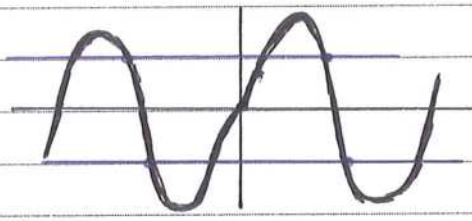
ex3 Which of the following is One-One ?



⇒ not One-One



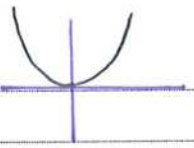
⇒ yes One-One



↳ not One-One.

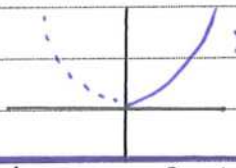
* Remark If $f$ is 1-1, then is invertiable (has inverse) : "$f^{-1}(x)$"

$x^2$ : not 1-1



* تنبيه !

but $x^2, x \geq 0 \Rightarrow$ تصبح في الربع ①



$x \geq 0$ ∴ One-One.

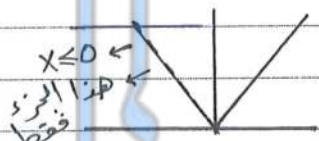
ex3 Which of the following is 1-1 ?

① $f(x) = x^4$.

② $f(x) = \sin x$.

③ $f(x) = x^2 - 1$.

④ $f(x) = |x|, x \leq 0$.



$x \leq 0$ ← فقط الجزء ده

$\rightarrow x \geq 0$ ← نرفضه

∴ ④ فرع

**Note :-** if $f(x)$ is a function, $f^{-1}(x)$ is the inverse of $f(x)$, then :-

① $(f \circ f^{-1})(x) = x$.

② $(f^{-1} \circ f)(x) = x$

**Note :-** $D_{f^{-1}(x)} = R_{f(x)}$

$R_{f^{-1}(x)} = D_{f(x)}$.

* Warning !!

↳ $f^{-1}(x) \neq \frac{1}{f(x)}$ اوعك تنسى.

**Note :-** $y = f(x)$ . $f^{-1}(x)$ ، طريقة ايجاد :-

① نضع مكان كل $y \leftarrow x$ ومكان كل $x \leftarrow y$ .

② نجعل $y$ هي محور الحل (موضع القانون)

③ $f^{-1}(x) = y$ .

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ex Find $f^{-1}(x)$ ?

① $y = \sqrt{3x-2}$ .

$\hookrightarrow x = \sqrt{3y-2}$ ... نجعل $y$ هي محور الحل

$\rightarrow x^2 = 3y - 2$

$3y = x^2 + 2$

$\therefore y = \frac{x^2+2}{3} \Rightarrow \therefore f^{-1}(x) = \frac{x^2+2}{3}$

② $f(x) = \sqrt[3]{2x-1}$ .

$\hookrightarrow y = \sqrt[3]{2x-1} \Rightarrow y^3 = 2x - 1$

$\Rightarrow y^3 + 1 = 2x \Rightarrow x = \frac{y^3+1}{2}$

$f^{-1}(x) = \frac{x^3+1}{2}$ .

③ $y = e^x$ .

$\hookrightarrow e^y = x \Rightarrow y = \ln x$

$\therefore f^{-1}(x) = \ln x$ .

④ $f(x) = \ln(x+3)$ .

$\hookrightarrow y = \ln(x+3) \Rightarrow x = \ln(y+3)$

$e^x = y + 3 \Rightarrow y = e^x - 3$

$\therefore f^{-1}(x) = e^x - 3$

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ex $f(1) = 5$ , $f(3) = 7$ , $f(8) = -10$

find :-

① $f^{-1}(7)$ ? ← ما العدد الذي صورته 7

$\therefore f^{-1}(7) = 3$ .

② $f^{-1}(5)$ ?

$\hookrightarrow f(1) = 5 \longrightarrow f^{-1}(5) = 1$ .

③ $(f \circ f^{-1})(-10)$ ?

$\hookrightarrow f(f^{-1}(-10)) = f(8) = -10$

OR $(f \circ f^{-1})(-10) = -10$

④ $(f^{-1} \circ f)(3)$ ?

$\hookrightarrow = 3$ .

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ex $f(x) = 2x^3 + 5x + 3$ , find $x$ such that $f^{-1}(x) = 1$ ?

$\hookrightarrow f^{-1}(x) = 1$

$f(f^{-1}(x)) = f(1)$

$x = f(1)$

$x = 2 + 5 + 3 = 10$

$\therefore x = 10$

---

ex let $f(x) = \frac{x^3}{x^2+1}$ , find $x$ if $f^{-1}(x) = 2$ ?

$\hookrightarrow f^{-1}(x) = 2 \Rightarrow f(f^{-1}(x)) = f(2)$

$x = f(2) \Rightarrow x = \frac{8}{4+1} = \frac{8}{5}$ .

5) $f(x) = \sqrt[3]{2x-1}$ ?

$y = \sqrt[3]{2x-1}$

$x = \sqrt[3]{2y-1}$

$x^3 = 2y-1 \Rightarrow 2y = x^3+1$

$\therefore y = \frac{1}{2}(x^3+1) \equiv f^{-1}(x)$

6) $y = e^{\sqrt{x}}$ ?

$x = e^{\sqrt{y}} \Rightarrow \ln x = \ln e^{\sqrt{y}}$

$\therefore \sqrt{y} = \ln x \Rightarrow f^{-1}(x) = (\ln x)^2$

7) $f(x) = \ln(x+3)$ ?

$y = \ln(x+3)$

$x = \ln(y+3) \Rightarrow e^x = e^{\ln(y+3)}$

$e^x = y+3 \Rightarrow y = e^x - 3$

$\therefore f^{-1}(x) = e^x - 3$

8) $f(x) = \frac{x+1}{x-1}$ ?

$y = \frac{x+1}{x-1} \Rightarrow \frac{x}{1} = \frac{y+1}{y-1}$ ضربنا تبادلي

$xy - x = y + 1 \Rightarrow xy - y = x + 1$

$y(x-1) = (x+1) \Rightarrow y = \frac{x+1}{x-1}$

$\therefore f^{-1}(x) = \frac{x+1}{x-1}$

9) $f(x) = \frac{4x-1}{2x+3}$ ?

$y = \frac{4x-1}{2x+3}$ بدّل الرموز

$\Rightarrow \frac{x}{1} = \frac{4y-1}{2y+3} \Rightarrow 2xy + 3x = 4y - 1$

$\Rightarrow 4y - 2xy = 1 + 3x$

$y(4-2x) = 1+3x \Rightarrow y = \frac{1+3x}{4-2x}$

8) $y = \frac{e^x}{1+2e^x}$ ?

$\frac{x}{1} = \frac{e^y}{1+2e^y}$

$\Rightarrow x + 2xe^y = e^y$

$x = e^y - 2xe^y$

$x = e^y(1-2x) \Rightarrow e^y = \frac{x}{1-2x}$

$\Rightarrow \ln e^y = \ln\left(\frac{x}{1-2x}\right)$

$\therefore f^{-1}(x) = \ln\left(\frac{x}{1-2x}\right)$

*homework*

$f(x) = \sqrt{e^{2x}+3}$ , find $f^{-1}(x)$ ?

الحل $f^{-1}(x) = \frac{1}{2}\ln(x^2-3)$

⑨ $f(x) = 4^{\sqrt{4-x}}$ , find:-

① Find $f^{-1}(x)$.

② R $f(x)$.

↳ ① $y = 4^{\sqrt{4-x}}$

$x = 4^{\sqrt{4-y}}$

$\log_4^x = \log_4^{4^{\sqrt{4-y}}}$

$\log_4^x = \sqrt{4-y}$

$(\log_4^x)^2 = 4-y$

$\therefore y = 4 - (\log_4^x)^2$

$\therefore f^{-1}(x) = 4 - (\log_4^x)^2$

$\Rightarrow R\,f(x) \equiv D\,f^{-1}(x) = (0,\infty)$.

⑩ $f(x) = \begin{cases} \frac{5}{2} - x & , x < 2 \\ \frac{1}{x} & , x \geq 2 \end{cases}$

find $f^{-1}(x)$?

↳ a) $y = \frac{5}{2} - x \Rightarrow x = \frac{5}{2} - y$

$\therefore f^{-1}(x) = \frac{5}{2} - x$

$* D f^{-1}(x) = R f(x) = (\frac{1}{2}, \infty)$

b) $y = \frac{1}{x} \Rightarrow x = \frac{1}{y}$

$\therefore f^{-1}(x) = \frac{1}{x}$

$* D f^{-1} = R f = (0, \frac{1}{2}]$

$\therefore f^{-1}(x) = \begin{cases} \frac{5}{2} - x & , x > \frac{1}{2} \\ \frac{1}{x} & , 0 < x \leq \frac{1}{2} \end{cases}$

⑪ $f(x) = \begin{cases} 2x & , x \leq 0 \\ x^2 & , x > 0 \end{cases}$ ?

↳ a) $y = 2x \Rightarrow x = \frac{y}{2}$

$\therefore f^{-1}(x) = \frac{x}{2}$

$\rightarrow D f^{-1} = R f = (-\infty, 0]$

b) $y = x^2 \Rightarrow x = \sqrt{y}$

$\therefore f^{-1}(x) = \sqrt{x}$

$\rightarrow D f^{-1} = R f = (0, \infty)$

$\Rightarrow \therefore f^{-1}(x) = \begin{cases} \frac{1}{2}x & , x \leq 0 \\ \sqrt{x} & , x > 0 \end{cases}$

ex3 $f(x) = 2x + \ln x$ , find $f^{-1}(2)$.

⇐ إنتبه ! لهذه الأسئلة لأنها تقلل دوره الحاجة لإيجاد $f^{-1}$ :-

$2x + \ln x = 2$

↳ ثم بالتخمين : ما القيمة التي يجب وضعها ليصبح الجواب = 2 .

* نجرب $x=1$ :- $2(1) + \ln(1) = 2$

∴ الجواب هو $x=1$ ⇐ $f^{-1}(2) = 1$

ex3 $f(x) = \ln(1+\ln x)$ , then $f^{-1}(\ln 2)$ ... (سؤال بوليتكنك)

ⓐ $e$ ⓑ $2$ ؟

ⓒ $e^{-2}$ ⓓ $e^{-1}$ .

↳ $\ln(1+\ln x) = \ln 2$ (نرفع القوة e)

$e^{\ln(1+\ln x)} = e^{\ln 2}$

$1 + \ln x = 2$

$\ln x = 1 \Rightarrow e^{\ln x} = e^1$

∴ $x = e$ ⓐ نختار

ex3 $g(x) = 3 + x + e^x$ , $g^{-1}(4)$ ؟

$x = g^{-1}(4)$

$g(x) = g(g^{-1}(4)) \Rightarrow g(x) = 4$

⇐ بالتجربة :- $4 = 3 + x + e^x$

$x = 0 \Rightarrow 3 + 0 + e^0 = 3 + 1 = 4$ ✓

∴ $x = 0 \Rightarrow g^{-1}(4) = 0$

ex3 If $f(x) = e^x + 3e^{-x} - 3$ , then $f^{-1}(1)$ is ...

↳ $e^x + 3e^{-x} - 3 = 1$

$x = 0 \Rightarrow e^0 + 3e^0 - 3 = 1$ ✓

∴ $f^{-1}(1) = 0$

ex3 *homework*

$f(x) = e^x + 3e^{-x} - 2$ , then $f^{-1}(2)$ ... ?

ex3 $f(x) = x + 4$ , $x \geq 0$ ... find domain of $f^{-1}(x)$ ?

↳ $D_{f^{-1}(x)} = R_{f(x)} = [4, \infty)$

⇒

| $x$ | 0 | 1 | 2 | ⟶ |
|---|---|---|---|---|
| $f(x)$ | 4 | 5 | 6 | ..... to $\infty$ |

اصغر صورة في المدى

ex3 $f(x) = x^2$ , find domain of $f^{-1}(x)$ ?

↳ $y = x^2 \Rightarrow x = y^2$ خذ الجذر .

$\sqrt{x} = y \Rightarrow \therefore f^{-1}(x) = \sqrt{x}$

$\Rightarrow D_{f^{-1}(x)} = R_{f(x)} = [0, \infty)$ .

$\pi = 180^\circ$

| $\theta$ | $\sin x$ | $\cos x$ | $\tan x$ |
|---|---|---|---|
| $0$ / $2\pi$ | $0$ | $1$ | $0$ |
| $\frac{\pi}{6}$ | $\frac{1}{2}$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{3}}$ |
| $\frac{\pi}{4}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{\sqrt{2}}$ | $1$ |
| $\frac{\pi}{3}$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{2}$ | $\sqrt{3}$ |
| $\frac{\pi}{2}$ | $1$ | $0$ | not defined |
| $\pi$ | $0$ | $-1$ | $0$ |
| $\frac{3\pi}{2}$ | $-1$ | $0$ | not defined |

$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$ ← زاوية ← عدد

$\sin$ (الزاوية) ⟹ عدد — عطي

$\sin^{-1}$ (عدد) ⟹ زاوية — عطي

$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ ← زاوية ← عدد

* سؤال جاي :- شو الزاوية اللي بتعطيني جوابها $\frac{1}{2}$ .

* Trigonometric Identities (مهم جداً) المتطابقات المثلثية حفظها .

① $\sin^2 x + \cos^2 x = 1$ .

② $1 + \tan^2 x = \sec^2 x$ .

③ $1 + \cot^2 x = \csc^2 x$ .

④ $\sin 2x = 2 \sin x \cos x$ .

⑤ $\cos 2x = \cos^2 x - \sin^2 x$ .
or $= 1 - 2\sin^2 x$ .
or $= 2\cos^2 x - 1$ .

⑥ $\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$
← متطابقة الجا : الاقترانات مختلفة لكن الإشارات متشابهة .

⑦ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
← متطابقة جتا : الاقترانات متشابهة لكن الإشارات مختلفة .

⑧ $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ . ($\sin$ ⊖)

⑨ $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ . ($\cos$ ⊕)

Q3 Circle the Correct answer :-

1) Let $f(x) = 1-x$, $g(x) = x^2$, $h(x) = \sin x$, then $(fogoh)(x)$:

a) $1+\sin x$ b) $\cos^2 x$

c) $\sin^2 x - 1$ d) $2\sin x$

↳ $fogoh = f(g(h(x)))$

$= f(g(\sin x)) = f(\sin^2 x)$

$= 1 - \sin^2 x \equiv \cos^2 x$ مطابقة

$* \sin^2 x + \cos^2 x = 1$

$\therefore \cos^2 x = 1 - \sin^2 x$

∴ فرع (b)

2) Let $f(x) = \cos x$ and $g(x) = 2x$ then $(fog)(x)$ ?

a) $\cos x$. b) $\cos^2 x - \sin^2 x$.

c) $2\sin x \cos x$. d) $2\cos x$.

↳ $fog = f(g(x)) = f(2x)$

$= \cos 2x = \cos^2 x - \sin^2 x$

$= 1 - 2\sin^2 x$

$= 2\cos^2 x - 1$

فكرها لمطابقة

∴ فرع (b)

*homework* If $f(x) = x^2 - 4x$, $x \le 1$

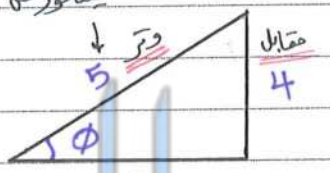
find : a) Find $f^{-1}(x)$.

b) R $f(x)$.

Q3 Given that $\phi = \tan^{-1}(\frac{4}{3})$, find the exact Value of $\sin\phi$, $\cos\phi$, $\cot\phi$, $\sec\phi$ and $\csc\phi$.

↳ $\phi = \tan^{-1}(\frac{4}{3}) \Rightarrow \tan\phi = \frac{4}{3}$ (مقابل / مجاور)

فيثاغورس



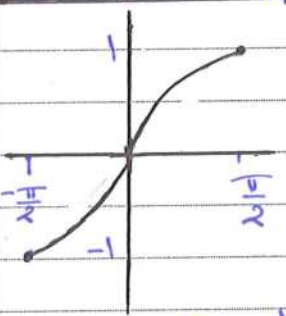

$\Rightarrow \sin\phi = \frac{\text{مقابل}}{\text{وتر}} = \frac{4}{5}$.

$\Rightarrow \cos\phi = \frac{\text{مجاور}}{\text{وتر}} = \frac{3}{5}$.

$\Rightarrow \cot\phi = \frac{\text{المجاور}}{\text{مقابل}} = \frac{3}{4}$.

$\Rightarrow \sec\phi = \frac{\text{الوتر}}{\text{المجاور}} = \frac{5}{3}$.

$\Rightarrow \csc\phi = \frac{\text{الوتر}}{\text{المقابل}} = \frac{5}{4}$




$\Rightarrow D_{\sin x} = [-\frac{\pi}{2}, \frac{\pi}{2}]$

$R_{\sin x} = [-1, 1]$

$\Rightarrow D_{\sin^{-1} x} = [-1, 1]$

$R_{\sin^{-1} x} = [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\Rightarrow \sin(\sin^{-1}(x)) = x$ if $x \in [-1, 1]$

$\Rightarrow \sin^{-1}(\sin(x)) = x$ if $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

* ملاحظة :- مجال sinx بشكل عام هو كل R لكنه اقتطعنا مجاله جزء لكي يصبح 1-1 وبكونه One-One يكون له inverse .

Student
- sinx ⊕
- cosx ⊖
- tanx ⊖

ALL
- sinx ⊕
- cosx ⊕
- tanx ⊕

Take
- tanx ⊕
- cosx ⊖
- sinx ⊖

Calculus
- cosx ⊕
- tanx ⊖
- sinx ⊖

$\frac{\pi}{4} \rightarrow \frac{3\pi}{4} \rightarrow \frac{5\pi}{4} \rightarrow \frac{7\pi}{4}$

في الربع الأول — ربع ثاني — ربع ثالث — ربع رابع

معامل البسط اقل من المقام بواحد — معامل البسط اكبر من المقام بواحد — معامل البسط اقل من ضعف المقام بواحد

$\frac{\pi}{3} \rightarrow \frac{2\pi}{3} \rightarrow \frac{4\pi}{3} \rightarrow \frac{5\pi}{3}$

① ② ③ ④

* Evaluate ?

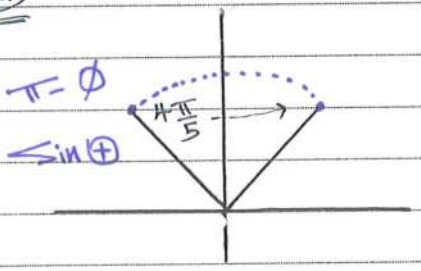
1) $\sin^{-1}(\sin 0) = 0$, b/c $0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

2) $\sin(\sin^{-1}\frac{3}{5}) = \frac{3}{5}$, $\frac{3}{5} \in [-1, 1]$

3) $\sin^{-1}(\sin\frac{\pi}{4}) = \frac{\pi}{4}$, $\frac{\pi}{4} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

4) $\sin^{-1}(\sin\frac{4\pi}{5})$ ?

↳ $\frac{4\pi}{5} \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$ — ربع ثاني



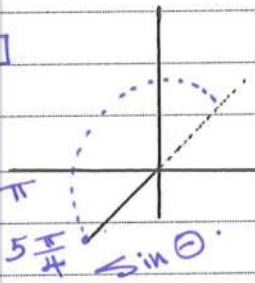

$\Rightarrow \pi - \emptyset = \pi - \frac{4\pi}{5} = \frac{5\pi}{5} - \frac{4\pi}{5}$

$= \frac{\pi}{5} \Rightarrow \sin^{-1}(\sin\frac{\pi}{5}) = \frac{\pi}{5}$

where $\frac{\pi}{5} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

5) $\sin^{-1}(\sin\frac{5\pi}{4})$ ?

↳ $\frac{5\pi}{4} \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$ — ربع ثالث



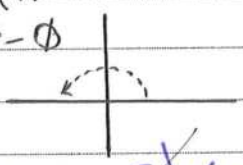

$\Rightarrow \frac{5\pi}{4} - \pi = \frac{\pi}{4}$

$\Rightarrow \sin^{-1}(\sin(-\frac{\pi}{4})) = -\frac{\pi}{4} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

في الربع الثالث sinx سالب

6) $\sin^{-1}(\sin\pi)$ ?

↳ $\pi \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$




$\Rightarrow \pi - \pi = 0$

$\Rightarrow \sin^{-1}(\sin(0)) = 0$

where $0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

7) $\sin^{-1}(\sin\frac{8\pi}{7})$ ؟

في الربع الثالث

$\Rightarrow \sin^{-1}(\sin(-\frac{\pi}{7})) = -\frac{\pi}{7}$.

8) $\sin^{-1}(\sin\frac{9\pi}{5})$ ؟

في الربع الرابع .

$\Rightarrow \emptyset - 2\pi = \frac{9\pi}{5} - 2\pi$

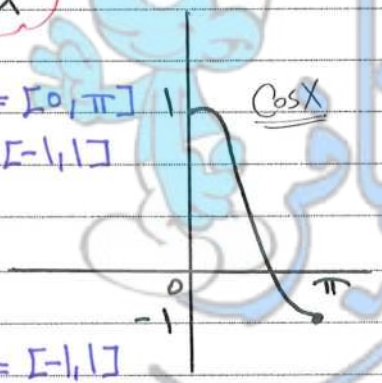
$\Rightarrow \frac{9\pi}{5} - \frac{10\pi}{5} = -\frac{\pi}{5}$

$\Rightarrow \sin^{-1}(\sin(-\frac{\pi}{5})) = -\frac{\pi}{5} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

* $\cos^{-1}x$

$\Rightarrow D_{\cos x} = [0,\pi]$

$R_{\cos x} = [-1,1]$




$\Rightarrow D_{\cos^{-1}x} = [-1,1]$

$R_{\cos^{-1}x} = [0,\pi]$

$\Rightarrow \cos(\cos^{-1}x) = x, \ x \in [-1,1]$

$\cos^{-1}(\cos x) = x, \ x \in [0,\pi]$

* Evaluate :—

$D\cos(\cos^{-1}\frac{11}{5}) = \frac{11}{5}, \ \frac{11}{5} \in [-1,1]$

2) $\cos(\cos^{-1}(-1)) = -1, \ -1 \in [-1,1]$

3) $\cos^{-1}(\cos\frac{\pi}{8}) = \frac{\pi}{8}, \ \frac{\pi}{8} \in [0,\pi]$

4) $\cos^{-1}(\cos 2\frac{\pi}{3}) = 2\frac{\pi}{3}, \ 2\frac{\pi}{3} \in [0,\pi]$

5) $\cos^{-1}(\cos\frac{20\pi}{3}) =$

↳ * اذا كانت الزاوية كبيرة جداً نطرح منها $n\pi$ حتى تعود إلى المجال المطلوب ...

$\Rightarrow \frac{20\pi}{3} - 6\pi = \frac{20\pi}{3} - \frac{18\pi}{3} = \frac{2\pi}{3}$

$\Rightarrow \cos^{-1}(\cos\frac{2\pi}{3}) = \frac{2\pi}{3}, \ \frac{2\pi}{3} \in [0,\pi]$

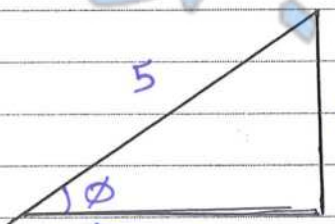
6) $\sin^{-1}(\sin\frac{21\pi}{2}) =$

↳ $\frac{21}{2} - 10\pi = \frac{21\pi}{2} - \frac{20\pi}{2}$

$= \frac{\pi}{2} \Rightarrow \sin^{-1}(\sin\frac{\pi}{2}) = \frac{\pi}{2}$ ✓

7) $\cos(\sin^{-1}\frac{3}{5}) =$

↳ $\sin^{-1}\frac{3}{5} = \emptyset \Rightarrow \sin\emptyset = \frac{3}{5}$.




↳ $\sqrt{25-9} = 4$

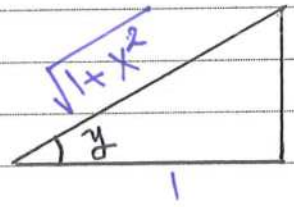
$\Rightarrow \cos\emptyset$ المطلوب

$\cos\emptyset = \frac{\text{مجاور}}{\text{وتر}} = \frac{4}{5}$

$\therefore \cos(\sin^{-1}\frac{3}{5}) = \frac{4}{5}$

⑧ $\cos(\tan^{-1}x) =$

$\tan^{-1}x = y \Rightarrow x = \tan y$




$\Rightarrow \cos(\tan^{-1}x) = \cos y = \frac{1}{\sqrt{1+x^2}}$

$\therefore \cos(\tan^{-1}x) = \frac{1}{\sqrt{1+x^2}}$ .

⑨ $\cos^{-1}(\cos\frac{7\pi}{6}) =$

$\frac{7\pi}{6} \Rightarrow$ (-) في الربع الثالث حيث الجتا

$\Rightarrow \emptyset - \pi = \frac{7\pi}{6} - \pi = \frac{\pi}{6}$

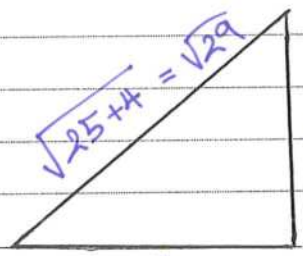
$\cos^{-1}(\cos(-\frac{\pi}{6})) = \cos^{-1}(\cos\frac{\pi}{6}) = \frac{\pi}{6}$ (تا) لا يعترف بالسالب

$\frac{\pi}{6} \in [0, \pi]$ ←

⑩ $\cos(\tan^{-1}(-\frac{2}{5})) =$

$\cos(-\tan^{-1}(\frac{2}{5})) = \cos(\tan^{-1}\frac{2}{5})$ (الجتا) يتعامل مع السالب .

$= \frac{5}{\sqrt{29}}$ .



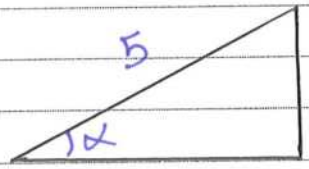

$\tan^{-1}\frac{2}{5}$ ...

③ Evaluate :- (يجب استخدام مطابقات)

① $\sin(2\sin^{-1}\frac{3}{5})$ ?

$\Rightarrow \sin^{-1}\frac{3}{5} = \alpha \Rightarrow \sin 2\alpha$ ليها مطابقة

$\sin 2\alpha = 2\sin\alpha\cos\alpha$




$\sin^{-1}\frac{3}{5}$

$\sqrt{25-9} = 4$

$\Rightarrow \sin 2\alpha = 2 * \frac{3}{5} * \frac{4}{5} = \frac{24}{25}$ .

② $\cos(2\tan^{-1}x)$ ?

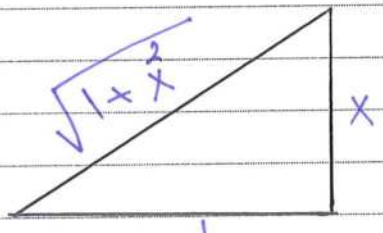
$\tan^{-1}x = a \Rightarrow \cos 2a$

$\cos 2a = \cos^2 a - \sin^2 a$

$\cos(2\tan^{-1}x) = (\cos(\tan^{-1}x))^2 - (\sin(\tan^{-1}x))^2$

$= \left(\frac{1}{\sqrt{1+x^2}}\right)^2 - \left(\frac{x}{\sqrt{1+x^2}}\right)^2$

$= \frac{1}{1+x^2} - \frac{x^2}{1+x^2} = \frac{1-x^2}{1+x^2}$ .




$\tan^{-1}\frac{x}{1}$ .

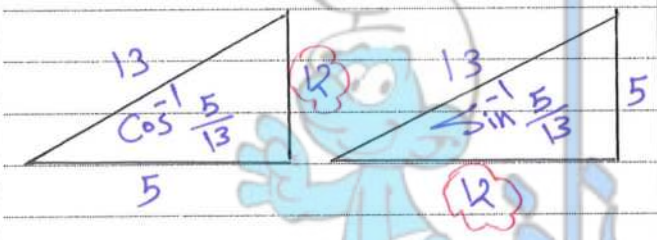
3) $\cos\left(\sin^{-1}\frac{5}{13} + \cos^{-1}\frac{5}{13}\right) = ?$

$\sin^{-1}\frac{5}{13}$ ↓ افرضها $\alpha$ — $\cos^{-1}\frac{5}{13}$ ↓ افرضها $\beta$

$\Rightarrow \cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$= \cos\left(\sin^{-1}\frac{5}{13}\right)\cdot\cos\left(\cos^{-1}\frac{5}{13}\right) - \sin\left(\sin^{-1}\frac{5}{13}\right)\sin\left(\cos^{-1}\frac{5}{13}\right)$

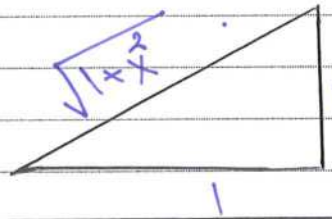
$= \left(\frac{12}{13}\right)\left(\frac{5}{13}\right) - \left(\frac{5}{13}\right)\left(\frac{12}{13}\right) = 0.$




4) $\cot^2(\tan^{-1}x)$ ? سؤال بوليتكنك

↳ $(\cot(\tan^{-1}x))^2$

$= \left(\frac{1}{\sqrt{1+x^2}}\right)^2 = \frac{1}{1+x^2}.$

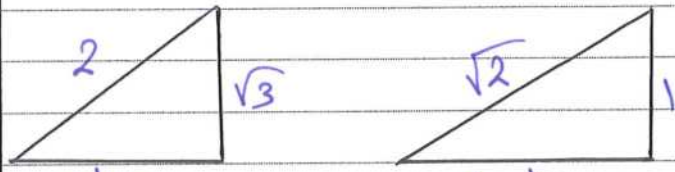



5) $\cos\left(\tan^{-1}(1) - \cos^{-1}\left(\frac{1}{2}\right)\right)$ ?

سؤال بوليتكنك

$\cos(\tan^{-1}(1))\cos(\cos^{-1}\frac{1}{2}) + \sin(\tan^{-1}(1))\sin(\cos^{-1}\frac{1}{2})$

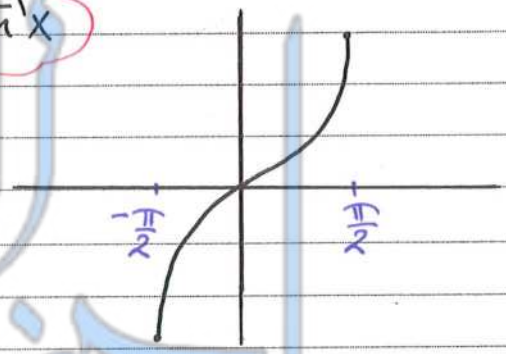
$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{1+\sqrt{3}}{2\sqrt{2}}.$




6) $\sin\left(\sin^{-1}\frac{1}{3} + \cos^{-1}\frac{2}{3}\right)$ ?

*homework*

* $\tan^{-1}x$




$\Rightarrow D_{\tan x} = (-\frac{\pi}{2}, \frac{\pi}{2})$.
$R_{\tan x} = \mathbb{R}$.

$\Rightarrow D_{\tan^{-1}x} = \mathbb{R}$.
$R_{\tan^{-1}x} = (-\frac{\pi}{2}, \frac{\pi}{2})$.

$\Rightarrow \tan^{-1}(\tan x) = x,\ x \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$\tan(\tan^{-1}x) = x,\ x \in \mathbb{R}$

∴ * متطابقات $\tan x$

① $\tan 2x = \frac{2\tan x}{1-\tan^2 x}$ إثبات كما هي

② $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ نعكس إشارة.

* Evaluate :-

1) $\tan(\tan^{-1} 9) = 9,\ 9 \in \mathbb{R}$

2) $\tan(\tan^{-1} \pi) = \pi,\ \pi \in \mathbb{R}$

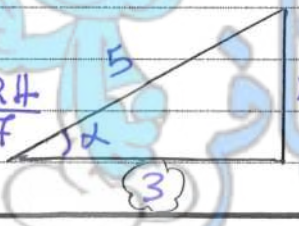
3) $\tan^{-1}(\tan \frac{\pi}{4}) = \frac{\pi}{4},\ \frac{\pi}{4} \in (-\frac{\pi}{2}, \frac{\pi}{2})$

4) $\tan(2 \sin^{-1} \frac{4}{5}) = \dots$

$\sin^{-1} \frac{4}{5} = \alpha \Rightarrow \tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$

$= \frac{2\tan(\sin^{-1}\frac{4}{5})}{1-(\tan(\sin^{-1}\frac{4}{5}))^2} = \frac{2(\frac{4}{3})}{1-(\frac{4}{3})^2}$

$= \frac{\frac{8}{3}}{1-\frac{16}{9}} = \frac{-24}{7}$




5) $\tan^{-1}(\tan \frac{23\pi}{7}) =$

$\hookrightarrow \frac{23\pi}{7} - 3\pi = \frac{23\pi}{7} - \frac{21\pi}{7}$

$= \frac{2\pi}{7}.$

$\Rightarrow \tan^{-1}(\tan \frac{2\pi}{7}) = \frac{2\pi}{7}$

where $\frac{2\pi}{7} \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

6) $\sec(4\frac{\pi}{3}) =$

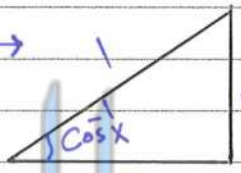
$\hookrightarrow \sec(4\frac{\pi}{3}) = \frac{1}{\cos(4\frac{\pi}{3})} = \frac{1}{-\cos(\frac{\pi}{3})}$

$\cos\theta$ — ربع ثالث

$= \frac{1}{-1/2} = -2.$

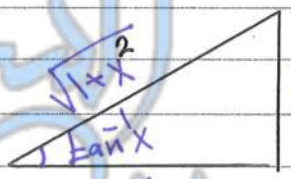
7) $\sin(\cos^{-1} x) =$

$\hookrightarrow$



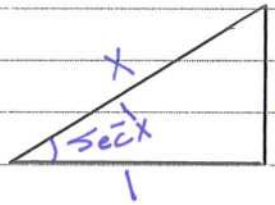

$\Rightarrow \therefore \sin(\cos^{-1} x) = \sqrt{1-x^2}$

8) $\csc(\tan^{-1} x) =$




$\Rightarrow \csc(\tan^{-1} x) = \frac{\sqrt{1+x^2}}{x}$

9) $\cot(\sec^{-1} x) =$




$\Rightarrow \therefore \cot(\sec^{-1} x) = \frac{1}{\sqrt{x^2-1}}.$

10) $\sec(\sin^{-1}(-\frac{3}{4})) =$ (homework)

$\hookrightarrow$ sol : $\frac{4}{\sqrt{7}}.$

مدى

Q domain $3 \le \sin^{-1}(2x-1) + 5$ ?

مجال

$\hookrightarrow -1 \le 2x-1 \le 1$

$0 \le 2x \le 2$

$0 \le x \le 1$

$\therefore x \in [0,1]$

تذكر
الداخلي ⇐ مجال
الخارجي ⇐ مدى

Q find Range for $2 \le \sin^{-1}(3x+1) + \pi$ ?

$\hookrightarrow -\frac{\pi}{2} \le \sin^{-1}(3x+1) \le \frac{\pi}{2}$

$-\pi \le 2\sin^{-1}(3x+1) \le \pi$

$0 \le 2\sin^{-1}(3x+1) + \pi \le 2\pi$

$\therefore$ Range : $[0, 2\pi]$.

Q find Range : $2\cos^{-1} 3x - \pi$ ?

$\hookrightarrow$ Range $\cos^{-1} x = [0, \pi]$

$\Rightarrow 0 \le \cos^{-1} 3x \le \pi$

$0 \le 2\cos^{-1} 3x \le 2\pi$

$-\pi \le 2\cos^{-1} 3x - \pi \le \pi$

$\therefore$ Range $[-\pi, \pi]$.

Q $f(x) = \cos^{-1}(2x+1)$ , find :

① domain $f(x)$ ?

② Range $f(x)$ ?

$\hookrightarrow$ ① $-1 \le 2x+1 \le 1$

$-2 \le 2x \le 0$

$-1 \le x \le 0$

$\therefore D_{f(x)} = [-1, 0]$.

② $0 \le \cos^{-1}(2x+1) \le \pi$

$\therefore R_{f(x)} = [0, \pi]$

العدى لا يتأثر

*homework*

Let $f(x) = 5 - 2\sin^{-1}(2x-1)$ , find:

① domain $f(x)$ ? → D : $[0,1]$

② Range $f(x)$ ? → R : $[5-\pi, 5+\pi]$

③ $f\left(\frac{1}{2} + \frac{1}{2}\sin 5\frac{\pi}{4}\right)$ ? → $= 5 - \frac{\pi}{2}$.

Q If $f(x) = 2x+1$ and $f(g(x)) = x^3 + x^2 + x + 1$, then $g(-1) = ?$ 2016

(a) 14 (b) 7

(c) $-\frac{1}{2}$ (d) $\frac{1}{2}$.

$\hookrightarrow f(g(-1)) = -1 + 1 - 1 + 1 = 0$

$\therefore f(g(-1)) = 0 \Rightarrow f(x) = 2x+1$

$f(g(-1)) = 2g(-1) + 1$

$0 = 2g(-1) + 1 \Rightarrow \therefore g(-1) = -\frac{1}{2}$



©

# Calculus "1"

## Homeworks :-

Q1) Find value of $9^{\log_3^5}$ ?

Q2) Solve :- ① $e^{e^x} = 2$

② $2 < \ln x < 9$

Q3) Find the domain and $f^{-1}$ for $f(x) = \ln(2 + \ln x)$ ?

Q4) The Range of $2\cos^{-1}(2x+1) + 2\pi$ ?

Q5) If the domain and range of $f(x)$ are $[-1,3]$ and $[0,4]$ respectively, then the Range of $g(x) = 1 - 4f^{-1}(3x)$ is ?

↳ Hint: $D_{f^{-1}} = R_f$ / $R_{f^{-1}} = D_f$

Q6) $\cos\left(2\tan^{-1}\frac{2}{3}\right)$ ?

Q7) If $16^x - 4^x - 6 = 0$ then, $x$ = ?

Q8) Determine wether the function $f$ is even, odd or neither ?

$$f(x) = \ln\left(\frac{4x+1}{1-4x}\right).$$

* find domain :-

① $f(x) = \frac{1}{e^{x-3}}$ / ② $f(x) = \frac{3}{\ln(2x-1)}$

Q9) $f(x) = \ln\left(\frac{4x+1}{1-4x}\right)$, find $f^{-1}(x)$ ?

Q10) $\log_{10}^{x^{\frac{3}{2}}} - \log_{10}^{\sqrt{x}} = 5$, then $x$ = ?

Q11) $f(x) = x^2 - 2x + 3$, find Rang $f(x)$ ?

Q12) if $g(x) = 2\cos^2 x + 4$, then Range $g$ is ?

Q13) If $f(x) = \frac{x^2}{\sqrt{x+1}}$, $g(x) =$ , the domain of $(f \circ g)(x)$ ?

Q14) let $f(x) = \frac{e^{2x} - 2}{e^{2x} + 2}$, find :

① $f^{-1}(x)$ = ?

② classify $f^{-1}(x)$ is even, odd or neither ?

Q15) Prove (شيء) that $f(x) = \frac{1 - e^{\frac{1}{x}}}{1 + e^{\frac{1}{x}}}$ is odd ?

Q16) $f(x)$ even, odd ?

$f(x) = \ln(x + \sqrt{1+x^2})$

# Chapter 1: LIMITS AND CONTINUITY

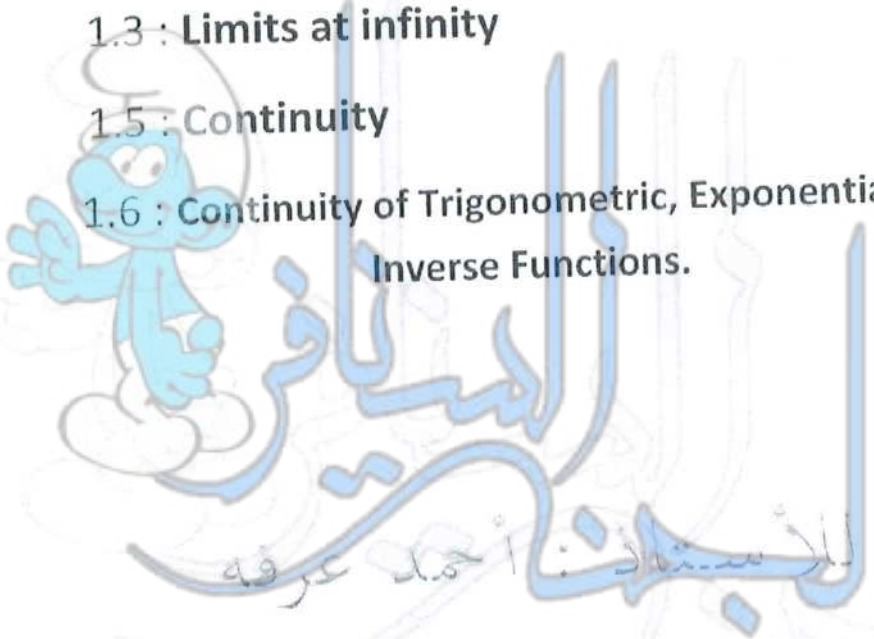
1.1 : **Limits**

1.2 : **Computing Limits**

1.3 : **Limits at infinity**

1.5 : **Continuity**

1.6 : **Continuity of Trigonometric, Exponential, and Inverse Functions.**

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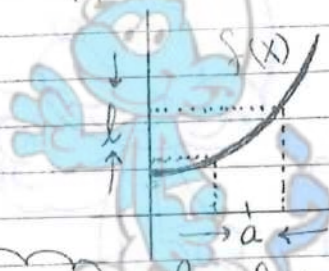
Calculus "1"

# Chapter 1 :-

## limits and Continuity

**1.1 : Limits , 1.2 Computing Limits.**

* Def :- $\lim_{x \to a} f(x) = l$ means if x approaches to a, then f(x) approaches to (l).




* Remark :- $\lim_{x \to a} f(x) = L$ exists If $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$.

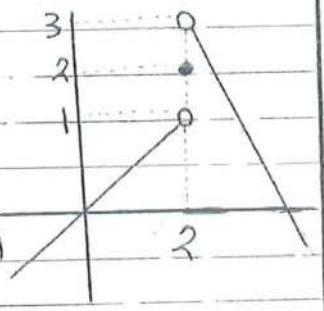
* If $\lim_{x \to a^+} f(x) \neq \lim_{x \to a^-} f(x)$ Then : $\lim_{x \to a} f(x)$ D. N. E.

ex Find :-

① $f(2) = ? = 2$

② $\lim_{x \to 2^+} f(x) = 3$

③ $\lim_{x \to 2^-} f(x) = ? = 1$




④ $\lim_{x \to 2} f(x) = ?$ = does not exist

$\lim_{x \to 2^+} f(x) \neq \lim_{x \to 2^-} f(x)$.

## * Properties * Theorem:-

Suppose that c is constant and the limit $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ exists, then:-

1. $\lim_{x \to a} (f(x) \pm g(x)) = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$
2. $\lim_{x \to a} (f(x) \cdot g(x)) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$
3. $\lim_{x \to a} c f(x) = c \cdot \lim_{x \to a} f(x)$
4. $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$, $\lim_{x \to a} g(x) \neq 0$ :

Calculus "1"

5) $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$

if n is even, must $f(x) > 0$

6) $\lim_{x\to a} c = c$

7) $\lim_{x\to a} (f(x))^n = \left(\lim_{x\to a} f(x)\right)^n$

**Ex** Given that $\lim_{x\to a} f(x) = 2$, $\lim_{x\to a} g(x) = -4$, $\lim_{x\to a} h(x) = 0$

Find :-

① $\lim_{x\to a} [f(x) + 2g(x)]$ ?

$\hookrightarrow \lim_{x\to a} f(x) + 2\lim_{x\to a} g(x)$

$= 2 + 2(-4) = 2 - 8 = -6$

② $\lim_{x\to a} [h(x) - 3g(x) + 1]$ ?

$\hookrightarrow 0 - 3\lim_{x\to a} g(x) + 1$

$= 0 - 3(-4) + 1$

$= 0 + 12 + 1 = 13$

③ $\lim_{x\to a} [f(x) \cdot g(x)]$ ?

$\hookrightarrow \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$

$= 2(-4) = -8$.

④ $\lim_{x\to a} (g(x))^2$ ?

$\hookrightarrow \left(\lim_{x\to a} g(x)\right)^2 = (-4)^2 = 16$.

⑤ $\lim_{x\to a} \sqrt[3]{6 + f(x)}$ ?

$\hookrightarrow \sqrt[3]{\lim_{x\to a} 6 + \lim_{x\to a} f(x)}$

$= \sqrt[3]{6+2} = \sqrt[3]{8} = 2$.

⑥ $\lim_{x\to a} \frac{2}{g(x)}$ ?

$\hookrightarrow \frac{\lim_{x\to a} 2}{\lim_{x\to a} g(x)} = \frac{2}{-4} = \frac{-1}{2}$.

* الأصل في النهايات هو التعويض المباشر

① $\lim_{x\to 2} x(x-1)(x+1)$ ?

$\hookrightarrow \lim_{x\to 2} x(x^2-1) = 2(4-1) = 6$

② $\lim_{x\to 3} \frac{x^2 - 2x}{x+1}$ ?

$\hookrightarrow \frac{9-6}{3+1} = \frac{3}{4}$.

③ $\lim_{x\to 0} \frac{6x-9}{x^3-12x+3}$ ? $\Rightarrow \frac{0-9}{0+3} = -3$.

④ $\lim_{x\to -1} \sqrt[3]{x^3-7}$ ?

$\hookrightarrow \sqrt[3]{-1-7} = \sqrt[3]{-8} = -2$

# Calculus "1"

* نأخذ النهايات من اليمين واليسار في الحالات الآتية ...

١) اقتراب التشعب (نقطة التشعب).

٢) اقتراب القيمة المطلقة (Abs Value)

٣) اقتراب أكبر عدد صحيح [عدد صحيح] ← ناتج التعويض

٤) الجذور الزوجية ← even ← 0 ← ناتج التعويض

٥) number ← 0

١) الحالة الأولى: الاقتراب المتشعب.

ex) $f(x) = \begin{cases} x+1 & , x \leq 1 \\ x-1 & , x > 1 \end{cases}$

find $\lim_{x \to 1} f(x) = ?$

← هنا $x=1$ هي نقطة تشعب :-

$\Rightarrow \lim_{x \to 1^+} (x-1) = 0$

$\Rightarrow \lim_{x \to 1^-} (x+1) = 2$

$\Rightarrow \lim_{x \to 1^+} \neq \lim_{x \to 1^-}$

$\therefore \lim_{x \to 1} f(x)$ d.n.e

ex) $f(t) = \begin{cases} t-2 & , t < 0 \\ t^2 & , 0 < t \leq 2 \\ 2t & , t > 2 \end{cases}$

Find :-

① $\lim_{t \to 3} f(t)$ ?

② $\lim_{t \to -1} f(t)$ ?

③ $\lim_{t \to 0} f(t)$ ?

④ $\lim_{t \to 2} f(t)$ ?

← ① $\lim_{t \to 3} 2t = 2(3) = 6$.

② $\lim_{t \to -1} (t-2) = -1-2 = -3$

③ $\lim_{t \to 0} f(t)$ ⟹ لدينا نقطة تشعب

$\lim_{t \to 0^+} t^2 = 0$ ، $\lim_{t \to 0^-} (t-2) = -2$

$\Rightarrow \lim_{t \to 0} f(t)$ : d.n.E.

④ $\lim_{t \to 2^+} 2t = 4$

$\lim_{t \to 2^-} t^2 = 4$

$\therefore \lim_{t \to 2} f(t) = 4$

# Calculus "1"

## ② اقتران القيمة المطلقة :

ex) $\lim_{x\to 0} |x|$ ؟

* يجب إعادة التعريف عند وجود القيمة المطلقة في السؤال.

| $(-x)$ | $(x)$ |
|---|---|
| − − − | + + + |
| | 0 |

$\Rightarrow \lim_{x\to 0^+} x = 0$

$\lim_{x\to 0^-} -x = 0$

$\therefore \lim_{x\to 0} |x| = 0$ .

ex) $\lim_{x\to 0} \frac{|x|}{x}$ ؟

| $(-x)$ | $0$ | $(x)$ |
|---|---|---|
| − − − | | + + + |

$\Rightarrow \lim_{x\to 0^+} \frac{x}{x} = 1$

$\lim_{x\to 0^-} \frac{-x}{x} = -1$

$\therefore \lim_{x\to 0^-} \frac{|x|}{x} \neq \lim_{x\to 0^+} \frac{|x|}{x}$

$\therefore \lim_{x\to 0} \frac{|x|}{x}$ D.N.E

## ③ اقتران أكبر عدد صحيح : $[\ ]$

ex) $\lim_{x\to 3} [x+0.5] = [3.5] = 3$

ex) $\lim_{x\to 2} [-x+0.25] = [-1.75] = -2$

ex) $\lim_{x\to 1} \left[\frac{x}{2}\right] = \left[\frac{1}{2}\right] = 0$

⇐ إذا كان ناتج التعويض عدد صحيح بشكل عام تكون النهاية غير موجودة لأن النقطة في هذه الحالة هي نقطة قول :-

$\lim_{x\to a^+} \neq \lim_{x\to a^-}$

ex) $\lim_{x\to 3} [x]$ ؟

$\lim_{x\to 3^+} [x] = [3]^+ = 3$

$\lim_{x\to 3^-} [x] = [3]^- = 2$

$\therefore \lim_{x\to 3} [x]$ d.n.e.

ex) $\lim_{x\to 1} [2-x]$ ؟

$\lim_{x\to 1^+} [2-x] = [1]^+ = 0$

$\lim_{x\to 1^-} [2-x] = [1]^- = 1$

$\therefore \lim_{x\to 1} [2-x]$ d.n.e.

Calculus "1"

4 الجذور الزوجية ($\sqrt{\ }$) .

ex) $\lim_{x\to 5} \sqrt{x+4} = \sqrt{9} = 3$ .

ex) $\lim_{x\to 2} \sqrt{x-1} = \sqrt{1} = 1$

ex) $\lim_{x\to 0} \sqrt{x-1} = \sqrt{-1}$ d.n.e .

ex) $\lim_{x\to 2} \sqrt{x-2}$ ← $\sqrt{0}$ !

انتبه ناتج الجذور $\sqrt{0}$

$\lim_{x\to 2^-} \sqrt{x-2} = \sqrt{0}$ d.n.e .

↓ يجعل ما داخل الجذر غير معرف على مجال

قيمة أقل من 2 بقليل

$\lim_{x\to 2^+} \sqrt{x-2} = \sqrt{0^+} = 0$

↓ معرف

↓ قيمة أكبر من 2 بقليل

$\therefore \lim_{x\to 2} \sqrt{x-2}$ d.n.e

ex) $\lim_{x\to 7} \sqrt{7-x}$ ?

↳ $\lim_{x\to 7^+} \sqrt{7-x} = \sqrt{0^-}$ d.n.e

$\lim_{x\to 7^-} \sqrt{7-x} = \sqrt{0^+}$

$\therefore \lim_{x\to 7} \sqrt{7-x}$ d.n.e .

ex) $\lim_{x\to 3} \sqrt[7]{x-3} = \sqrt[7]{0} = 0$

↓ الجذر فردي معرف على $\mathbb{R}$

5 عدد $\Leftarrow$ $\frac{\text{number}}{0}$ صفر .

* Remark :-

* $\frac{\text{عدد موجب}}{0} = +\infty$ ، * $\frac{\text{عدد سالب}}{0} = -\infty$

ex) $\lim_{x\to 0^-} \frac{1}{x^3} = \frac{1}{0^-} = -\infty$

* $\lim_{x\to 1^+} \frac{2}{x-1} = \frac{2}{0^+} = +\infty$

* $\lim_{x\to 0^-} \frac{1}{x^2} = \frac{1}{0^+} = +\infty$ لأنها تربيعي

* $\lim_{x\to 1^-} \frac{2}{x-1} = \frac{2}{0^-} = -\infty$

* $\lim_{x\to 1^+} \frac{-2}{x-1} = \frac{-2}{0^+} = -\infty$

* $\lim_{x\to 0^-} \frac{x^3-1}{x^4-x^3} = \frac{-1}{0^+ + 0^+} = \frac{-1}{0^+} = -\infty$

# Calculus "1"

* إذا طلب السؤال النهاية دون تحديد الجهة نرسلها من الجهتين :-

ex) $\lim_{x\to 1} \frac{5}{1-x}$ ؟

$\lim_{x\to 1^+} \frac{5}{1-x} = \frac{5}{0^-} = -\infty$

$\lim_{x\to 1^-} \frac{5}{1-x} = \frac{5}{0^+} = +\infty$

$\therefore \lim_{x\to 1} \frac{5}{1-x}$ d.n.e.

ex) $\lim_{x\to 0^+} \left(\frac{1}{x} - \frac{1}{x^2}\right)$ ؟

$\lim_{x\to 0^+} \frac{x-1}{x^2} = \frac{-1}{0^+} = -\infty$

ex) $\lim_{x\to 3^+} \frac{x}{x-3} = \frac{3}{0^+} = +\infty$

ex) $\lim_{x\to 2} \frac{x}{x^2-4}$ ؟

$\lim_{x\to 2^+} \frac{x}{x^2-4} = \frac{2}{0^+} = \infty$

$\lim_{x\to 2^-} \frac{x}{x^2-4} = \frac{2}{0^-} = -\infty$

$\therefore \lim_{x\to 2} \frac{x}{x^2-4}$ d.n.e.

ex) $\lim_{x\to 3^-} \frac{1}{|x-3|}$ ؟

$\lim_{x\to 3^-} \frac{1}{(3-x)}$ — (3-x) — $\frac{1}{3}$ — (x-3) + + +

$= \frac{1}{0^+} = \infty$

ex) $\lim_{x\to -2^+} \frac{1}{\sqrt{x^2+4x+4}}$ ؟

$\lim_{x\to -2^+} \frac{1}{\sqrt{(x+2)^2}}$

$= \lim_{x\to -2^+} \frac{1}{|x+2|}$ — (x+2) | (x+2) + + — $-2$

$\Rightarrow \lim_{x\to -2^+} \frac{1}{x+2} = \frac{1}{0^+} = \infty$

ex) $\lim_{x\to 2} \sqrt[4]{4-x^2}$ ؟ الجذور الزوجية تعامل معاملة الجذر التربيعي ...

+ + + | − − − $-2$ (1) نعرف (2) غير نعرف

$\Rightarrow \lim_{x\to 2^-} \sqrt[4]{4-x^2} = \sqrt[4]{0^+} = 0$.

$\lim_{x\to 2^+} \sqrt[4]{4-x^2} = \sqrt[4]{0^-} =$ d.n.e

$\therefore \lim_{x\to 2} \sqrt[4]{4-x^2}$ d.ne.

Calculus "1"

ex) $\lim_{x\to 1} \frac{\sqrt{x^2-1}}{\sqrt{x-1}}$ ?

لا نهتم بحال كل من البسط و المقام على حدة ...

$\sqrt{x^2-1}$ : − − − | + + +  (عند $-1$ و $1$)

$\sqrt{x-1}$ : − − − | + + +  (عند $1$)

الجزء الذي عند $x=1$ من البسط معرف لكنه غير معرف من المقام ...

$$\Rightarrow \lim_{x\to 1^+} \frac{\sqrt{x^2-1}}{\sqrt{x-1}} = \lim_{x\to 1^+} \frac{\sqrt{x-1}\sqrt{x+1}}{\sqrt{x-1}}$$

$$= \sqrt{2}.$$

$$\Rightarrow \lim_{x\to 1^-} \frac{\sqrt{x^2-1}}{\sqrt{x-1}} \quad \text{d.n.e}$$

$$\therefore \lim_{x\to 1} \frac{\sqrt{x^2-1}}{\sqrt{x-1}} \quad \text{d.ne.}$$

ex) $\lim_{x\to 3} \sqrt{\frac{9-x^2}{3-x}}$ ?

$$\hookrightarrow \lim_{x\to 3} \sqrt{\frac{(3-x)(3+x)}{(3-x)}}$$ بنزل دالة

$$= \lim_{x\to 3} \sqrt{3+x} = \sqrt{6}.$$

* Indeterminates :-
القيم المرفوضة في تعويض النهايات

$$\frac{0}{0}, \frac{\pm\infty}{\pm\infty}, 0\cdot\pm\infty, \infty-\infty$$

$$1^\infty, \infty^0, 0^0 \ldots.$$

But :-

① $\infty+\infty=\infty$ ② $-\infty-\infty=-\infty$

③ $\infty\cdot\infty=\infty$ ④ $(-\infty)\cdot(-\infty)=\infty$

⑤ $(-\infty)\cdot\infty=-\infty$ ⑥ $(+a)\cdot\infty=+\infty$

⑦ $(-a)\infty=-\infty$ ⑧ $a^\infty=\infty$ if $a>1$

⑨ $a^\infty=0$, $0<a<1$ ⑩ $\frac{\text{number}}{\pm\infty}=0$

⑪ $\frac{0}{\text{number}}=0$ ⑫ $\frac{\pm\infty}{\text{number}}=\pm\infty$

* Evaluate :-

1) $\lim_{x\to 7} \frac{x^2-49}{x-7}$ ?

$\hookrightarrow \frac{0}{0}! \Rightarrow \lim_{x\to 7} \frac{(x-7)(x+7)}{(x-7)} = 14.$

2) $\lim_{x\to 1} \frac{x^2-x}{x-1}$ ?

$\hookrightarrow \frac{0}{0}! \Rightarrow \lim_{x\to 1} \frac{x(x-1)}{(x-1)} = 1$

3) $\lim_{x\to 1} \frac{x^2-x^3}{(1-x)}$ ?

$\hookrightarrow \frac{0}{0}! \Rightarrow \lim_{x\to 1} \frac{x^2(1-x)}{(1-x)} = 1$

Calculus "1"

(4) $\lim_{x\to 1} \frac{(x^2-2x+1)^5}{(x^2-1)^{10}}$ ?

$\lim_{x\to 1} \frac{((x-1)(x-1))^5}{(x-1)^{10}(x+1)^{10}}$

$= \lim_{x\to 1} \frac{((x-1)^2)^5}{(x-1)^{10}(x+1)^{10}}$

$= \lim_{x\to 1} \frac{(x-1)^{10}}{(x-1)^{10}} \cdot \frac{1}{(x+1)^{10}} = \frac{1}{2^{10}}$

(5) $\lim_{x\to 3} \frac{3-x}{x^2-9}$ ?

$\frac{0}{0}! \Rightarrow \lim_{x\to 3} \frac{(-1)(3-x)}{(x-3)(x+3)}$

$= \frac{-1}{6}$

(6) $\lim_{x\to -4} \frac{2x^2+8x}{x^2+x-12}$ ?

$\lim_{x\to -4} \frac{x(2x+8)}{(x+4)(x-3)}$

$= \lim_{x\to -4} \frac{2x(x+4)}{(x+4)(x-3)} = \frac{-8}{-7} = \frac{8}{7}$

* Rules:-

① $a^3+b^3 = (a+b)(a^2-ab+b^2)$

② $a^3-b^3 = (a-b)(a^2+ab+b^2)$

(7) $\lim_{x\to -1} \frac{x^3+1}{x^2-1}$ ?

$\frac{0}{0}! \Rightarrow \lim_{x\to -1} \frac{(x+1)(x^2-x+1)}{(x-1)(x+1)}$

$= \lim_{x\to -1} \frac{x^2-x+1}{x-1} = \frac{-3}{2}$

(8) $\lim_{x\to 2} \frac{x^2-4x+4}{x^2+x-6}$ ? (homework)

(9) $\lim_{x\to 2} \frac{x^3-8}{x-2}$ ?

$\lim_{x\to 2} \frac{(x-2)(x^2+2x+4)}{(x-2)} = 12$

كيف نحللها ؟ $* 2x^2+x-1=0$

عددان حاصل ضربهما (-2) ومجموعهما (1) هما -1 ، 2

$2x^2+(2+(-1))x-1 = 0$

$2x^2+2x-x-1 = 0$

$2x(x+1)-1(x+1)$ (نأخذ (x+1) عامل مشترك)

$* 3x^2-11x+6=0$ ؟

عددين حاصل ضربهما (3×6 = 18) ومجموعهما 11 هما -9 ، -2

$(x-3)(3x-2)=0$

(10) $\lim_{x\to -1} \frac{2x^2+x-1}{(x+1)} = \frac{0}{0}!$

$\lim_{x\to -1} \frac{(x+1)(2x-1)}{(x+1)} = -3$

Calculus "1"

(12) $\lim_{x\to 4} \frac{|x^2-4x+4|-4}{(x-4)}$ ؟

$\lim_{x\to 4} \frac{|(x-2)^2|-4}{x-4}$

← القوة الزوجية تلغي المطلق لكن القوة الفردية لا تلغيه ...

← وبما أن $(x-2)^2$ دائماً موجبة فلا داعي للمطلق ...

$\Rightarrow \lim_{x\to 4} \frac{(x-2)^2-4}{x-4}$

$\lim_{x\to 4} \frac{(x^2-4x+4)-4}{x-4}$

$\lim_{x\to 4} \frac{x(x-4)}{(x-4)} = 4$ .

* في حالة الجذور نستخدم عادة الضرب بالمرافق ...

(ex) $\lim_{x\to 9} \frac{x-9}{\sqrt{x}-3}$ ؟

$\lim_{x\to 9} \frac{x-9}{\sqrt{x}-3} * \frac{\sqrt{x}+3}{\sqrt{x}+3}$

$= \lim_{x\to 9} \frac{x-9}{x-9}(\sqrt{x}+3) = 6$ .

(ex) $\lim_{x\to 0} \frac{x}{\sqrt{x+1}-1}$ ؟

$\frac{0}{0}$ ! $\Rightarrow \lim_{x\to 0} \frac{x}{\sqrt{x+1}-1} * \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}$

$= \lim_{x\to 0} \frac{(x)(\sqrt{x+1}+1)}{x+x-x} = 2$ .

(ex) $\lim_{y\to 4} \frac{4-y}{2-\sqrt{y}}$ ؟

$\lim_{y\to 4} \frac{4-y}{2-\sqrt{y}} * \frac{2+\sqrt{y}}{2+\sqrt{y}}$

$= \lim_{y\to 4} \frac{4-y}{4-y} * 2+\sqrt{y} = 4$ .

(homework) $\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$ ؟

$\frac{1}{4}$

(ex) $\lim_{x\to 0} \frac{\sqrt{x^2+4}-2}{x}$ ؟

$\lim_{x\to 0} \frac{\sqrt{x^2+4}-2}{x} * \frac{\sqrt{x^2+4}+2}{\sqrt{x^2+4}+2}$

$= \lim_{x\to 0} \frac{x^2+4-4}{x(\sqrt{x^2+4}+2)} = \frac{0}{4} = 0$ .

(ex) $\lim_{x\to 2} \frac{\sqrt{6-x}-2}{\sqrt{3-x}-1}$ ؟

$\lim_{x\to 2} \frac{\sqrt{6-x}-2}{\sqrt{3-x}-1} * \frac{\sqrt{6-x}+2}{\sqrt{6-x}+2} * \frac{\sqrt{3-x}+1}{\sqrt{3-x}+1}$

$= \lim_{x\to 2} \frac{6-x-4}{3-x-1} * \frac{\sqrt{3-x}+1}{\sqrt{6-x}+2}$

$= \lim_{x\to 2} \frac{2-x}{2-x} * \frac{2}{4} = \frac{1}{2}$ .

# Calculus "1"

ex) $\lim_{x\to 0} \frac{\sqrt{x^2+4}-2+x}{x}$ ?

$\hookrightarrow \lim_{x\to 0} \frac{\sqrt{x^2+4}-2}{x} + \lim_{x\to 0} \frac{x}{x}$

نوزع المقام على البسط حيث يكون ناتج التعويض كما نهاية $\frac{0}{0}$.

$\hookrightarrow \lim_{x\to 0} \frac{\sqrt{x^2+4}-2}{x} \times \frac{\sqrt{x^2+4}+2}{\sqrt{x^2+4}+2} + 1$

$= \lim_{x\to 0} \frac{x^2+4-4}{x} \times \frac{1}{4} + 1$

$= 0 + 1 = 1.$

* $a^3 - b^3 = (a-b)(a^2+ab+b^2)$

مرافق تكعيبي

*ex) $\lim_{x\to 2} \frac{\sqrt[3]{x+6}-2}{x-2}$ ?

$\hookrightarrow \lim_{x\to 2} \frac{\sqrt[3]{x+6}-2}{x-2} \times \frac{\sqrt[3]{(x+6)^2}+2\sqrt[3]{x+6}+4}{\sqrt[3]{(x+6)^2}+2\sqrt[3]{x+6}+4}$

$= \lim_{x\to 2} \frac{x+6-8}{x-2} \times \frac{1}{4+4+4}$

$= \lim_{x\to 2} \frac{x-2}{x-2} \times \frac{1}{12} = \frac{1}{12}$

Remark $\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \cdot \frac{d}{c}$

ex) $\lim_{x\to 4} \frac{\frac{1}{x}-\frac{1}{4}}{x-4}$ ?

$\hookrightarrow$ توحيد المقامات : $\lim_{x\to 4} \frac{4-x}{(4x)(x-4)} \overset{(-1)}{=} \frac{-1}{16}$

ex) $\lim_{x\to 3} \frac{\frac{1}{2+x}-\frac{1}{5}}{2x-6}$ ?

$\hookrightarrow \lim_{x\to 3} \frac{5-2-x}{2(x-3)(10+5x)}$

$= \lim_{x\to 3} \frac{3-x}{2(x-3)(25)} = \frac{-1}{50}$

* homework *

$\hookrightarrow \lim_{x\to 6} \left(\frac{1}{7}-\frac{1}{x+1}\right)\left(\frac{1}{6-x}\right)$ ?

$\rightarrow$ الحل $\frac{-1}{49}$

* مثال على العوامل *

If $\lim_{x\to 1} \frac{x^2+4x+a}{x+3} = 3$, find (a)

لـ ناتج التعويض المباشر مقبول ...

$\frac{1+4+a}{1+3} = 3$

$\hookrightarrow 12 = 5 + a$

$\therefore a = 7$

# Calculus "1"

* قاعدة مهمة جداً :-

اذا كانت النهاية موجودة و نهاية المقام = صفر

فإن نهاية البسط = صفر

ex) If $\lim_{x\to 1} \frac{f(x)-8}{x-1} = 10$ (exists), find $\lim_{x\to} f(x)$ ?

↳ بما ان نهاية المقام صفر

$\lim_{x\to 1} x-1 = 0$

فإن نهاية البسط = صفر أيضا

$\lim_{x\to 1} f(x)-8 = 0$

$\therefore \lim_{x\to 1} f(x) = 8$

ex) If $\lim_{x\to 2} \frac{\sqrt{4x^2+a}-5}{x^2-4}$ exist, find (a) ?

↳ $\lim_{x\to 2} x^2-4 = 0$ نهاية المقام صفر

$\Rightarrow \lim_{x\to 2} \sqrt{4x^2+a} - 5 = 0$

$\sqrt{16+a} - 5 = 0$

$\sqrt{16+a} = 5$

$16+a = 25$

$\therefore a = 9$

Rules :-

① $\lim_{x\to 0} \frac{\sin(ax)}{ax} = 1 = \frac{a}{a}$

② $\lim_{x\to 0} \frac{\tan(ax)}{ax} = 1 = \frac{a}{a}$

③ $\lim_{x\to 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$

④ $\lim_{x\to 0} \frac{\tan(ax)}{bx} = \frac{a}{b}$

⑤ $\lim_{x\to 0} \frac{\sin(ax)}{\sin(bx)} = \frac{a}{b}$.

⑥ $\lim_{x\to 0} \frac{\sin(ax)}{\tan(bx)} = \frac{a}{b}$.

⑦ $\lim_{x\to 0} \frac{\tan(ax)}{\tan(bx)} = \frac{a}{b}$.

EX)

① $\lim_{x\to 0} \frac{\sin(3x)}{3x} = \frac{3}{3} = 1$

② $\lim_{x\to 0} \frac{\sin(3x)}{5x} = \frac{3}{5}$

③ $\lim_{x\to 0} \frac{\sin(3x)}{x} = \frac{3}{1}$

# Calculus "1"

④ $\lim_{x\to 5} \frac{\sin(x-5)}{(x-5)}$ ؟

↳ $\frac{0}{0}$ ! $\Rightarrow$ $y = x-5$, $y \to 0$, $x \to 5$

$\Rightarrow \lim_{y\to 0} \frac{\sin y}{y} = 1$ حسب القاعدة

⑤ $\lim_{x\to 0} \frac{\sin(3x)}{\sin(5x)} = \frac{3}{5}$

⑥ $\lim_{x\to 0} \frac{\tan x}{x} = \frac{1}{1} = 1$

⑦ $\lim_{x\to 0} \frac{\sin(\sqrt{2}x)}{x} = \frac{\sqrt{2}}{1} = \sqrt{2}$.

⑧ $\lim_{x\to 0} \frac{\sin x}{x + \tan x}$ ؟

← نقسم كل من البسط والمقام على x ثم نوزع النهايات ..

$\Rightarrow \lim_{x\to 0} \frac{\sin x}{x}$

$\lim_{x\to 0} \frac{x}{x} + \lim_{x\to 0} \frac{\tan x}{x}$

$= \frac{1}{1+1} = \frac{1}{2}$.

⑨ $\lim_{x\to 0} \frac{\sin x^2}{x}$ ؟ $\Rightarrow \frac{0}{0}$ !

* لاستخدام القاعدة يجب ان تكون زاوية sin نفس المقسوم عليها ...

* نضرب كل من البسط والمقام بـ x :-

$\lim_{x\to 0} \frac{\sin x^2}{x} \cdot \frac{x}{x}$

$= \lim_{x\to 0} \frac{\sin(x^2)}{x^2} * \lim_{x\to 0} x$

$x^2 = y \Rightarrow x \to 0, y \to 0$

$= \lim_{y\to 0} \frac{\sin y}{y} * \lim_{x\to 0} x$

$= 1 \cdot 0 = 0$.

⑩ $\lim_{x\to 1} \frac{\sin(x-1)}{x^2+x-2}$ ؟

↳ $\lim_{x\to 1} \frac{\sin(x-1)}{(x-1)(x+2)}$

$= \lim_{x\to 1} \frac{\sin(x-1)}{(x-1)} * \lim_{x\to 1} \frac{1}{x+2}$

let $x-1 = y$, $x \to 1$, $y \to 0$ بتعويض

$\Rightarrow \lim_{y\to 0} \frac{\sin y}{y} * \frac{1}{3}$

$= \frac{1}{1} * \frac{1}{3} = \frac{1}{3}$.

Calculus "1"

(11) $\lim_{x\to -2} \frac{\sin(x^2+3x+2)}{x+2}$ ؟

إذا كان ناتج تعويض ما داخل ال sin , tan هيز نقسم على ما داخلها .

$$\lim_{x\to -2} \frac{\sin(x^2+3x+2)}{\frac{x^2+3x+2}{x+2}} \qquad \frac{x+2}{x^2+3x+2}$$

$y = x^2+3x+2$

$x \to -2 \Rightarrow y \to 0$

$$\therefore \lim_{y\to 0} \frac{\sin y}{y} = 1$$

$$\lim_{x\to -2} \frac{(x+2)}{(x+2)(x+1)} = -1$$

(12) $\lim_{x\to 0} x^2(1+\cot^2 3x)$ ؟

$$\lim_{x\to 0} x^2 + \lim_{x\to 0} x^2 \cot^2 3x$$

$$\lim_{x\to 0} x^2 + \lim_{x\to 0} \frac{x^2}{\tan^2 3x}$$

$$\lim_{x\to 0} x^2 + \left(\lim_{x\to 0} \frac{x}{\tan 3x}\right)^2$$

$$= 0 + \left(\frac{1}{3}\right)^2 = \frac{1}{9} .$$

(13) $\lim_{x\to 2} (x^2-2x)\cot(x^2-4)$ ؟

$$\lim_{x\to 2} \frac{x^2-2x}{\tan(x^2-4)} = \lim_{x\to 2} \frac{x^2-2x}{x^2-4} \cdot \frac{x^2-4}{\tan(x^2-4)}$$

$$= \lim_{x\to 2} \frac{x(x-2)}{(x-2)(x+2)} = \frac{1}{2} .$$

$$\lim_{y\to 0} \frac{\tan y}{y}$$

* اذا كانت ناتج تعويض النهاية $\frac{0}{0}$ وكان زاوية sin, tan لفظ المقسوم عليها يكون الناتج 1 دون الحاجة لأي اشتقاق .

ex) $\lim_{x\to 1} \frac{\sin(x-1)}{x-1} = 1$

ex) $\lim_{x\to 0} \frac{\sin(1-\cos x)}{1-\cos x} = 1$

(14) $\lim_{x\to 3} \frac{\sin(x-3)}{x^2-2x-3}$ (homework)

(15) $\lim_{x\to 2} \frac{\tan(x-2)}{x^2-4}$ ؟

$$\lim_{x\to 2} \frac{\tan(x-2)}{(x-2)} = 1$$

$$\lim_{x\to 2} \frac{(x^2-4)}{(x-2)} \qquad \lim_{x\to 2} \frac{(x-2)(x+2)}{(x-2)}$$

$= \frac{1}{4}$ . فاضل بوليتكنك .

Calculus "1"

(حالات الضرب بالمرافق)

$1 \pm$ دائريا

المرافق

| | |
|---|---|
| $1+\sin x$ | $1-\sin x$ |
| $3-\cos x$ | $3+\cos x$ |
| $1-\sqrt{2}\sin x$ | $1+\sqrt{2}\sin x$ |
| $\sin 4x+1$ | $\sin 4x-1$ |
| $\sin x+\cos x$ | $\sin x-\cos x$ |

(3) $\lim_{x\to 0} \frac{\cos x-1}{x}$ ? $\frac{0}{0}$ !

$= \lim_{x\to 0} \frac{\cos x-1}{x} * \frac{\cos x+1}{\cos x+1}$

$= \lim_{x\to 0} \frac{\cos^2 x-1}{x} \cdot \frac{1}{2}$

$= \lim_{x\to 0} \frac{-\sin^2 x}{2x} = \lim_{x\to 0} -\frac{\sin x}{2}\frac{\sin x}{x}$

$= \frac{0}{2} \cdot 1 = 0$

(1) $\lim_{x\to 0} \frac{x^2}{1-\cos x}$ ? $= \frac{0}{0}$ !

$= \lim_{x\to 0} \frac{x^2}{1-\cos x} * \frac{1+\cos x}{1+\cos x}$ (ضرب تبادلي)

$= \lim_{x\to 0} \frac{x^2}{1-\cos^2 x} * (1+1)$

$= \lim_{x\to 0} \frac{x^2}{\sin^2 x} \cdot (2)$

$= \lim_{x\to 0} \left(\frac{x}{\sin x}\right)^2 \cdot (2) = 2$

(2) $\lim_{x\to \pi} \frac{1+\cos x}{\sin x}$ ? $\frac{0}{0}$ !

$\hookrightarrow \lim_{x\to \pi} \frac{1+\cos x}{\sin x} * \frac{1-\cos x}{1-\cos x}$

$= \lim_{x\to \pi} \frac{1-\cos^2 x}{\sin x} \cdot \frac{1}{(2)}$

$= \lim_{x\to \pi} \frac{\sin^2 x}{\sin x} \cdot \frac{1}{2}$

$= 0 \cdot \frac{1}{2} = 0$

(4) $\lim_{x\to 0} \frac{1-\cos 5x}{\cos 7x-1}$ ?

$\hookrightarrow \lim_{x\to 0} \frac{1-\cos 5x}{\cos 7x-1} * \frac{1+\cos 5x}{1+\cos 5x} * \frac{\cos 7x+1}{\cos 7x+1}$

$= \lim_{x\to 0} \frac{1-\cos^2 5x}{\cos^2 7x-1} \cdot \frac{1+1}{1+1}$

$= \lim_{x\to 0} \frac{\sin^2 5x}{-\sin^2 7x} = \lim_{x\to 0} -\left(\frac{\sin 5x}{\sin 7x}\right)^2$

$= -\frac{25}{49}$

$\Rightarrow \sin^2 5x+\cos^2 5x=1$

$1-\cos^2 5x = \sin^2 5x$

$\Rightarrow \sin^2 7x+\cos^2 7x=1$

$\therefore \cos^2 7x-1 = -\sin^2 7x$.

Calculus "1"

5) $\lim_{x\to 0} \frac{-2\sin(1-\cos x)}{x^2}$ ?

$\downarrow$ $\lim_{x\to 0} \frac{-2\sin(1-\cos x)}{1-\cos x} * \frac{x^2}{1-\cos x} * \frac{1+\cos x}{1+\cos x}$

$\lim_{y\to 0} \frac{-2\sin y}{y}$

$\lim_{x\to 0} \frac{2x^2}{\sin^2 x}$

$= \frac{-1}{\lim_{x\to 0}\left(\frac{x}{\sin x}\right)^2} = \frac{-1}{1} = -1$

6) $\lim_{x\to 1} \frac{\sin \pi x}{x-1}$ ?

$\downarrow$ $y = x-1 \rightarrow x = y+1$

$y\to 0$ , $x\to 1$

$\lim_{y\to 0} \frac{\sin \pi(y+1)}{y}$

$= \lim_{y\to 0} \frac{\sin(\pi y + \pi)}{y}$

$= \lim_{y\to 0} \frac{\sin \pi y \cos \pi + \sin \pi \cos \pi y}{y}$

$= -\lim_{y\to 0} \frac{\sin \pi y}{y} = -\pi$ .

طريقة أخرى لمثال رقم (6) :-

$\lim_{x\to 1} \frac{\sin(\pi - \pi x)}{(x-1)}$ بالدين زاوية

$= \lim_{x\to 1} \frac{\sin \pi(1-x)}{(x-1)}$

$1-x = y$ $\quad y\to 0$

$x-1 = -y$

$\therefore \lim_{y\to 0} \frac{\sin \pi y}{-y} = -\pi$

7) $\lim_{x\to \frac{\pi}{4}} \frac{\cos x - \sin x}{x - \frac{\pi}{4}}$ ?

$\downarrow$ Sol :- $-\sqrt{2}$ (homework)

* Theorem :- (Squeezing) (Sandwish)

$\Rightarrow$ If $f(x) \le g(x) \le h(x)$ and

$\lim_{x\to a} f(x) = l$ and $\lim_{x\to a} h(x) = l$

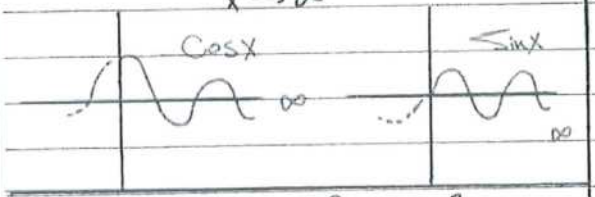
Then $\lim_{x\to a} g(x) = l$ .

* نستخدم هذه النظرية في حالات تعويض النهايات (o.d.n.e)

# Calculus "1"

* Note:- 1) $\lim_{x\to\infty} \sin x$ = d.n.e

2) $\lim_{x\to\infty} \cos x$ = d.n.e




ex) If $4x-9 \le f(x) \le x^2-4x+7$ find $\lim_{x\to 4} f(x)$ ?

$4x-9 \le f(x) \le x^2-4x+7$

$\lim_{x\to 4} 4x-9 = 7$ , $\lim_{x\to 4} x^2-4x+7 = 7$

$\therefore \lim_{x\to 4} f(x) = 7$

ex) $2\cos(x-1) \le \frac{f(x)}{(x+1)} \le \frac{x^2+2x-3}{x^2-1}$

find $\lim_{x\to 1} f(x)$ ?

↳ $\lim_{x\to 1} 2\cos(x-1) = 2\cdot 1 = 2$

$\Rightarrow \lim_{x\to 1} \frac{(x-1)(x+3)}{(x-1)(x+1)} = \frac{4}{2} = 2$

$\therefore \lim_{x\to 1} \frac{f(x)}{x+1} = 2$

$\frac{\lim_{x\to 1} f(x)}{\lim_{x\to 1} x+1} = 2$

$\therefore \lim_{x\to 1} f(x) = 4$

ex) $\lim_{x\to 0^+} x^2 \sin\left(\frac{1}{x}\right)$ ?

↳ $0 \cdot \sin(\infty)$ !! Squeezing

$\Rightarrow -1 \le \sin\left(\frac{1}{x}\right) \le 1$

$-x^2 \le x^2 \sin\left(\frac{1}{x}\right) \le x^2$

$\lim_{x\to 0^+} -x^2 = 0$ , $\lim_{x\to 0^+} x^2 = 0$

$\lim_{x\to 0^+} = 0$

ex) $\lim_{x\to 0} x \sin^2\left(\frac{1}{x}\right)$ ? = 0 (d.n.e) !

$0 \le \sin^2\left(\frac{1}{x}\right) \le 1$

$0 \le x \sin^2\left(\frac{1}{x}\right) \le x$

$\lim_{x\to 0} 0 = 0$ , $\lim_{x\to 0} x = 0$

$\therefore \lim_{x\to 0} = 0$

ex) $\lim_{x\to 0^+} x^2 \sin\left(\frac{\pi}{\sqrt{x}}\right)$ ?

$-1 \le \sin\left(\frac{\pi}{\sqrt{x}}\right) \le 1$

$-x^2 \le x^2 \sin\left(\frac{\pi}{\sqrt{x}}\right) \le x^2$

$\lim_{x\to 0} = 0$ , $\therefore \lim_{x\to 0} = 0$ , $\lim_{x\to 0} = 0$

* homework * $\lim_{x\to 0^+} x \cos^2\left(\frac{1}{\sqrt{x}}\right)$ ?

↳ Sol : 0.

# Calculus "1"

ex) If $2x+2 \le f(x) \le x^2+3$
Find $\lim_{x\to 1} \frac{f(x)-4}{x-1}$ ?

$\hookrightarrow 2x+2 \le f(x) \le x^2+3$

$2x-2 \le f(x)-4 \le x^2-1$

$\frac{2x-2}{x-1} \le \frac{f(x)-4}{x-1} \le \frac{x^2-1}{x-1}$

$\lim_{x\to 1} \frac{2(x-1)}{(x-1)} = 2$ ، $\lim_{x\to 1} \frac{(x-1)(x+1)}{(x-1)} = 2$

$\therefore \lim_{x\to 1} \frac{f(x)-4}{(x-1)} = 2$

## # Vertical Asymptote:- (الخط المقارب العمودي)

$\Rightarrow x=a$ is said to be vertical Asymptote happens if $\lim_{x\to a} f(x) = \pm\infty$, $\lim_{x\to a^+} f(x) = \pm\infty$ or $\lim_{x\to a^-} f(x) = \pm\infty$.

* ممكن المتوقع انه نجد (V.A) عند أصفار المقام.
* ليس كل أصفار المقام هي (V.A)
* يفضل الإختصار قبل اخراج اصفار المقام.

Ex) Find (V.A) ?

① $f(x) = \frac{1}{x}$ ؟ أصفار المقام.

$\hookrightarrow \lim_{x\to 0} \frac{1}{x} = \pm\infty$ $\therefore (x=0)$

② $f(x) = \frac{3}{x^2+1}$ ؟

$\hookrightarrow x^2+1=0$ أصفار المقام لا يحل فلا يوجد أصفار.

$\therefore$ No (V.A)

③ $f(x) = \frac{x-1}{|x|-1}$ ؟

$\hookrightarrow$ أصفار مقام $\Rightarrow |x|-1=0$

$|x|=1 \Rightarrow (x=\pm 1)$

$\lim_{x\to 1} \frac{x-1}{x-1} = 1 \Rightarrow \therefore x=1$ not V.A

$\lim_{x\to -1} \frac{(x-1)}{-(x+1)} = \frac{2}{0} = +\infty \therefore (x=-1)$ (V.A)

④ $f(x) = \frac{x-2}{x^2-4}$ ؟

$\hookrightarrow \frac{(x-2)}{(x-2)(x+2)} = \frac{1}{x+2} \Rightarrow x=-2$

$\hookrightarrow \lim_{x\to -2} \frac{x-2}{x^2-4} = \frac{-4}{0} = \pm\infty$

$\therefore (x=-2$ is V.A)

⑤ $f(x) = \frac{x^2-1}{x+1}$ ؟

$\hookrightarrow \frac{(x-1)(x+1)}{(x+1)} = (x-1)$ $\therefore$ No V.A

⑥ $f(x) = \frac{x+1}{x^2-1}$ ؟

$\hookrightarrow \frac{(x+1)}{(x+1)(x-1)} = \frac{1}{x-1} \Rightarrow x=1$

$\hookrightarrow \lim_{x\to 1} \frac{(x+1)}{(x+1)(x-1)} = \frac{1}{0} = \pm\infty$

$\therefore x=1$ is V.A

Calculus "1"

⑦ $f(x) = \ln x$ ?

↓

* الحالات الوحيدة V.A بدون مقام: $x=0$

* اللي بجعل ما داخل ln = صفر

$\Rightarrow \lim_{x\to 0^+} \ln x = \ln 0^+ = -\infty$

$\therefore x=0$ V.A

⑧ $f(x) = \ln(x-2)$ ?

↳ $x-2=0 \Rightarrow x=2$

$\Rightarrow \lim_{x\to 2^+} \ln(x-2) = \ln 0^+ = -\infty$

$\therefore x=2$ V.A

⑨ $f(x) = \ln(3x+1)$ ?

↳ $3x+1=0 \Rightarrow x=-\frac{1}{3}$

$\Rightarrow \lim_{x\to -\frac{1}{3}^+} \ln(3x+1) = \ln 0^+ = -\infty$

$\therefore x=-\frac{1}{3}$ V.A

⑩ $f(x) = \frac{\sin(x-2)}{(x-2)}$ ?

↳ أصفار المقام $\Rightarrow x-2=0 \Rightarrow x=2$

$\lim_{x\to 2} \frac{\sin(x-2)}{(x-2)} = 1$

$\therefore x=2$ Not V.A

$\therefore$ No V.A

⑪ $f(x) = \frac{\sin(x-2)}{x^2-4}$ ?

↳ أصفار المقام $x^2-4=0$

$x=\pm 2$

$(x=2)$

↳ $\lim_{x\to 2} \frac{\sin(x-2)}{x^2-4} = \frac{0}{0}$ !

$= \lim_{x\to 2} \frac{\sin(x-2)}{(x-2)} * \lim_{x\to 2} \frac{1}{(x+2)}$

$= 1 * \frac{1}{4} = \frac{1}{4}$ $\therefore x=2$ not V.A

$(x=-2)$

↳ $\lim_{x\to -2} \frac{\sin(x-2)}{x^2-4}$

$= \frac{\sin(-4)}{0} = \pm\infty$ $\therefore x=-2$ is V.A

⑫ $f(x) = \frac{\sin x}{x}$ ? (homework)

⑬ $f(x) = \tan x$ ?

↳ $f(x) = \frac{\sin x}{\cos x}$

* أصفار المقام ($\cos x$) ... V.A

$\therefore x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}$, ..... are V.A's

⑭ $f(x) = \csc x$ ?

↳ $f(x) = \frac{1}{\sin x}$ (أصفار $\sin x$)

$x = 0, \pm\pi, \pm 2\pi, \pm 3\pi$ ....

$\therefore x = \pm n\pi$ are V.A(s)

# Calculus "1"

## # Limits at infinity :-

**Theorem :-**

① $\lim_{x \to \pm\infty} c = c$

② $\lim_{x \to \pm\infty} \frac{1}{x^r} = 0, \quad r > 0$

③ $\lim_{x \to \infty} x^n = \infty, \quad n = 1, 2, 3, 4, \dots$

④ $\lim_{x \to -\infty} x^n = \begin{cases} -\infty, & n = 1, 3, 5, \dots \\ +\infty, & n = 2, 4, 6, \dots \end{cases}$

⑤ $\lim_{x \to \pm\infty} (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0)$

$= \lim_{x \to \pm\infty} a_n x^n$ (أكبر قوة مع معاملها)

⑥ $\lim_{x \to \pm\infty} \left( \frac{a_n x^n + a_{n-1} x^{n-1} \dots + a_1 x}{b_m x^m + b_{m-1} x^{m-1} \dots + b_1 x} \right)$

$= \lim_{x \to \pm\infty} \frac{a_n x^n}{b_m x^m}$ (أكبر أس بأكبر أس)

**ex) Evaluate :-**

① $\lim_{x \to \infty} 8 = 8$

② $\lim_{x \to -\infty} \frac{3}{x} = \frac{3}{-\infty} = 0$

③ $\lim_{x \to \infty} (1 + 2x^3 - 2x^5) =$

$= \lim_{x \to \infty} -2x^5 = -2(\infty)^5 = -\infty$

(H.W)

④ $\lim_{x \to -\infty} (2x^5 + x^3 + x^2 + 1)$ ؟!

⑤ $\lim_{x \to -\infty} (x-1)^4 \cdot (x+2)^8 (x+1)$ ؟!

$\lim_{x \to -\infty} x^4 \cdot x^8 \cdot x = \lim_{x \to -\infty} x^{13}$

$= (-\infty) = -\infty$.

⑥ $\lim_{x \to -\infty} \frac{|x|}{x}$ ؟! $= \frac{\infty}{-\infty}$ !.

If $x \to -\infty$

$|x| \to -x$

$= \lim_{x \to -\infty} \frac{-x}{x} = -1$

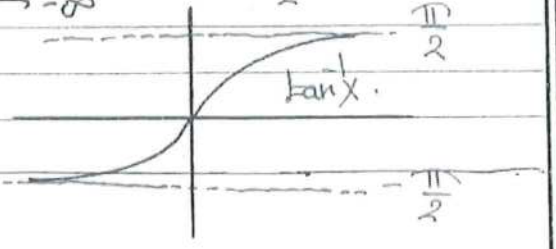
⑦ $\lim_{x \to \infty} e^x = e^{\infty} = \infty$

⑧ $\lim_{x \to -\infty} e^x = e^{-\infty} = \frac{1}{\infty} = 0$

⑨ $\lim_{x \to 0^-} e^{\frac{1}{x}} = e^{-\infty} = 0$

⑩ $\lim_{x \to \infty} \tan^{-1} x = \frac{\pi}{2}$

⑪ $\lim_{x \to -\infty} \tan^{-1} x = -\frac{\pi}{2}$



Calculus "1"

(12) $\lim_{x\to 2^+} \tan^{-1}\left(\frac{1}{x-2}\right)$ ?

↳ $\tan^{-1}\left(\frac{1}{0^+}\right) = \tan^{-1}(\infty) = \frac{\pi}{2}$.

(13) $\lim_{x\to 0^+} \tan^{-1}(\ln x)$ ?

↳ $\tan^{-1}(\ln 0^+) = \tan^{-1}(-\infty) = -\frac{\pi}{2}$

(14) $\lim_{x\to\infty} x^2 - 2x$ ?

↳ $\lim_{x\to\infty} x^2 = (\infty)^2 = \infty$

(15) $\lim_{x\to\infty} \tan^{-1}(x^2 - x^4)$ ?

↳ $\tan^{-1}\left(\lim_{x\to\infty}(x^2 - x^4)\right)$

$= \tan^{-1}\left(\lim_{x\to\infty}(-x^4)\right)$

$= \tan^{-1}(-\infty) = -\frac{\pi}{2}$.

(16) $\lim_{x\to\infty} e^{-2x}\cos x$ ?

↳ $0 \cdot \cos(\infty)$ !

$-1 \le \cos x \le 1$

$-e^{-2x} \le e^{-2x}\cos x \le e^{-2x}$

$\lim_{x\to\infty} -e^{-2x} = 0$ , $\lim_{x\to\infty} e^{-2x} = 0$

$\therefore \lim_{x\to\infty} e^{-2x}\cos x = 0$

(17) $\lim_{x\to\infty} e^{x^2 - x^5}$ ?

↳ $e^{\lim_{x\to\infty} x^2 - x^5} = e^{\lim_{x\to\infty} -x^5}$

$= e^{-\infty} = 0$.

(18) $\lim_{x\to\infty} \frac{\sin x}{x}$ ?

↳ $-1 \le \sin x \le 1$

$\frac{-1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$

$\lim_{x\to\infty} = 0$ , $\lim_{x\to\infty} = 0$

$\therefore \lim_{x\to\infty} \frac{\sin x}{x} = 0$

(19) $\lim_{x\to\infty} \frac{\sin x}{2x^3}$ ? (homework)

↳ sol : 0 . نفس حل (17)

(20) $\lim_{x\to-\infty} \sqrt[3]{\frac{4x+1}{2x-3}}$ ?

↳ $\sqrt[3]{\lim_{x\to-\infty} \frac{4x+1}{2x-3}}$

$= \sqrt[3]{\lim_{x\to-\infty} \frac{4x}{2x}} = \sqrt[3]{\frac{4}{2}} = \sqrt[3]{2}$.

(21) (homework)

$\lim_{x\to\infty} \sqrt[4]{\frac{32x^2}{1+2x^2}}$

↳ sol 2 .

Calculus "1"

(22) $\lim_{x\to\infty} \frac{1-7x^6}{3x^6-1}$ ؟

$\lim_{x\to\infty} \frac{-7x^6}{3x^6} = -\frac{7}{3}$ .

(23) $\lim_{x\to\infty} \frac{x^2+5}{1-x^4}$ ؟

$\lim_{x\to\infty} \frac{x^2}{-x^4} = \lim_{x\to\infty} \frac{-1}{x^2}$

$= \frac{-1}{\infty} = 0$

(24) $\lim_{x\to-\infty} \frac{x^5-7}{1-x^3}$ ؟

$\lim_{x\to-\infty} \frac{x^5}{-x^3} = \lim_{x\to-\infty} -x^2$

$= -(-\infty)^2 = -\infty$

(25) $\lim_{x\to\infty} \frac{1+3x^5+8x^{102}}{1+2x^{100}-x^5}$ ؟

$\lim_{x\to\infty} \frac{8x^{102}}{2x^{100}} = \lim_{x\to\infty} 4x^2$

$= 4(\infty)^2 = \infty$

(26) $\lim_{x\to\infty} \frac{|x^4|+x}{x+1}$ ؟

$\lim_{x\to\infty} \frac{x^4}{x} = \lim_{x\to\infty} x^3$

$= (\infty)^3 = \infty$ .

ex) $\lim_{x\to\infty} x\sin\left(\frac{1}{x}\right)$ ؟

$\hookrightarrow (0\cdot\infty)$ !

* في حالة ظهور $(0\cdot\infty)$ نستخدم طريقة الإستبدال .

$\Rightarrow$ let $y=\frac{1}{x}$ , $x=\frac{1}{y}$

$y\to 0$ , $x\to\infty$

$\Rightarrow \lim_{y\to 0} \frac{\sin y}{y} = \frac{1}{1} = 1$

ex) $\lim_{x\to\infty} 4x\tan\left(\frac{2}{x}\right)$ ؟ (اختيارات)

(a) 8 (b) $\frac{1}{2}$ (c) 0 (d) 2

$\hookrightarrow (0\cdot\infty)$ ! $\Rightarrow \frac{2}{x}=y \Rightarrow x=\frac{2}{y}$

$x\to\infty \Rightarrow y\to 0$

$\Rightarrow \lim_{y\to 0} \frac{8\tan y}{y} = 8$ (a) ∴ الحل :

ex) $\lim_{x\to\infty} \frac{e^{3x}+4}{e^x-3}$ ؟

$\hookrightarrow e^x = z$ $\Rightarrow \lim_{z\to\infty} \frac{z^3+4}{z-3}$

$x\to\infty$

$z\to\infty$

$= \lim_{z\to\infty} \frac{z^3}{z} = \infty$ .

* Def : $y = a$, is said to be Horizontal Asymptote ( خط تقارب عامودي ) ( H.A ) for $f(x)$ , if

$\lim_{x \to \infty} f(x) = a$ or $\lim_{x \to -\infty} f(x) = a \quad (a \neq \pm\infty)$

Note : To find H.A , you must find :

$\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$

Ex : Find (H.A) :-

1) $f(x) = \frac{x+1}{x^2+4x}$

Sol :-

$\lim_{x \to \infty} \frac{x+1}{x^2+4x} = \lim_{x \to \infty} \frac{x}{x^2} = \lim_{x \to \infty} \frac{1}{x} = 0$

$\lim_{x \to -\infty} \frac{x+1}{x^2+4x} = \lim_{x \to -\infty} \frac{x}{x^2}$

$= \lim_{x \to -\infty} \frac{1}{x} = 0$

$\therefore y = 0$ is H.A

2) $f(x) = \frac{3x^2+1}{x-2x^2}$

Sol :-

$\lim_{x \to \infty} \frac{3x^2+1}{x-2x^2} = \lim_{x \to \infty} \frac{3x^2}{-2x^2} = \frac{-3}{2}$

$\lim_{x \to -\infty} \frac{3x^2+1}{x-2x^2} = \lim_{x \to -\infty} \frac{3x^2}{-2x^2} = -3/2$

$\therefore y = -3/2$ is H.A

3) $f(x) = e^x$

Sol :-

$\lim_{x \to \infty} e^x = e^{\infty} = \infty$ ( No H.A)

$\lim_{x \to -\infty} e^x = e^{-\infty} = 0$

$\therefore y = 0$ is H.A

4) $f(x) = \tan^{-1} x$

Sol :-

$\lim_{x \to \infty} \tan^{-1} x = \tan^{-1}(\infty) = \frac{\pi}{2}$

$\lim_{x \to -\infty} \tan^{-1} x = \tan^{-1}(-\infty) = \frac{-\pi}{2}$

$\therefore y = \frac{-\pi}{2}, \frac{\pi}{2}$ are H.A

5) $f(x) = \frac{x^2+5x}{x+1}$

Sol :-

$\lim_{x \to \infty} \frac{x^2+5x}{x+1} = \lim_{x \to \infty} \frac{x^2}{x} = \lim_{x \to \infty} x = \infty$

$\lim_{x \to -\infty} \frac{x^2+5x}{x+1} = \lim_{x \to -\infty} \frac{x^2}{x} = \lim_{x \to -\infty} x = -\infty$

( No H.A) .

6) $f(x) = \frac{1}{x}$ [H.W]

Sol :- $y = 0$ H.A

7) $f(x) = x^2 - x$

Sol :-

$\lim_{x\to\infty} x^2 - x = \lim_{x\to\infty} x^2 = \infty$

$\lim_{x\to-\infty} x^2 - x = \lim_{x\to-\infty} x^2 = \infty$

∴ No H.A

8) $f(x) = \frac{5x^{100} + 4x}{1 + 2x^{100}}$ [H.W]

Sol :- $y = \frac{5}{2}$ H.A

9) $f(x) = \sqrt{x^2+1} - x$

Sol :-

$\lim_{x\to\infty} \sqrt{x^2+1} - x = \infty - \infty$ !

$\lim_{x\to\infty} \sqrt{x^2+1} - x \cdot \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1} + x}$

$= \lim_{x\to\infty} \frac{x^2+1-x^2}{\sqrt{x^2+1} + x} = \lim_{x\to\infty} \frac{1}{\sqrt{x^2+1} + x}$

$= \frac{1}{\infty} = 0$

$\lim_{x\to-\infty} \sqrt{x^2+1} - x = (\infty) - (-\infty)$

$= \infty + \infty = \infty$ ($\frac{\text{No}}{\text{H.A}}$)

∴ $y = 0$ is H.A

10) $f(x) = \frac{\sqrt{x^2+6}}{3x-6}$

Sol :-

$\lim_{x\to\infty} \frac{\sqrt{x^2+6}}{3x-6} = \lim_{x\to\infty} \frac{|x|}{3x}$

$x \to \infty$ , $|x| \to x$

$= \lim_{x\to\infty} \frac{x}{3x} = \frac{1}{3}$

$\lim_{x\to-\infty} \frac{\sqrt{x^2+6}}{3x-6} = \lim_{x\to-\infty} \frac{|x|}{3x}$

$x \to -\infty$ , $|x| \to -x$

$= \lim_{x\to\infty} \frac{-x}{3x} = \frac{-1}{3}$

$y = -\frac{1}{3}$ , $\frac{1}{3}$ are H.A

11) $f(x) = \sqrt{x^6 + 5x^3} - x^3$ [H.W]

12) $f(x) = \frac{1 - e^x}{1 + 2e^x}$

Sol :-

$\lim_{x\to-\infty} \frac{1-e^x}{1+2e^x} = \frac{1-e^{-\infty}}{1+2e^{-\infty}} = \frac{1-0}{1+0} = 1$

$\lim_{x\to\infty} \frac{1-e^x}{1+2e^x} = \frac{\infty}{\infty}$ !

نأخذ $e^x$ عامل مشترك من البسط والمقام ⟺ أو ⟺ نقسم على $e^x$ كل من البسط والمقام

$= \lim \frac{e^x(\frac{1}{e^x} - 1)}{e^x(\frac{1}{e^x} + 2)} = \frac{\frac{1}{\infty} - 1}{\frac{1}{\infty} + 2}$

$= \frac{0-1}{0+2} = \frac{-1}{2}$

$y = 1, \frac{-1}{2}$ are H.A

## أسئلة بنوات على النهايات
## (بوليتكنك)

Q1) The limit

$$\lim_{x\to\infty}\left[\sqrt{x^2+(a+2)x}-\sqrt{x^2+2x}\right]$$

$$=\frac{a^a}{2}$$

then the value of the constant a equals to

Sol :-

بالضرب في المرافق الأيمن

$$\lim_{x\to\infty}\left[\sqrt{x^2+(a+2)x}-\sqrt{x^2+2x}\right]\cdot\frac{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}$$

$$=\lim_{x\to\infty}\frac{x^2+ax+2x-x^2-2x}{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}$$

$$=\lim_{x\to\infty}\frac{ax}{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}$$

$$=\lim_{x\to\infty}\frac{ax}{|x|+|x|}=\lim_{x\to\infty}\frac{ax}{2x}$$

$$=\frac{a}{2}$$

$$\Rightarrow \frac{a}{2}=\frac{a^a}{2}$$

$$\Rightarrow a^1=a^a \Rightarrow (a=1)$$

Q2) The H.A for $f(x)=e^{2-x^2}$ is (are)

Sol :-

$$\lim_{x\to\infty} e^{2-x^2}=e^{2-\infty}=e^{-\infty}=0$$

$$\lim_{x\to-\infty} e^{2-x^2}=e^{2-(-\infty)^2}=e^{2-\infty}=e^{-\infty}=0$$

$\therefore y=0$ is H.A.

Q3) $\lim_{x\to\infty}\frac{2x+1}{\sqrt{3x^2+5}-7}$

Sol :- $\lim_{x\to\infty}\frac{2x+1}{\sqrt{3x^2+5}-7}=\frac{\infty}{\infty}$ |.

$$\lim_{x\to\infty}\frac{2x}{\sqrt{3}\,|x|}=\lim_{x\to\infty}\frac{2x}{\sqrt{3}\,x}=\frac{2}{\sqrt{3}}$$

Q4) $\lim_{x\to0}\frac{\sin(2x)}{x+\tan(3x)}$

Sol :- $\lim_{x\to0}\frac{\sin(2x)}{x+\tan(3x)}=\frac{0}{0}$ |.

بقسمة كل من البسط والمقام على $x$

$$\Rightarrow \lim_{x\to0}\frac{\frac{\sin(2x)}{x}}{\frac{x}{x}+\frac{\tan(3x)}{x}}$$

$$=\frac{2}{1+3}=\frac{2}{4}=\frac{1}{2}$$

Q5) $\lim_{x\to-\infty}\frac{3x-5x^3+1}{x-x^5+1}$

$$\lim_{x\to-\infty}\frac{-5x^3}{-x^5}=\lim_{x\to-\infty}\frac{5}{x^2}$$

$$=\frac{5}{\infty}=0$$

Q6) The limit

$$\lim_{x\to0} x^2\left[\sin^2\left(\cos\left(\frac{5}{x}\right)\right)+\frac{2}{\sin^2(2x)}\right]$$

Sol :-

$$\lim_{x\to 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) + \frac{2x^2}{\sin^2(2x)}$$

(I) (II)

(I) $\lim_{x\to 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right)$

$0 \le \sin^2\left(\cos\left(\frac{5}{x}\right)\right) \le 1$

$0 \le x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) \le x^2$

$0 \qquad\qquad 0$

$0$

(II) $\lim_{x\to 0} \frac{2x^2}{\sin^2(2x)}$

$= 2\left(\lim_{x\to 0} \frac{x}{\sin(2x)}\right)^2$

$= 2 \cdot \left(\frac{1}{2}\right)^2 = \frac{2}{4} = \frac{1}{2}$

$\therefore \lim_{x\to 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) + \frac{2x^2}{\sin^2(2x)}$

$= 0 + \frac{1}{2} = 1/2$

Q7) The V.A of the function $f(x) = \frac{2}{x} + \frac{1}{x^2}$ is

Sol :- $(x=0)$

$\lim_{x\to 0^+} \frac{2}{x} + \frac{1}{x^2} = \frac{2}{0^+} + \frac{1}{0^+}$

$= \infty + \infty = \infty$

$\therefore x=0$ is V.A

Q8) The number of H.A for the function $f(x) = \sqrt{x^2-x} - x^7$ is ?

Sol :-

$\lim_{x\to -\infty} \sqrt{x^2-x} - x^7 =$

$\sqrt{\infty+\infty} - (-\infty)^7 =$

$\infty + \infty = \infty$ $(\mathcal{N}_0$ H.A$)$

$\lim_{x\to \infty} \sqrt{x^2-x} - x^7 = \infty - \infty$ !

$\lim_{x\to \infty} \sqrt{x^2-x} - x^7 \cdot \frac{\sqrt{x^2-x}+x^7}{\sqrt{x^2-x}+x^7}$

$= \lim_{x\to \infty} \frac{x^2-x-x^{14}}{\sqrt{x^2-x}+x^7}$

$= \lim_{x\to \infty} \frac{-x^{14}}{x^7} = \lim_{x\to \infty} -x^7$

$= -\infty$ $(\mathcal{N}_0$ H.A$)$

The number of H.A (s) is 0.

Q9) $\lim_{x\to 1} \frac{e^{2\ln x} - 9}{x^3 - 3x^2 + x - 3}$

Sol :- $= \frac{e^{2\ln 1} - 9}{1-3+1-3}$

$= \frac{e^0 - 9}{-4} = \frac{1-9}{-4} = \frac{-8}{-4} = 2$

Q10) $\lim_{x\to 2^-} \left(\frac{x+2}{2-x}\right) = \frac{4}{0^+} = \infty$

Q11) $\lim_{x\to\infty} \sqrt{x^2-5} - x = \infty - \infty$ !

Sol :- $\lim_{x\to\infty} \sqrt{x^2-5} - x \cdot \frac{\sqrt{x^2-5}+x}{\sqrt{x^2-5}+x}$

$= \lim_{x\to\infty} \frac{x^2-5-x^2}{\sqrt{x^2-5}+x}$

$= \lim_{x\to\infty} \frac{-5}{\sqrt{x^2-5}+x} =$

$= \frac{-5}{\infty} = 0$ .

Q12) $\lim_{x\to\infty} \sqrt{x^2+4x} - \sqrt{x^2+x}$

Sol :- $= \infty - \infty$ !

$\lim_{x\to\infty} \sqrt{x^2+4x} - \sqrt{x^2+x} \cdot \frac{\sqrt{x^2+4x}+\sqrt{x^2+x}}{\sqrt{x^2+4x}+\sqrt{x^2+x}}$

$= \lim_{x\to\infty} \frac{x^2+4x-x^2-x}{\sqrt{x^2+4x}+\sqrt{x^2+x}}$

$= \lim_{x\to\infty} \frac{3x}{|x|+|x|} = \lim_{x\to\infty} \frac{3x}{2x}$

$= \frac{3}{2}$ .

Q13) If $f(x) = \frac{x^2-1}{x^2-2x+1}$ , then H.A and V.A are :-

Sol :- H.A $\lim_{x\to\pm\infty} \frac{x^2-1}{x^2-2x+1}$

$= \lim_{x\to\pm\infty} \frac{x^2}{x^2} = 1$

∴ $y = 1$ is H.A

V.A $x = 1$ أصفار المقام .

$\lim_{x\to1} \frac{x^2-1}{x^2-2x+1} = \lim_{x\to1} \frac{(x-1)(x+1)}{(x-1)(x-1)}$

$= \frac{2}{0} = \pm\infty$

∴ $x = 1$ is V.A

Q14) $\lim_{x\to\infty} \sqrt{x+3\sqrt{x}} - \sqrt{x}$

Sol :- $= \infty - \infty$ !

$\lim_{x\to\infty} \sqrt{x+3\sqrt{x}} - \sqrt{x} \cdot \frac{\sqrt{x+3\sqrt{x}}+\sqrt{x}}{\sqrt{x+3\sqrt{x}}+\sqrt{x}}$

$= \lim_{x\to\infty} \frac{x+3\sqrt{x}-x}{\sqrt{x+3\sqrt{x}}+\sqrt{x}}$

$= \lim_{x\to\infty} \frac{3\sqrt{x}}{\sqrt{x}\left(\sqrt{1+\frac{3}{\sqrt{x}}}+1\right)}$

$= \frac{3}{\sqrt{1+0}+1} = 3/2$

Q15) If $\lim_{x\to0} \frac{\sqrt{ax+b}-2}{x} = 4$

then the values of a and b are 8

Sol :- إذا كانت النهاية موجودة و
نهاية المقام = صفر فإن
نهاية البسط = صفر .

$\lim_{x\to0} x = 0 \Rightarrow \lim_{x\to0} \sqrt{ax+b}-2 = 0$

$\sqrt{b} = 2$

$\Rightarrow b = 4$

$$\lim_{x\to 0}\frac{\sqrt{ax+4}-2}{x}\cdot\frac{\sqrt{ax+4}+2}{\sqrt{ax+4}+2}$$

$$=\lim_{x\to 0}\frac{ax+4-4}{x(\sqrt{ax+4}+2)}$$

$$=\lim_{x\to 0}\frac{a}{\sqrt{ax+4}+2}=\frac{a}{4}$$

$$\Rightarrow \frac{a}{4}=4$$

$$\Rightarrow a=16$$

Q16) If $\lim_{x\to a}[f(x)+g(x)]=3$ and $\lim_{x\to a}[f(x)-g(x)]=2$ then $\lim_{x\to a}\frac{f(x)}{g(x)}=$

Sol :

(I) $\lim_{x\to a}[f(x)+g(x)]+\lim_{x\to a}[f(x)-g(x)]=3+2$

$\Rightarrow \lim_{x\to a}[f(x)+g(x)+f(x)-g(x)]=5$

$\Rightarrow 2\lim_{x\to a}f(x)=5$

$\Rightarrow \lim_{x\to a}f(x)=\frac{5}{2}$

(II) $\lim_{x\to a}[f(x)+g(x)]-\lim_{x\to a}[f(x)-g(x)]$

$=3-2$

$\Rightarrow \lim_{x\to a}[f(x)+g(x)-f(x)+g(x)]$

$=1$

$\Rightarrow 2\lim_{x\to a}g(x)=1$

$\lim_{x\to a}g(x)=\frac{1}{2}$

$\therefore \lim_{x\to a}\frac{f(x)}{g(x)}=\frac{5/2}{1/2}=\frac{5}{1}=5$

Q17) $\lim_{x\to\infty}\frac{3x^2+\cos^2 x}{x^2+1}$

Sol :

$$\lim_{x\to\infty}\frac{\frac{3x^2}{x^2}+\frac{\cos^2 x}{x^2}}{\frac{x^2}{x^2}+\frac{1}{x^2}}$$

$$=\frac{3+0}{1+0}=3$$

* $\lim_{x\to\infty}\frac{\cos^2 x}{x^2}$

$0\le \cos^2 x\le 1$

$0<\frac{\cos^2 x}{x^2}\le\frac{1}{x^2}$

$0\to 0$, $\frac{1}{x^2}\to 0$

Q18) The H.A(s) of $f(x)=\sqrt{x^2+x+1}-\sqrt{x^2-4x}$ are (H.W)

Sol :

$y=\frac{-5}{2}, \frac{5}{2}$ are H.A(s)

Q19) Let $f(x) = \frac{5}{x^2 - Kx + \frac{9}{4}}$

then the values of K such that f has no vertical ?

Sol:~ no vertical = لا أصفار للمقام

∴ لا يكون للمعادلة التربيعية أصفار إذا كان مميزها أقل من صفر .

$b^2 - 4ac < 0$

$K^2 - 4(1)\left(\frac{9}{4}\right) < 0$

$K^2 - 9 < 0 \Rightarrow K^2 < 9$

$\Rightarrow \sqrt{K^2} < \sqrt{9}$

$\Rightarrow |K| < 3$

$-3 < K < 3$

$K \in (-3, 3)$

Q20) $\lim_{x \to \infty} \frac{x + \sin 5x}{5x + \sin x}$

Sol:~ x على المقام

$$\lim_{x \to \infty} \frac{\frac{x}{x} + \frac{\sin 5x}{x}}{\frac{5x}{x} + \frac{\sin x}{x}}$$

$$= \lim_{x \to \infty} \frac{1 + \frac{\sin 5x}{x}}{5 + \frac{\sin x}{x}} = \frac{1+0}{5+0}$$

$$= \frac{1}{5}$$

$-1 \le \sin x \le 1$ , $-1 \le \sin 5x \le 1$

$\frac{-1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$ , $\frac{-1}{x} \le \frac{\sin 5x}{x} \le \frac{1}{x}$

$\downarrow 0$ , $\downarrow 0$ , $\downarrow 0$ , $\downarrow 0$ , $\downarrow 0$

$\to 0$ , $\to 0$

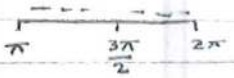
Q21) The vertical asy. of $f(x) = \ln(x-2)$ is

Sol:~ $x = 2$ .

$\lim_{x \to 2^+} \ln(x-2) = \ln 0^+ = -\infty$

$x = 2$ V.A

Q22) $\lim_{x \to \frac{3\pi}{2}^+} \frac{|\sin x|}{\sin x}$




Sol:~ $\lim_{x \to \frac{3\pi}{2}^+} \frac{-\sin x}{\sin x} = -1$

Q23) $\lim_{x \to 2^+} (x-2) \sin\left(\frac{1}{x^2-4}\right)$

Sol:~ = 0 . (d.n.e) !

$-1 \le \sin\left(\frac{1}{x^2-4}\right) \le 1$

$-(x-2) \le (x-2) \sin\left(\frac{1}{x^2-4}\right) \le (x-2)$

$\lim_{x \to 2^+} -(x-2) = 0$ , $\lim_{x \to 2^+} (x-2) = 0$

$\to 0$

Q24) If $f(x) = \frac{2x^2+1}{(2x-1)^2-1}$, then H.A and V.A are ؟ (H.W)

Sol : H.A $y = \frac{1}{2}$

V.A $x=1$ and $x=0$

Q25) $\lim_{x\to 3} (x^2-9)\cot(x^2-x-6)$

Sol : $0 \cdot \infty$ !

$$\lim_{x\to 3} (x^2-9)\frac{\cos(x^2-x-6)}{\sin(x^2-x-6)}$$

$$= \lim_{x\to 3} \cos(x^2-x-6) \cdot \lim_{x\to 3} \frac{x^2-9}{\sin(x^2-x-6)}$$

$$= 1 \cdot \lim_{x\to 3} \frac{x^2-9}{\frac{\sin(x^2-x-6)}{x^2-x-6}}$$

$$= \lim_{x\to 3} \frac{(x-3)(x+3)}{(x-3)(x+2)}$$

$$= \frac{5}{6}$$

Q26) $\lim_{x\to -\infty} \sqrt{x^2+4x} - \sqrt{x^2-x}$

Sol : $\lim_{x\to -\infty} \sqrt{x^2+4x} - \sqrt{x^2-x} \cdot \frac{\sqrt{x^2+4x}+\sqrt{x^2-x}}{\sqrt{x^2+4x}+\sqrt{x^2-x}}$

$$= \lim_{x\to -\infty} \frac{5x}{\sqrt{x^2+4x}+\sqrt{x^2+x}}$$

$$= \lim_{x\to -\infty} \frac{5x}{|x|+|x|}$$

$x \to -\infty$ , $|x| \to -x$

$$= \lim_{x\to -\infty} \frac{5x}{-2x} = \frac{-5}{2}$$

Q27) If $\lim_{x\to 1} \frac{x^3+2x^2-x+a}{x^2-1}$ exists, then $a=$ ؟

Sol : بما أن النهاية موجودة ونهاية البسط = صفر ⟸ نهاية المقام = صفر

$$\Rightarrow \lim_{x\to 1} x^3+2x^2-x+a = 0$$

$$1+2-1+a = 0$$

$\therefore a = -2$

Q28) a) $\lim_{x\to 0} \frac{\sqrt{x+1}-3}{x}$ (H.W)

Sol : $\frac{1}{6}$

b) $\lim_{x\to 0^-} 3x \sin(\frac{2}{x})$ [H.W]

Sol : 0

(Q29) H.A for

$$f(x) = \frac{2x^2+1}{x+2x^2} + \frac{x-2}{3-|x|}$$

Sol :

$$\lim_{x\to\infty} \frac{2x^2+1}{x+2x^2} + \frac{x-2}{3-|x|}$$

$$= \lim_{x\to\infty} \frac{2x^2}{2x^2} + \frac{x}{-x}$$

$$= 1 - 1 = 0$$

$\therefore y = 0$ H.A

$$\lim_{x\to-\infty} \frac{2x^2}{2x^2} + \frac{x}{x}$$

$$= 1 + 1 = 2$$

$\therefore y = 2$ H.A

$y = 0, 2$ are H.A.

(Q30) $\lim_{x\to 3^-} \frac{x^2-3x}{x^2-6x+9}$

Sol :

$$= \frac{0}{0}\ !$$

$$\lim_{x\to 3^-} \frac{x(x-3)}{(x-3)(x-3)} = \frac{3}{0^-}$$

$$= -\infty$$

(Q31) $\lim_{x\to 3} \frac{\sin(x-3)}{x^2-2x-3}$

Sol :

$$= \frac{0}{0}\ !$$

$$\lim_{x\to 3} \frac{\sin(x-3)}{(x-3)(x+1)}$$

$$= \lim_{x\to 3} \frac{\sin(x-3)}{x-3} \cdot \lim_{x\to 3} \frac{1}{x+1}$$

$$= 1 \cdot \frac{1}{4} = \frac{1}{4}$$

(Q32) 6/12/2016

$$\lim_{x\to\infty} \sqrt{ax^2+bx} - 3x = 2$$

then the values of a and b are :-

Sol :

$$\lim_{x\to\infty} \sqrt{ax^2+bx} - 3x \cdot \frac{\sqrt{ax^2+bx}+3x}{\sqrt{ax^2+bx}+3x} = 2$$

$$\Rightarrow \lim_{x\to\infty} \frac{ax^2+bx-9x^2}{\sqrt{ax^2+bx}+3x} = 2$$

$$\Rightarrow \lim_{x\to\infty} \frac{(a-9)x^2+bx}{\sqrt{ax^2+bx}+3x} = 2$$

بما أن النهاية موجودة وتساوي عدد ثابت
يجب أن تكون أعلى قوة في البسط = أعلى
قوة في المقام . لهذا يجب أن يكون
معامل $x^2$ في البسط = صفر

$$\Rightarrow a = 9$$

$$\Rightarrow \lim_{x\to\infty} \frac{bx}{3x+3x} = 2$$

$$\lim_{x\to\infty} \frac{bx}{6x} = 2$$

$$\frac{b}{6} = 2 \Rightarrow b = 12$$

Q33) $\lim_{x\to 1} \frac{(x-1)}{\sqrt{x^2-2x+1}}$ 2016

Sol: $\frac{0}{0}$ !

$$\lim_{x\to 1} \frac{x-1}{\sqrt{(x-1)^2}} = \lim_{x\to 1} \frac{(x-1)}{|x-1|}$$

$|x-1|$: $-$ for $x<1$, $+$ for $x>1$, $0$ at $x=1$

* $\lim_{x\to 1^+} \frac{x-1}{x-1} = 1$
* $\lim_{x\to 1^-} \frac{x-1}{-(x-1)} = -1$

$\lim_{x\to 1} \frac{x-1}{|x-1|}$ d.n.e.

Q34) $\lim_{x\to\infty} 3x \tan\left(\frac{2}{x}\right)$ 2016

Sol: $\infty \cdot 0$ ! استبدال

$y = \frac{1}{x} \Rightarrow x = \frac{1}{y}$

$x \to \infty$ , $y \to 0$

$$\lim_{y\to 0} \frac{3\tan(2y)}{y}$$
$$= 3 \lim_{y\to 0} \frac{\tan(2y)}{y}$$
$$= 3 \cdot \frac{2}{1} = 6$$

Q35) The H.A and V.A for $f(x) = e^{\left(\frac{x+\sin 3x}{2x}\right)}$ 2016

Sol: V.A $x=0$

* $\lim_{x\to 0} e^{\left(\frac{x+\sin 3x}{2x}\right)}$
$= e^{\lim_{x\to 0}\left(\frac{x}{2x} + \frac{\sin 3x}{2x}\right)}$
$= e^{\frac{1}{2}+\frac{3}{2}} = e^2 \neq \pm\infty$

$\therefore x=0$ (Not V.A)

* $\lim_{x\to \pm\infty} e^{\left(\frac{x+\sin 3x}{2x}\right)}$
$= e^{\lim_{x\to \pm\infty}\left(\frac{x}{2x} + \frac{\sin 3x}{2x}\right)}$ ← Squezing
$= e^{\frac{1}{2}+0}$
$= \sqrt{e}$

$\therefore y = \sqrt{e}$ H.A

## 1.5 Continuity

**Def:** $f(x)$ is said to be continuous (متصل) at $a$ if :-

1) $f(a)$ is defiend
2) $\lim_{x\to a} f(x)$ exists
3) $\lim_{x\to a} f(x) = f(a)$

** Not continuous ≡ Dis continuous.

**Ex:** If $f(x) = \begin{cases} \frac{x+3}{|x^2+3x|} & , x<-3 \\ \sqrt[3]{\frac{x+2}{27}} & , x>-3 \\ -\frac{1}{3} & , x=-3 \end{cases}$

[1] Is $f$ cont at $x=-3$ ??

**Sol:** ⊛ $f(-3) = -\frac{1}{3}$

⊛ $\lim_{x\to -3^+} \sqrt[3]{\frac{x+2}{27}} = \sqrt[3]{\frac{-1}{27}} = -\frac{1}{3}$

⊛ $\lim_{x\to -3^-} \frac{x+3}{|x^2+3x|} =$

$\lim_{x\to -3^-} \frac{x+3}{x^2+3x} = \lim_{x\to -3^-} \frac{x+3}{x(x+3)}$

$= \frac{-1}{3}$

$\therefore f$ is cts at $x=-3$.

[2] Is $f$ cont. at $x=0$??

**Sol:** $f(0) = \sqrt[3]{\frac{2}{27}}$.

$\lim_{x\to 0} \sqrt{\frac{x+2}{27}} = \sqrt{\frac{2}{27}}$

$\therefore f(0) = \lim_{x\to 0} f(x)$

$\rightarrow f$ is cont. at $x=0$

**Ex:** If $f(x) = \begin{cases} x^2-2kx+k^2 & , x\neq 0 \\ 2k & , x=0 \end{cases}$ ← الصورة — الدالة

is cont at $x=0$ ; find the value(s) of $K$ ?

**Sol:** أي أن cont. عند

$f(0) = \lim_{x\to 0} f(x)$

$\rightarrow 2k = \lim_{x\to 0} x^2-2kx+k^2$

$2k = k^2 \rightarrow k^2-2k=0$

$k(k-2)=0$

$k=0, \quad k=2$.

**Thm:** If $f(x)$ and $g(x)$ cont., then :-

1) $f \pm g$ is cont.
2) $f.g$ is cont.
3) $\frac{f}{g}$ is cont ; $g\neq 0$.

**Ex:** Find the interval of continuity of

1) $f(x) = \sqrt{x}$ ; $[0, \infty)$
2) $f(x) = e^{\frac{1}{x-3}}$ ; $\mathbb{R}-\{3\}$
3) $f(x) = \ln x$ ; $(0, \infty)$
4) $f(x) = \frac{x+5}{x^2-4}$ ; $\mathbb{R}-\{-2, 2\}$.

5) $f(x) = \ln x + \tan^{-1}(x)$ ;

$(0,\infty) \cap \mathbb{R} = (0,\infty)$

6) $f(x) = 1 - \ln(\ln x)$

Sol $\ln x > 0$

$e^{\ln x} > e^0 \rightarrow x > 1$

7) $f(x) = \ln(x-3)$

Sol $x - 3 > 0$

$\Rightarrow x > 3$ ; $(3,\infty)$

8) $f(x) = x^2 + 5x - 7$

Sol $\mathbb{R}$

9) $f(x) = \sin^{-1}(2x)$

Sol $-1 \le 2x \le 1$

$-\frac{1}{2} \le x \le \frac{1}{2}$ ;

$[-\frac{1}{2}, \frac{1}{2}]$ .

10) $f(x) = \cos^{-1}(\ln x)$

Sol $-1 \le \ln x \le 1$

$e^{-1} \le e^{\ln x} \le e^1$

$\frac{1}{e} \le x \le e \rightarrow [\frac{1}{e}, e]$

11) $f(x) = e^{(\frac{\sin x}{x})}$

Sol $\mathbb{R} - \{0\}$ .

12) $f(x) = \frac{\ln(\tan^{-1} x)}{x^2 - 9}$

Sol $(0,3)$ , $(3,\infty)$

13) $f(x) = \frac{\ln x + \tan^{-1}(x)}{x^2 - 1}$

Sol $((0,\infty) \cap \mathbb{R}) - \{-1, 1\}$

$\Rightarrow (0,\infty) - \{1\}$

14) $f(x) = \sqrt{x^2 - 4}$ (H.W)

Sol $(-\infty, -2]$ , $[2, \infty)$

Ex/ Find the Discontinuity points of the following :

(نقاط عدم الاتصال ← أصفار المقام)

- لا نهتم عند الفراجي

1- $f(x) = \frac{x^2 - 9}{x^2 - 3x}$

Sol $f(x) = \frac{x^2 - 9}{x(x-3)}$

discts at $x = 0$ , $x = 3$

2- $f(x) = \frac{x}{|x| - 3}$

Sol $|x| - 3 = 0$

$|x| = 3 \Rightarrow x = \pm 3$

discts at $x = \pm 3$

3- $f(x) = \frac{5}{x} + \frac{2x}{x+4}$

discts at $x = 0$ , $x = -4$

4) $f(x) = \dfrac{|x|-2}{|x-2|-1}$

Sol $|x-2|-1=0$

$|x-2|=1$

$x-2=1$ ⇒ $x=3$ ; $x-2=-1$ ⇒ $x=1$

discts at $x=1,3$.

5) $f(x) = |\cot x|$

Sol $f(x) = \left|\dfrac{\cos x}{\sin x}\right|$

discts at $x=n\pi$ ; $n=0,\pm1,\dots$

6) $f(x) = \csc x$

Sol نفس

7) $f(x) = \sec x$

Sol $f(x) = \dfrac{1}{\cos x}$

discts at $x=\dfrac{\pi}{2}+n\pi$ ; $n=0,\pm1,\pm2,\dots$

8) $f(x) = \dfrac{1}{1+\sin^2 x}$

Sol $1+\sin^2 x = 0$

$\rightarrow \sin^2 x = -1$ !!

ليس للمقام أصفار

∴ $f(x)$ cts. every where.

9) $f(x) = \cos\left(\dfrac{x}{x-\pi}\right)$

discts at $x=\pi$

10) $f(x) = \dfrac{1}{1-2\sin x}$

Sol ⇒ $1-2\sin x=0$

$1=2\sin x$

$\rightarrow \dfrac{1}{2} = \sin x$

$\Rightarrow x = \dfrac{\pi}{6} + 2n\pi$ and

$x = \dfrac{5\pi}{6} + 2n\pi$.

11) $f(x) = \sin(x^2-2)$

Sol cts every where.

12) $f(x) = \sqrt{|\ln(x^2+4-4x)|}$

Sol ⇒ $\sqrt{|\ln(x^2-4x+4)|}$

$= \sqrt{|\ln(x-2)^2|}$

* هنا داخل ln موجب دائماً ما داخل

discts at $x=2$.

13) $f(x) = \dfrac{x^2-16}{x^2+9x+20}$ (H.W)

Sol discts at $-4,-5$.

14) $f(x) = \frac{1}{1-\sin x}$ , $x \in [0, 2\pi]$

Sol $1 - \sin x = 0$

$\sin x = 1$

$x = \boxed{\frac{\pi}{2}}$ , $\frac{5\pi}{2}$ , $\frac{9\pi}{2}$ , ...

$\therefore x = \frac{\pi}{2}$ is dis cts .

15) $f(x) = \frac{\cot x}{\sqrt{x^2-4} - 1}$

Sol : $f(x) = \frac{\cos x}{\sin x(\sqrt{x^2-4} - 1)}$

أوجد القيم

- $\sin x = 0$ , $x = \pm n\pi$
- $\sqrt{x^2-4} - 1 = 0$

$\Rightarrow \sqrt{x^2-4} = 1$

$\Rightarrow x^2 - 4 = 1$

$\Rightarrow x^2 = 5 \Rightarrow x = \pm\sqrt{5}$

$\sqrt{x^2-4}$ موجبة $(-\infty, -2], [2, \infty)$

$\sqrt{x^2-4}$ غير معرفة $(-2, 2)$

$\therefore$ dis cts at $(-2,2)$, $\pm n\pi$, $\pm\sqrt{5}$

Thm : If $\lim_{x\to a} g(x) = b$ and $f$ is cont at $b$, then

$\lim_{x\to a} f \circ g(x) = \lim_{x\to a} f(g(x)) = f(\lim_{x\to a} g(x)) = f(b)$

Ex If $\lim_{x\to 2} g(x) = 4$ , $g(2) = 5$ ,

$f$ is cont on $(-\infty, \infty)$ , $f(5) = 8$ and $f(4) = -3$ Find $\lim_{x\to 2} f(g(x))$

Sol $\lim_{x\to 2} f(g(x)) = f(\lim_{x\to 2} g(x))$ لأن $f$ متصل

$= f(4) = -3$

Remark :

1) If $f(x)$ cont $\rightarrow f^{-1}$ is cont .
2) $\sin x$ , $\cos x$, are cont $\rightarrow \sin^{-1} x$, $\cos^{-1} x$ are cont .
3) If $f(x)$ is cont , then $\lim_{x\to a} f \circ g(x) = f(\lim_{x\to a} g(x))$ .

Ex : Evaluate :

1) $\lim_{x\to\infty} \sec\left(\frac{1-\pi x^2}{3x^2+1}\right)$

Sol : $\sec(\lim_{x\to\infty} \frac{1-\pi x^2}{3x^2+1})$

$= \sec(\lim_{x\to\infty} \frac{-\pi x^2}{3x^2})$

$= \sec(-\frac{\pi}{3}) = \sec(\frac{\pi}{3}) = 2$ .

2) $\lim_{x\to-\infty} \cos\left(\frac{\pi x}{2-3x}\right)$

Sol $\cos(\lim_{x\to-\infty} \frac{\pi x}{2-3x}) = \cos(\lim_{x\to-\infty} \frac{\pi x}{-3x})$

$= \cos(-\frac{\pi}{3}) = \cos(\frac{\pi}{3}) = \frac{1}{2}$

3) $\lim_{x\to 1} \sin^{-1}\left(\frac{\sqrt{x}-1}{x-1}\right)$

Sol $\sin^{-1}\left(\lim_{x\to 1} \frac{\sqrt{x}-1}{x-1}\right)$

$= \sin^{-1}\left(\lim_{x\to 1} \frac{\sqrt{x}-1}{(\sqrt{x}-1)(\sqrt{x}+1)}\right)$

$= \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

4) $\lim_{x\to\infty} \tan\left(\frac{\pi x}{2-3x}\right)$

Sol $\tan\left(\lim_{x\to\infty} \frac{\pi x}{2-3x}\right)$

$= \tan\left(\lim_{x\to\infty} \frac{\pi x}{-3x}\right)$

$= \tan\left(\frac{-\pi}{3}\right) = -\tan\left(\frac{\pi}{3}\right) = -\sqrt{3}$

5) $\lim_{x\to\infty} \ln(x^2+x+1) - \ln(2x^2+4)$

Sol $\infty - \infty$ !.

$\lim_{x\to\infty} \ln\left(\frac{x^2+x+1}{2x^2+4}\right)$

$= \ln\left(\lim_{x\to\infty} \frac{x^2}{2x^2}\right) = \ln\left(\frac{1}{2}\right)$

$= \ln 1 - \ln 2 = -\ln 2$ .

6) $\lim \sin\left(\frac{1-\cos x}{x}\right)$ [H.W]

Sol 0 .

7) $\lim_{x\to 3^+} [\ln(\sin(x^2-9)) - \ln(x-3)]$

Sol $-\infty + \infty$ !.

$\lim_{x\to 3^+} \ln\left(\frac{\sin(x^2-9)}{x-3}\right)$

$= \lim_{x\to 3^+} \ln\left(\frac{\frac{\sin(x^2-9)}{x^2-9}}{\frac{x-3}{x^2-9}}\right)$

$= \ln\left(\frac{1}{6}\right) = \ln 1 - \ln 6$

$= -\ln 6$ .

8) $\lim_{x\to 2} 3\sin^{-1}\left(\frac{x^2-5x+6}{x^2-2x}\right)$

Sol $3\sin^{-1}\left(\lim_{x\to 2} \frac{(x-3)(x-2)}{x(x-2)}\right)$

$= 3\sin^{-1}\left(\frac{-1}{2}\right) = -3\sin^{-1}\left(\frac{1}{2}\right)$

$= -3 \cdot \frac{\pi}{6}$

$= \frac{-\pi}{2}$

9) $\lim_{x\to\infty} [\ln(x^2+2) - 2\ln(2x-1)]$

Sol $\infty - \infty$ !.

$\lim_{x\to\infty} [\ln(x^2+2) - \ln(2x-1)^2]$

$= \lim_{x\to\infty} \ln\left(\frac{x^2+2}{(2x-1)^2}\right)$

$= \ln\left(\lim_{x\to\infty} \frac{x^2+2}{(2x-1)^2}\right)$

$= \ln\left(\lim_{x\to\infty} \frac{x^2}{4x^2}\right) = \ln\left(\frac{1}{4}\right)$

$= \ln 1 - \ln 4 = -\ln 4$ .

## أسئلة الجاهيل

Q1) If $F(x) = \begin{cases} 4ax & , x<1 \\ \frac{x^2-1}{x-1} & , x>1 \end{cases}$ and $\lim_{x\to 1} F(x)$ exists, find a

Sol نبا أن النهاية موجودة

$\lim_{x\to 1^+} F(x) = \lim_{x\to 1^-} F(x)$ ←

$\lim_{x\to 1^+} \frac{x^2-1}{x-1} = \lim_{x\to 1^-} 4ax$

$\Rightarrow 2 = 4a$

$\Rightarrow a = \frac{2}{4}$

$\Rightarrow \boxed{a = \frac{1}{2}}$

Q2) If $F(x) = \begin{cases} \frac{x^2-a^2}{x-a} & , x>a \\ 4x+6 & , x\le a \end{cases}$ and $\lim_{x\to a} F(x)$ exists, find a ..

Sol $\lim_{x\to a^+} F(x) = \lim_{x\to a^-} F(x)$

$\lim_{x\to a^+} \frac{x^2-a^2}{x-a} = \lim_{x\to a^-} 4x+6$

$\lim_{x\to a^+} \frac{(x-a)(x+a)}{x-a} = 4a+6$

$2a = 4a+6$

$\Rightarrow -2a = 6 \Rightarrow \boxed{a=-3}$

Q3) Find a value of (k) that makes the function

$F(x) = \begin{cases} \frac{\sin(3x)}{2x} & , x\neq 0 \\ x+2k & , x=0 \end{cases}$

continuous at $x=0$.

Sol cts $\Rightarrow F(0) = \lim_{x\to 0} F(x)$

$0+2k = \lim_{x\to 0} \frac{\sin 3x}{2x}$

$\Rightarrow 2k = \frac{3}{2} \Rightarrow \boxed{k = \frac{3}{4}}$

Q4 (اختبار 2017) $F(x) = \begin{cases} cx^2 + 2\cos(x-1) & , x\le 1 \\ 4x^3 + \frac{3x}{c} & , x>1 \end{cases}$

and f(x) is cont at $x=1$, Then $c = ??$

Sol cts $\Rightarrow \lim_{x\to 1} f(x)$ exists.

$\Rightarrow \lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x)$

$\Rightarrow \lim_{x\to 1^+} 4x^3 + \frac{3x}{c} = \lim_{x\to 1^-} cx^2 + 2\cos(x-1)$

$\Rightarrow 4 + \frac{3}{c} = c+2$

$\Rightarrow -4c+3 = c^2+2c$

$\Rightarrow c^2 - 2c - 3 = 0$

$(c-3)(c+1) = 0$

$c=3 \quad , \quad c=-1$

Q5) $F(x) = \begin{cases} \frac{\tan(kx)}{x} & , x<0 \\ 3x+2k^2 & , x\ge 0 \end{cases}$

Find (k) so that F is is cont at $x=0$ ?

Sol F cts at $x=0$ ⇒

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x)$$

$$\Rightarrow \lim_{x\to 0^+} 3x + 2k^2 = \lim_{x\to 0^-} \frac{\tan(kx)}{x}$$

$$\Rightarrow 2k^2 = k$$

$$\Rightarrow 2k^2 - k = 0$$

$$\Rightarrow k(2k-1) = 0$$

$k=0$ , $k=\frac{1}{2}$

Q6) $f(x) = \begin{cases} x & , x\le 1 \\ kx+h & , 1<x\le 4 \\ -2x & , x>4 \end{cases}$

[H.W]

Sol : $k=-3$, $h=4$ .

* حل أسئلة سؤال

Q) $\lim_{x\to 0} \cos\left(\frac{\sin(x+\pi)}{2x}\right)$

Sol. $\cos\left(\lim_{x\to 0} \frac{\sin(x+\pi)}{2x}\right)$ (0)

$$\cos\left(\lim_{x\to 0} \frac{\sin x\cos\pi + \sin\pi\cos x}{2x}\right)$$

$$\cos\left(\lim_{x\to 0} \frac{-\sin x}{2x}\right) = \cos\left(\frac{-1}{2}\right)$$

$$= \cos\left(\frac{1}{2}\right)$$

Q) $\lim_{x\to\infty} [\ln(3e^x - e^{-x}) - \ln(e^x - e^{-x})]$

Sol. $\infty - \infty$ !

$$\lim_{x\to\infty} \ln\left(\frac{3e^x - e^{-x}}{e^x - e^{-x}}\right)$$

$$\ln\left(\lim_{x\to\infty} \frac{3e^x - e^{-x}}{e^x - e^{-x}}\right)$$

$$= \ln\left(\lim_{x\to\infty} \frac{e^x(3 - e^{-2x})}{e^x(1 - e^{-2x})}\right)$$

$$= \ln\left(\frac{3}{1}\right) = \ln 3$$

Q) The continuity points for $f(x) = \frac{\ln(x-1)}{x^2+x+1}$ are

Sol. $\ln(x-1)$ is cts on $(1,\infty)$

$x^2+x+1$ is cts on $\mathbb{R}$

$b^2-4ac = -3 < 0$ ليس للمقام أصفار

$f(x)$ cts on $(1,\infty)\cap\mathbb{R} = (1,\infty)$

Q) $\lim_{x\to-\infty} \cos^{-1}\left(\frac{1-x^2}{2x^2+1}\right)$

Sol. $\cos^{-1}\left(\lim_{x\to-\infty} \frac{-x^2}{2x^2}\right)$

$$= \cos^{-1}\left(\frac{-1}{2}\right) = \frac{2\pi}{3}$$

Q) The limit
$\lim_{x\to 2} 3\sin^{-1}\left(\frac{x^2-5x+6}{x^2-2x}\right)$

Sol.
$3\sin^{-1}\left(\lim_{x\to 2}\frac{x^2-5x+6}{x^2-2x}\right)$

$3\sin^{-1}\left(\lim_{x\to 2}\frac{(x-3)(x-2)}{x(x-2)}\right)$

$= 3\sin^{-1}\left(\frac{-1}{2}\right) = -3\sin^{-1}\left(\frac{1}{2}\right)$

$= -3\cdot\frac{\pi}{6} = \frac{-\pi}{2}$

---

Q) H.W. $f(x) = \begin{cases} \frac{\tan kx}{x}, & x<0 \\ 2x+3k^2, & x\geq 0 \end{cases}$

all values of k that let $\lim_{x\to 0} f(x)$ exists are

Sol. $k = 0, \frac{1}{3}$

---

Q) $\lim_{x\to 0} \tan\left(\frac{1-\cos x}{x}\right)$

Sol. $\tan\left(\lim_{x\to 0}\frac{1-\cos x}{x}\right)$

$\tan\left(\lim_{x\to 0}\frac{1-\cos x}{x}\cdot\frac{1+\cos x}{1+\cos x}\right)$

$= \tan\left(\lim_{x\to 0}\frac{1-\cos^2 x}{2x}\right)$

$= \tan\left(\lim_{x\to 0}\frac{\sin^2 x}{2x}\right)$

$= \tan\left(\lim_{x\to 0}\frac{\sin x}{2x}\cdot\lim_{x\to 0}\sin x\right)$

$= \tan(0) = 0$

---

Q) Let $f(x) = \sqrt{|x|-4}$, then f continuous on

Sol. $|x|-4 \geq 0$

$|x| \geq 4 \Rightarrow x \geq 4$ or $x \leq -4$

$[4,\infty)$ , $(-\infty,-4]$

---

Q) All values of a such that

$f(x) = \begin{cases} x+4, & x\leq a \\ x^2, & x>a \end{cases}$

cts on $\mathbb{R}$ are

Sol. $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$

$\lim_{x\to a^+} x^2 = \lim_{x\to a^-} x+4$

$a^2 = a+4$

$a^2 - a - 4 = 0$

$a = \frac{1+\sqrt{1-4(1)(-4)}}{2(1)} = \frac{1\pm\sqrt{17}}{2}$

---

Q) $f(x) = \frac{\csc x}{|x-1|-2}$ is discontinuous at

Sol.
$f(x) = \frac{1}{\sin x(|x-1|-2)}$

$\sin x = 0$ , $x = \pm n\pi$ أصفار المقام

$|x-1|-2=0 \rightarrow x-1=2 \Rightarrow x=3$
$\rightarrow x-1=-2 \Rightarrow x=-1$

discts at $x = \pm n\pi, -1, 3$

# Chapter 2 : The Derivative

**2.1 : Tangent Lines and Rates of Change.**

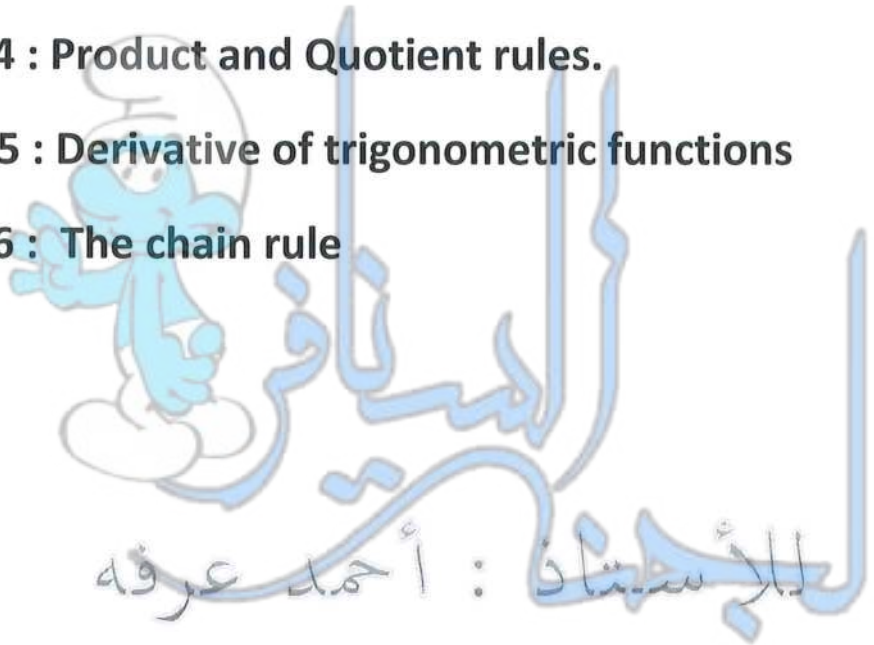
**2.2 : The Derivative Function.**

**2.3 : Techniques of differentiation.**

**2.4 : Product and Quotient rules.**

**2.5 : Derivative of trigonometric functions**

**2.6 : The chain rule**

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# Chapter 2: The Derivative

## (2.1) Tangent lines and rates of change

Def: $f$ is differentiable (قابل للاشتقاق) at $x=a$, If

$$\lim_{x\to a} \frac{f(x)-f(a)}{x-a} \text{ exists } = f'(a)$$

Note: $$f'(x) = \lim_{y\to x} \frac{f(y)-f(x)}{y-x}$$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

# Equation of the tangent line is

$$y - y_0 = m(x - x_0)$$

where $m$ is slope (الميل) and $m = f'(x_0)$

(حيث الميل = المشتقة الأولى عند $x_0$)

Def: the Derivative of a function $f$ at $a$, denoted by

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

OR

$$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$$

دائماً عندما تلاحظ أن النهاية $x\to a$ و المقام $x-a$
فهذا يعني تعريف المشتقة $f'(a)$

ex) If $f(x) = x^2$, then $f'(x) =$

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

$= \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h}$

$= \lim_{h\to 0} \frac{x^2+2xh+h^2-x^2}{h}$

$= \lim_{h\to 0} \frac{h(2x+h)}{h} = 2x$

ex) If $f(x) = \sqrt{x}$, then $f'(x) =$

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

$= \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h}$

ضرب بالمرافق

$= \lim_{h\to 0} \frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})}$

$= \lim_{h\to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}}$

$= \frac{1}{2\sqrt{x}}$

H.W

ex) If $f(x) = x^2$, then $f'(1) =$

Sol. $\lim_{x\to 1} \frac{f(x)-f(1)}{x-1} = \lim_{x\to 1} \frac{x^2-1}{x-1}$

$= 2$

## (2.3) Techniques of differentiation

## (2.4) Product and Quotient rules.

## (2.5) Derivative of trigonometric functions

* Differentiation Rules:

① $\frac{d}{dx} c = 0$ c: constant

② $\frac{d}{dx} x^n = n x^{n-1}$

③ $\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$

④ $\frac{d}{dx} c f(x) = c f'(x)$

⑤ $\frac{d}{dx} (f(x).g(x)) = f(x).g'(x) + g(x).f'(x)$

⑥ $\frac{d}{dx} \left(\frac{f(x)}{g(x)}\right) = \frac{g(x).f'(x) - f(x).g'(x)}{(g(x))^2}$

⑦ $\frac{d}{dx} \left(\frac{c}{f(x)}\right) = \frac{-c.f'(x)}{(f(x))^2}$

⑧ $\frac{d}{dx} \sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}$

* الاقترانات المثلثية كل اقتران يبدأ بحرف c تشتق سالبة

(9) $\frac{d}{dx}\sin x = \cos x$

(10) $\frac{d}{dx}\cos x = -\sin x$

(11) $\frac{d}{dx}\tan x = \sec^2 x$

(12) $\frac{d}{dx}\cot x = -\csc^2 x$

(13) $\frac{d}{dx}\sec x = \sec x \cdot \tan x$

(14) $\frac{d}{dx}\csc x = -\csc x \cdot \cot x$

Ex) Differentiate the function.

1) $f(x) = 186$

Sol. $f'(x) = 0$

2) $f(x) = x^7 - 5x + \pi$

Sol. $f'(x) = 7x^6 - 5 + 0$

3) $f(x) = x^3 + 2x^6 - x^{-3} + \sqrt{5}$

Sol. $f'(x) = 3x^2 + 12x^5 + 3x^{-4} + 0$

4) $f(x) = \frac{3}{4}$ $f'(x) = 0$

5) $f(x) = \frac{1}{\sqrt[3]{x}} + \frac{1}{x^2}$

Sol. قبل الاشتقاق

$f(x) = x^{-\frac{1}{3}} + x^{-2}$

$f'(x) = -\frac{1}{3}x^{-\frac{4}{3}} - 2x^{-3}$

6) $f(x) = \sqrt{x^2+4}$

Sol. $f'(x) = \frac{2x}{2\sqrt{x^2+4}}$ مشتقة ما داخل الجذر / $x^2$ الجذر نفسه

$= \frac{x}{\sqrt{x^2+4}}$

7) $f(x) = \sqrt{x}$

Sol. $f'(x) = \frac{1}{2\sqrt{x}}$

8) $f(x) = x - 3\sin x$

Sol. $f'(x) = 1 - 3\cos x$

9) $f(x) = x \sin x$ * ضرب

Sol. $f'(x) = x\cos x + \sin x \cdot 1$

الأول × م. الثاني + الثاني × م. الأول

10) $y = (x^2+1)\tan x$

Sol. $y' = (x^2+1)\sec^2 x + \tan x \cdot 2x$

11) $y = \frac{x-1}{2x+1}$ قسمة

Sol. $y' = \frac{(2x+1).1 - (x-1).2}{(2x+1)^2}$

$= \frac{2x+1-2x+2}{(2x+1)^2}$

$= \frac{3}{(2x+1)^3}$

12) $y = \frac{\sin x}{x^2}$

Sol. $y' = \frac{x^2\cos x - \sin x . 2x}{(x^2)^2}$

$= \frac{x^2\cos x - 2x\sin x}{x^4}$

13) $f(x) = \frac{1}{x}$

Sol. $f'(x) = \frac{-1.1}{x^2} = \frac{-1}{x^2}$

14) $f(x) = x^5 + x^3 + x^2 + 1$ , find $f'''(1) =$

Sol. $f'(x) = 5x^4 + 3x^2 + 2x$

$f''(x) = 20x^3 + 6x + 2$

$f'''(x) = 60x^2 + 6$

$f'''(1) = 60 + 6 = 66$

Notation

$f'(x) = \frac{dy}{dx}$ $\quad f'''(x) = \frac{d^3y}{dx^3}$

$f''(x) = \frac{d^2y}{dx^2}$

* المشتقات العليا

$f, f', f'', f^{(3)}, f^{(4)}, f^{(5)}, \dots\dots$

15) $f(x) = 3\cot x - 2\csc x$

Sol. $f'(x) = -3\csc^2 x + 2\csc x \cdot \cot x$

H.W

16) $y = \frac{1+\sin x}{x+\cos x}$ , $y' =$

H.W

17) $y = \sqrt[4]{x^3}$ , $y' =$

18) $f(\frac{\pi}{3}) = 4$ and $f'(\frac{\pi}{3}) = -2$

and $g(x) = f(x)\sin x$

and $h(x) = \frac{\cos x}{f(x)}$

Find

i) $g'(\frac{\pi}{3})$

Sol. $g'(x) = f(x)\cos x + \sin x \cdot f'(x)$

$g'(\frac{\pi}{3}) = f(\frac{\pi}{3})\cos(\frac{\pi}{3}) + \sin(\frac{\pi}{3}) f'(\frac{\pi}{3})$

$= 4 \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot -2$

$= 2 - \sqrt{3}$

ii) $h'(\frac{\pi}{3})$

Sol. $h'(x) = \frac{-f(x)\sin x - \cos x \cdot f'(x)}{(f(x))^2}$

$h'(\frac{\pi}{3}) = \frac{(-4)\cdot(\frac{\sqrt{3}}{2}) - (\frac{1}{2})\cdot(-2)}{(4)^2}$

$= \frac{-2\sqrt{3} + 1}{16}$

H.W

19) If $f(x) = \sec x$, Find $f''(\frac{\pi}{4})$ =

Sol. $3\sqrt{2}$

20) If $f(x) = \sqrt{x}\, g(x)$, $g(4) = 2$

$g'(4) = 3$, Find $f'(4)$ ?

Sol. $f'(x) = \sqrt{x} \cdot g'(x) + g(x) \cdot \frac{1}{2\sqrt{x}}$

$f'(4) = \sqrt{4} \cdot g'(4) + g(4) \cdot \frac{1}{2\sqrt{4}}$

$= 2 \cdot 3 + 2 \cdot \frac{1}{4}$

$= 6 + \frac{2}{4} = \frac{13}{2}$

21) $f(x) = \sin^2 x + \cos^2 x$

Sol. $f(x) = 1$ مقدار ثابت

$f'(x) = 0$

H.W

22) $f(x) = (x^2+1)\sec x$

23) $f(x) = \frac{x}{2 - \tan x}$

24) $f(x) = \csc x \cot x$

25) $g(x) = \frac{3x-1}{2x+1}$

26) $f(x) = \sqrt{\sin x + 1}$

27) $f(x) = \frac{\sin x}{x^2 + \sin x}$

28) If $h(2) = 4$ and $h'(2) = -3$

find $\frac{d}{dx}\left(\frac{h(x)}{x}\right)\Big|_{x=2}$

Note:
differentibility ⇒ Continuoity
الاشتقاق ⇒ الاتصال

cts ⇏ diff.

not cts ⇒ not diff.

ex) $f(x)=|x|$, Is $f(x)$ diff. at $x=0$ ?

sol. $f(x)=|x|=\begin{cases} x, & x\geq 0 \\ -x, & x<0 \end{cases}$

. $f(x)$ is cts at $x=0$

$f(0)=\lim_{x\to 0} f(x)$

إذا كان شيء نشتقت

$f'(x)=\begin{cases} 1, & x>0 \\ -1, & x<0 \\ d.n.e, & x=0 \end{cases}$

$f'_+(0)=1$ , $f'_-(0)=-1$

⇒ f isn't diff. at $x=0$.

ex) $f(x)=\begin{cases} -1-2x, & x<-1 \\ x^2, & x\geq -1 \end{cases}$

Is f diff. at $x=-1$ ?

sol. f is cts at $x=-1$

$\lim_{x\to -1^+} f(x)=\lim_{x\to -1^-} f(x)=f(-1)=1$

$f'(x)=\begin{cases} -2, & x<-1 \\ 2x, & x>-1 \end{cases}$

$f'_-(-1)=-2$

$f'_+(-1)=2(-1)=-2$

$f'_-(-1)=f'_+(-1)$

⇒ f is diff. at $x=-1$

ex) $f(x)=\begin{cases} x^2, & x\geq 1 \\ 2x+3, & x<1 \end{cases}$

Is f(x) diff. at $x=1$ ?

sol. cts $\lim_{x\to 1^+} x^2=1$ $\lim_{x\to 1^-} 2x+3=5$

∴ $\lim_{x\to 1} f(x)$ (d.n.e) ∴ f isn't cts

⇒ f(x) isn't diff. at $x=1$.

ex) $f(x)=\begin{cases} x^2, & x\leq 2 \\ mx+b, & x>2 \end{cases}$

Calculate the value of (m) and (b) If f(x) is diff. at $x=2$

sol. f diff at $x=2$.

$f'_+(2)=f'_-(2)$

$m=2x\big|_{x=2} \Rightarrow m=4$

. f is cts at $x=2$

$\lim_{x\to 2^+} f(x)=\lim_{x\to 2^-} f(x)$

$\lim_{x\to 2^+} 4x+b=\lim_{x\to 2^-} x^2$

$8+b=4 \Rightarrow b=-4$

ex) $f(x)=\begin{cases}3x^2, & x\le 1\\ ax+b, & x>1\end{cases}$

find $a,b$ so that $f$ is diff. at $x=1$.

Sol. $f$ is diff. at $x=1$

$f'_+(1)=f'_-(1)$

$a=6x\big|_{x=1} \Rightarrow a=6$

$\Rightarrow f$ is cts at $x=1$

$\lim_{x\to 1^+} f(x)=\lim_{x\to 1^-} f(x)$

$\lim_{x\to 1^+} 6x+b=\lim_{x\to 1^-} 3x^2$

$6+b=3 \Rightarrow b=-3$

---

* إذا اشتقنا كثير الحدود عدد مرات بأكثر من درجته يكون الناتج صفراً.

ex) $f(x)=x^2$ , $f'''(x)=$

Sol. $f'(x)=2x$

$f''(x)=2$

$f'''(x)=0$

---

* إذا اشتقنا كثير الحدود بنفس درجته يكون الناتج (مضروب قوته)

ex) $f(x)=x^2$

Sol. $f'(x)=2x$

$f''(x)=2$

$=2!$

ex) $f(x)=x^3$

Sol. $f'(x)=3x^2$

$f''(x)=6x$

$f'''(x)=6=3!$

---

ex) $f(x)=x^{40}+2x^{30}+9x^{10}+7$

find $f^{(40)}(x)=$

Sol. $f^{(40)}(x)=40!$

---

1, 3, 5, 7, 9, ..... $(2n+1)$ sinx

or $(2n-1)$ cosx

0, 2, 4, 6, 8, ----- $(2n)$

الصيغة العامة للأعداد الزوجية $2n$

الصيغة العامة للأعداد الفردية لـ sin $(2n+1)$

الصيغة العامة للأعداد الفردية لـ cos $(2n-1)$

$2n+1$ ← مشتقها الأولى موجبة ← sinx

$2n-1$ ← مشتقها الأولى سالبة ← cosx

إذا اشتقنا sinx عدد فردي من المرات سنصبح cosx بغض النظر عن الإشارة

وإذا اشتقنا sinx عدد زوجي من المرات سنبقى sinx بغض النظر عن الإشارة

وكذلك الحال في الـ cosx

ex) If $f(x)=\sin x$, $f^{(26)}(x)=$

Sol. $\sin x$ ← تبقى ← عدد زوجي

26 عدد زوجي — الإشارة (+,-)

$2n=26$

$n=13$

$f^{(26)}(x)=(-1)^{13}\sin x=-\sin x$

---

ex) $f(x)=\cos x$ $\dfrac{d^{100}f}{dx^{100}}=$

Sol. $\cos x$ ← تبقى ← 100 زوجي

$2n=100$

$n=50$

$f^{(100)}(x)=(-1)^{50}\cos x=\cos x$

---

ex) If $f(x)=\cos x$, $f^{(27)}(x)=$

Sol. $\sin x$ ستصبح $\cos x$ عدد فردي

$2n-1=27$

$2n=28$

$n=14$

$f^{(27)}(x)=(-1)^{14}\sin x=\sin x$

---

ex) $f(x)=2x^{100}+\sin x$, find $f^{(101)}(x)=?$

Sol. $2x^{100}$ ← مرة 101 ← $0$

$2n+1=101$

$2n=100$

$n=50$

$f^{(101)}(x)=(-1)^{50}\cos x=\cos x$

---

H.W ex) $f(x)=\cos x$, find $f^{(87)}(x)=$

H.W ex) $f(x)=\sin x$, find $f^{(206)}(x)=$

---

(2.6) The chain rule

① $\dfrac{d}{dx}(f(x))^n=n(f(x))^{n-1}\cdot f'(x)$

ex) find $f'(x)$:

1) $f(x)=(x^2+3)^5$

Sol. $f'(x)=5(x^2+3)^4\cdot 2x$

---

2) $f(x)=\sqrt[3]{x^2+1}$

Sol. $f(x)=(x^2+1)^{1/3}$

$f'(x)=\dfrac{1}{3}(x^2+1)^{-2/3}\cdot 2x$

3) $f(x) = \tan^2 x$

Sol. $f(x) = (\tan x)^2$

$f'(x) = 2(\tan x)^1 \cdot \sec^2 x$

$= 2\tan x \sec^2 x$

4) $f(x) = \sin^2 x$

Sol. $f(x) = (\sin x)^2$

$f'(x) = 2\sin x \cdot \cos x$

$= \sin(2x)$

◇2 $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

ex) find $f'(x)$:

1) $f(x) = \sin(x^2)$

Sol. $f'(x) = \cos(x^2) \cdot 2x$

2) $f(x) = \tan(\sqrt{x})$

Sol. $f'(x) = \sec^2(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$

3) $f(x) = \sec(3x)$

Sol. $f'(x) = \sec(3x)\tan(3x) \cdot 3$

4) $y = \sin(\sin x)$

Sol. $y' = \cos(\sin x) \cdot \cos x$

5) $f(x) = \tan(\sin(x^3))$

Sol. $f'(x) = \sec^2(\sin(x^3)) \cdot \cos(x^3) \cdot 3x^2$

6) $f(x) = \sec^3(x^2)$

Sol. $f(x) = (\sec(x^2))^3$

$f'(x) = 3(\sec(x^2))^2 \cdot \sec(x^2)\tan(x^2) \cdot 2x$

$= 6x\sec^3(x^2)\tan(x^2)$

7) If $f(x) + x^2(f(x))^3 = 10$, $f(1) = 2$

find $f'(1) = ?$

Sol. $f'(x) + x^2 \cdot 3(f(x))^2 \cdot f'(x) + (f(x))^3 \cdot 2x = 0$

الآن نعوض $x = 1$

$f'(1) + 3(f(1))^2 \cdot f'(1) + (f(1))^3 \cdot 2 = 0$

$f'(1) + 12f'(1) + 16 = 0$

$13f'(1) = -16 \Rightarrow f'(1) = \frac{-16}{13}$

8) If $\frac{d}{dx} f(3x) = 6x$, find $f'(x)$ ?

Sol. $\frac{d}{dx} f(3x) = f'(3x) \cdot 3$

$f'(3x) \cdot 3 = 6x$

$f'(3x) = 2x$

let $y = 3x \Rightarrow x = \frac{y}{3}$

$f'(y) = 2\left(\frac{y}{3}\right)$

$f'(x) = \frac{2}{3}x$

9) If $\frac{d}{dx} f(2x+1) = 4x$, $f'(x) =$

Sol. $f'(2x+1) \cdot 2 = 4x$

$f'(2x+1) = 2x$

let $y = 2x+1 \Rightarrow x = \frac{y-1}{2}$

$f'(y) = 2\left(\frac{y-1}{2}\right)$

$f'(y) = y-1$

$\therefore f'(x) = x-1$

H.W

10) $\frac{d}{dx} f(6x) = x^3$, $f'(6) =$

H.W

11) $f(x) = \sin(2x)$, $f^{(41)}(x) =$

12) $f(x) = (1-x^3)^7$, $f'(x) =$

(3) If $y = f(u)$, $u = g(x)$, then

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$y \to u \to x$ حفظ أن

ex) If $y = \sin(u)$, $u = x^4$, find $\frac{dy}{dx}$ ?

$y \to u \to x$

Sol. $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= \cos u \cdot 4x^3$

$= \cos(x^4) \cdot 4x^3$ نعيد تعويض u

ex) If $y = \tan(w)$ , $w = \sqrt{x}$

find $\frac{dy}{dx}$ ? $y \to w \to x$

Sol. $\frac{dy}{dx} = \frac{dy}{dw} \cdot \frac{dw}{dx}$

$= \sec^2(w) \cdot \frac{1}{2\sqrt{x}}$

$= \sec^2(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$

---

* $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$

ex) Evaluate

1) $\lim_{x \to \frac{\pi}{4}} \frac{\tan x - 1}{x - \frac{\pi}{4}}$

Sol. L'dp!

$\lim_{x \to \frac{\pi}{4}} \frac{\tan x - \tan(\frac{\pi}{4})}{x - \frac{\pi}{4}}$

$= \frac{d}{dx} \tan x \Big|_{x = \frac{\pi}{4}}$

$= \sec^2 \frac{\pi}{4} = 2$

∴ $\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = f'(a)$

---

2) If $f(x) = x^{50}$ , $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1}$

Sol. $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = f'(1)$

$f'(x) = 50x^{49}$

$f'(1) = 50$

---

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3) If $f(x) = kx^3 + 2x$ and

$\lim_{x \to 2} \frac{f(x^2) - f(4)}{x - 2} = 56$ ,

find $k$ ?

Sol. $\lim_{x \to 2} \frac{f(x^2) - f(2^2)}{x - 2}$

$= \frac{d}{dx} f(x^2) \Big|_{x=2}$

$f'(x^2) \cdot 2x \Big|_{x=2} = 56$

$f'(4) \cdot 4 = 56$

$f'(4) = 14$

$48k + 2 = 14 \Rightarrow 48k = 12$

$\Rightarrow k = \frac{1}{4}$

# L'Hopitals Rule:

قاعدة لوبيتال تستخدم عندما يكون ناتج تعويض النهاية $\frac{0}{0}$ , $\frac{\pm\infty}{\pm\infty}$

عندها نشتق كل من البسط والمقام على حدة ثم نعيد تعويض النهاية

ex) 1) $\lim_{x\to 3} \frac{x^2-9}{x-3} = \frac{0}{0}!$

Sol. L'H

$\lim_{x\to 3} \frac{2x}{1} = 6$

2) $\lim_{x\to 1} \frac{x^{100}-1}{x-1} = \frac{0}{0}!$

Sol. $\lim_{x\to 1} \frac{100x^{99}}{1} = 100$

3) $\lim_{x\to 2} \frac{\frac{1}{x}-\frac{1}{2}}{x-2} = \frac{0}{0}!$

Sol. $\lim_{x\to 2} \frac{\frac{-1}{x^2}}{1} = \frac{-1}{4}$

4) $\lim_{x\to 0} \frac{\sin x}{x} = \frac{0}{0}!$

Sol. $\lim_{x\to 0} \frac{\cos x}{1} = \cos 0 = 1$

5) $\lim_{x\to\infty} \frac{x^2}{e^x} = \frac{\infty}{\infty}!$

Sol. $\lim_{x\to\infty} \frac{2x}{e^x} = \frac{\infty}{\infty}!$

$\lim_{x\to\infty} \frac{2}{e^x} = \frac{2}{\infty} = 0$ مرة أخرى

6) $\lim_{x\to\infty} \frac{e^x}{x^2} = \frac{\infty}{\infty}!$

Sol. $\lim_{x\to\infty} \frac{e^x}{2x} = \frac{\infty}{\infty}!$

$\lim_{x\to\infty} \frac{e^x}{2} = \frac{\infty}{2} = \infty$

H.W

7) $\lim_{x\to 0} \frac{1-\cos x}{x}$

Sol. 0

H.W

8) $\lim_{x\to 2} \frac{\sqrt{x+2}-2}{x-2}$

Sol. $\frac{1}{4}$

# equation of the tangent line

$$y - y_0 = m(x - x_0)$$

Ex) Find the equation of the tangent line to the function $y = \sqrt{x}$ at point $(4,1)$ ?

$x_0$ $y_0$

Sol. $(4, 1)$

$$m = \left.\frac{dy}{dx}\right|_{x=4}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

eq. of tangent

$$y - 1 = \frac{1}{4}(x - 4)$$

$$y = \frac{1}{4}x - 1 + 1$$

$$y = \frac{1}{4}x$$

Ex) Find an equation of the tangent line to the curve

$$y = \frac{1}{\sin x + \cos x} \quad \text{at } x = 0$$

Sol. إذا لم نعطى زوجاً مرتباً $(x_0, y_0)$ وأعطيت x نعوضها في معادلة y فيصبح عندنا $(x_0, y_0)$

$$x = 0 \longrightarrow y = \frac{1}{\sin 0 + \cos 0} = 1$$

$$(x_0, y_0) = (0, 1)$$

$$m = \left.\frac{dy}{dx}\right|_{x=0}$$

$$\frac{dy}{dx} = \frac{-1.(\cos x - \sin x)}{(\sin x + \cos x)^2}$$

$$m = \frac{-1}{1} = -1$$

equ. of tangent $y - 1 = -1(x - 0)$

$$y = -x + 1$$

بوليتكنيك

Ex) Equation of tangent line to the curve $y = \sec^2 2x$ at $x = \frac{\pi}{8}$ is

Sol. $x = \frac{\pi}{8} \rightarrow y = \sec^2\left(\frac{\pi}{4}\right) = 2$

$$\left(\frac{\pi}{8}, 2\right)$$

$$\frac{dy}{dx} = 2\sec(2x).\sec(2x)\tan(2x).2$$

$$= 4\sec^2(2x)\tan(2x)$$

$$m = \left.\frac{dy}{dx}\right|_{x=\frac{\pi}{8}} = 4.2.1 = 8$$

equ. of tangent line

$$y-2=8\left(x-\frac{\pi}{8}\right)$$

$$y=8x-\pi+2$$

Remark:

① Tangent line is horizontal if

$$\frac{dy}{dx}=0$$

② Tangent line is vertical if:

$$\frac{dy}{dx}=\pm\infty \quad \left(\frac{\text{عدد}}{0}\right)$$

ex) The vertical tangent of

$y=x^{2/3}$ at $x=$

Sol. $\frac{dy}{dx}=\frac{2}{3}x^{-1/3}=\frac{2}{3x^{1/3}}$

$\frac{dy}{dx}=\pm\infty$ ملاقية x القيم تجعل

$$\frac{2}{3x^{1/3}}\stackrel{?}{=}\pm\infty$$

$\therefore x=0$ is Vertical tangent

ex) The horizontal tangent of

$y=\frac{1}{3}x^3-9x$ at $x=$

Sol.

$$\frac{dy}{dx}=x^2-9$$

$\frac{dy}{dx}=0$ قيم x القيم تجعل

$$x^2-9=0$$

$$(x-3)(x+3)=0$$

$$x=3,\ x=-3$$

H.W ex) If $f(x)=x^2-6\sqrt[3]{x}$, find a vertical tangent of f

ex) If $f(x)=\frac{\sec x}{1+\tan x}$. Find the values of x such that f has horizontal tangent at these values ?

Sol. $\frac{dy}{dx}=\frac{(1+\tan x)\sec x\tan x-\sec x\sec^2 x}{(1+\tan x)^2}$

$$=\frac{\sec x\tan x+\sec x\tan^2 x-\sec^3 x}{(1+\tan x)^2}$$

$$= \frac{\sec x \tan x + \sec^3 x - \sec x - \sec^3 x}{(1+\tan x)^2}$$

$$= \frac{\sec x \tan x - \sec x}{(1+\tan x)^2}$$

$$\frac{dy}{dx} = 0$$

$$\frac{\sec x \tan x - \sec x}{(1+\tan x)^2} = 0$$

$$\Rightarrow \sec x \tan x - \sec x = 0$$

$$\sec x \tan x = \sec x$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}$$

ch-2 أسئلة سنوات سابقة *

Q1) $f(x) = \frac{g(x^2)}{h(3x)}$ , $f(2) = 1$

$h(6) = 2$ , $h'(6) = 4$ , $g'(4) = 6$ ,

then $f'(2) =$

Sol.

$$f'(x) = \frac{h(3x) \cdot g'(x^2) \cdot 2x - g(x^2) \cdot h'(3x) \cdot 3}{(h(3x))^2}$$

$$= \frac{h(6) \cdot g'(4) \cdot 4 - g(4) \cdot h'(6) \cdot 3}{(h(6))^2}$$

$$= \frac{2.6.4 - 4.4.3}{2^2} = \frac{48-48}{2} = 0$$

Q2) $f(x) = \sec^2(x^2)$ , then $f'(x) =$

Sol. $f(x) = (\sec(x^2))^2$

$$f'(x) = 2(\sec(x^2))^1 . \sec(x^2)\tan(x^2) . 2x$$

$$= 4\sec^2(x^2)\tan(x^2)$$

Q3) If $y = \sqrt{\cot(\sqrt{x})}$ , then $\frac{dy}{dx} =$

Sol. $y = (\cot(\sqrt{x}))^{1/2}$

$$\frac{dy}{dx} = \frac{1}{2}(\cot(\sqrt{x}))^{-1/2} . (-\csc^2(\sqrt{x})) . \frac{1}{2\sqrt{x}}$$

$$= \frac{-\csc^2(\sqrt{x})}{4\sqrt{x}\sqrt{\cot(\sqrt{x})}}$$

Q4) Given that $f'(x) = \sqrt{x^2+1}$ , $g(x) = x^4+1$ , then $(fog)'(x) =$

Sol. $(fog)'(x) =$

$$f'(g(x)) \cdot g'(x)$$

$$= f'(x^4+1) \cdot g'(x)$$

$$= \sqrt{(x^4+1)^2+1} \cdot 4x^3$$

$$= 4x^3\sqrt{x^8+2x^4+2}$$

Q5) Given $G(x) = x f^4(5x)$, $f(5) = -1$, $f'(5) = 2$, then $G'(1) =$

Sol.
$$G'(x) = x \cdot 4 f^3(5x) \cdot f'(5x) \cdot 5 + f^4(5x) \cdot 1$$
$$G'(1) = 4 f^3(5) \cdot f'(5) \cdot 5 + f^4(5)$$
$$= 4 \cdot (-1)^3 \cdot 2 \cdot 5 + (-1)^4$$
$$= -40 + 1 = -39$$

Q6) An equation of the tangent line to the curve $y = \tan(\sin x)$ at the point $(4\pi, 0)$ is

Sol. $(4\pi, 0)$ ($x_0$, $y_0$)

$$m = \left.\frac{dy}{dx}\right|_{x=4\pi}$$

$$\frac{dy}{dx} = \sec^2(\sin x) \cdot \cos x$$

$$m = \left.\frac{dy}{dx}\right|_{x=4\pi} = 1 \cdot 1 = 1$$

equ. of tangent $y - 0 = 1(x - 4\pi)$

$$y = x - 4\pi$$

Q7) Suppose $f$ is a differentiable function such that $f(g(x)) = 4x$ and $f'(x) = 1 + [f(x)]^2$, then $g'(x) =$

Sol
$$f(g(x)) = 4x$$ نشتق الطرفين

من المعطى $f'(g(x)) \cdot g'(x) = 4$

$$1 + [f(g(x))]^2 \cdot g'(x) = 4$$

$$\Rightarrow g'(x) = \frac{4}{1 + (f(g(x)))^2}$$

$$= \frac{4}{1 + (4x)^2} = \frac{4}{1 + 16x^2}$$

Q8) $y = \cos^3(\sqrt{x})$, then $\frac{dy}{dx} =$

Sol.
$$y = (\cos(\sqrt{x}))^3$$

$$\frac{dy}{dx} = -3(\cos(\sqrt{x}))^2 \cdot \sin(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$$
$$= \frac{-3 \sin(\sqrt{x}) \cos(\sqrt{x}) \cos(\sqrt{x})}{2\sqrt{x}}$$

$$= \frac{-3}{4\sqrt{x}} \sin(2\sqrt{x}) \cos(\sqrt{x})$$

H.W

Q9) Equation of tangent line to the curve $y = \sec^2(2x)$ at $x = \frac{\pi}{8}$ is

Sol. $y = 8x - \pi + 2$

Q10) If $f(x) = x^4$, then $\lim_{x \to -2} \frac{f(x) - 16}{x + 2} =$

Sol. $\lim_{x \to -2} \frac{f(x) - f(-2)}{x - (-2)} = f'(-2)$

$f'(x) = 4x^3$

$f'(-2) = 4(-2)^3 = 4.(-8) = -32$

Q11) If $f'(x) = \frac{x}{x^2+1}$, $g(x) = \sqrt{x}$ and $h(x) = f(g(x))$, then $h'(3) =$

Sol. $\frac{1}{8}$

Q12) If $f(x) = \sqrt{x^3+5}$, $f'(x) =$

Sol. $f'(x) = \frac{3x^2}{2\sqrt{x^3+5}}$

Q13) let $g(x) = f(6x + h(x))$ and $h(1) = 2$, $f'(8) = 4$, $h'(1) = 0$, then $g'(1) =$

Sol.

$g'(x) = f'(6x + h(x)) (6 + h'(x))$

$g'(1) = f'(6 + h(1)) . (6 + h'(1))$

$= f'(8) . 6 = 4 . 6 = 24$

Q14) If $y = \cos(3x)$, then $\frac{d^{22} y}{dx^{22}}$

Sol.

$y' = -3 \sin(3x)$

$y'' = -3^2 \cos(3x)$

$y''' = +3^3 \sin(3x)$

$2n = 22$

$n = 11$

$\frac{d^{22} y}{dx^{22}} = (-1)^{11} 3^{22} \cos(3x)$

$= -3^{22} \cos(3x)$

H.W

Q15) The equation of the tangent line to the curve $y = \sqrt{1 + 6\sin x}$ at the point $(0,1)$ is

Sol. $y - 1 = 3(x - 0)$

Q16) $\lim_{x \to \frac{\pi}{4}} \frac{\cot x - 1}{x - \frac{\pi}{4}}$

Sol.

$$\lim_{x \to \frac{\pi}{4}} \frac{\cot x - \cot\left(\frac{\pi}{4}\right)}{x - \frac{\pi}{4}}$$

$$= \left(\frac{d}{dx} \cot(x)\right)\Big|_{x=\frac{\pi}{4}}$$

$$= -\csc^2\left(\frac{\pi}{4}\right) = -2$$

another way :- (L'H)

$$\lim_{x \to \frac{\pi}{4}} \frac{-\csc^2 x}{1} = -\csc^2\left(\frac{\pi}{4}\right) = -2$$

Q17) Let $f$ be any function such that $f(2) = 2$ and $f'(2) = 2$, then

$$\lim_{x \to 2} \frac{x f(x) - 4}{x - 2} =$$

Sol.

$$\lim_{x \to 2} \frac{x f(x) - 2f(2)}{x - 2}$$

$$= \left(\frac{d}{dx} x f(x)\right)\Big|_{x=2}$$

$$= \left(x f'(x) + f(x) \cdot 1\right)\Big|_{x=2}$$

$$= 2f'(2) + f(2)$$

$$= 4 + 2 = 6$$

Q18) If $\frac{d}{dx} f(\tan x) = \frac{\sin^2 x}{\cos^4 x}$,

then $f'(x) =$

Sol. نشتغل الطرف الأيسر فقط

$$f'(\tan x) \cdot \sec^2 x = \frac{\sin^2 x}{\cos^4 x}$$

$$f'(\tan x) = \cos^2 x \, \frac{\sin^2 x}{\cos^4 x}$$

$$f'(\tan x) = \frac{\sin^2 x}{\cos^2 x}$$

$$f'(\tan x) = \tan^2 x$$

let $y = \tan x \Rightarrow f'(y) = y^2$

therefore, $f'(x) = x^2$

Q19) If $\frac{d}{dx}(f(4x)) = 8x^2$, then

$f'(x) =$

Sol.

$$f'(x) = \frac{x^2}{8}$$

Q20) $f(x) = x^8 - 2x + 3$, find

$\lim_{h \to 0} \frac{f'(1+h) - f'(1)}{h}$ ?

Sol. $= f''(1)$

$f'(x) = 8x^7 - 2$

$f''(x) = 56x$

$= 56$

Q21) Consider the function

$$f(x)=\frac{(x+1)(\sin x+1)}{\sqrt{4+x^2}}$$

Then $f'(0)$ equals $\pm 0$

Sol.

$$f'(x)=\frac{(\sqrt{4+x^2})[(x+1)\cos x+(\sin x+1).1]-(x+1)(\sin x+1)\frac{2x}{2\sqrt{4+x^2}}}{(4+x^2)}$$

$$f'(0)=\frac{2(1+1)-(1)(1).0}{4+0}$$

$$=\frac{4}{4}=1$$

Q22) $f(1)=3$, $f'(1)=-3$, $g(3)=3$

2017 $g'(3)=4$, $g(1)=-1$, $g'(1)=2$

Find $\left(\frac{gof}{f}\right)'(1)=$

Sol $\frac{f(x).g'(f(x)).f'(x)-g(f(x)).f'(x)}{f^2(x)}$

$$\frac{f(1)\,g'(f(1))\,f'(1)-g(f(1))\,f'(1)}{f^2(1)}$$

$$\frac{(3)(4)(-3)-(3)(-3)}{9}$$

$$=\frac{-36+9}{9}=\frac{-27}{9}=-3$$

Q23) $f(x)=\sec^2(x^3)$, Find $f'(x)=$

2017

Sol $f(x)=(\sec(x^3))^2$

$$f'(x)=2(\sec(x^3))^1\sec(x^3)\tan(x^3).3x^2$$

$$=6x^2\sec^2(x^3)\tan(x^3)$$

Final

Q24) $f(g(x))=x$, $f'(x)=1+(f(x))^2$

2017

Then $g'(x)=$

Sol. $f(g(x))=x$

$$f'(g(x)).g'(x)=1$$

$$1+[f(g(x)]^2\,g'(x)=1$$

$$g'(x)=\frac{1}{1+[f(g(x)]^2}$$

$$=\frac{1}{1+x^2}$$

Final

Q25) $f(x)=kx^3+2x$,

2017

$$\lim_{x\to 2}\frac{f(x^2)-f(4)}{x-2}=56$$

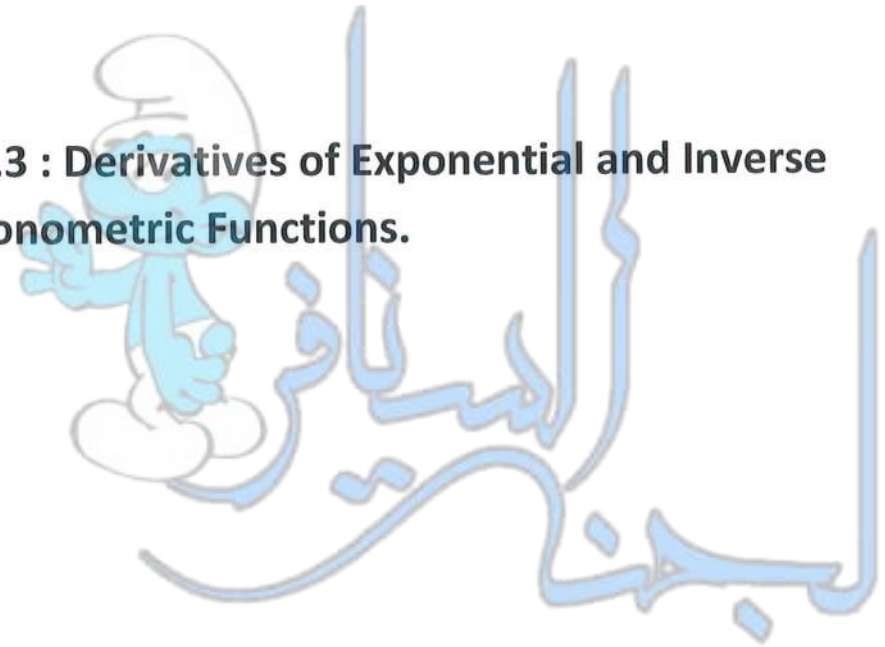
Sol $k=\frac{1}{4}$

# Chapter 3 : Topics In Differentiation

**3.1 : Implicit differentiation.**

**3.2 : Derivative of Logarithmic Functions.**

**3.3 : Derivatives of Exponential and Inverse Trigonometric Functions.**



للأستاذ : أحمد عرفه

٠٧٨٦٥٤٣٦٦٥

## 3.1 Implicit Differentiation
## الاشتقاق الضمني

Note: $y = \sin x$ explicit form اشتقاق صريح

$y = \sin(xy)$ implicit form ضمني — لا يمكن فصلهم

* $\frac{d}{dx} x^2 = 2x$ $\quad \frac{d}{dx} y^2 = 2y \frac{dy}{dx}$

$\frac{d}{dx} \sin x = \cos x$ $\quad \frac{d}{dx} \sin y = \cos y \frac{dy}{dx}$

* الاقتران الذي يكون لا يمكن فصل x عن y يكون اشتقاق ضمني ونشتق كالقاعدة لكن عند اشتقاق y نضع $\frac{dy}{dx}$ في كل مرة

ex) Find $\frac{dy}{dx}$ ?

① $x^3 + y^3 = 8$

Sol. نشتق المعادلة كاملة

$3x^2 + 3y^2 \frac{dy}{dx} = 0$

نجعل الحدود التي تحوي $\frac{dy}{dx}$ في طرف والتي لا تحوي $\frac{dy}{dx}$ في طرف

$3y^2 \frac{dy}{dx} = -3x^2 \Rightarrow \frac{dy}{dx} = \frac{-3x^2}{3y^2}$

② $5y^2 + \sin y = x^2$

Sol. $10y \frac{dy}{dx} + \cos y \frac{dy}{dx} = 2x$

نأخذ $\frac{dy}{dx}$ عامل مشترك

$\frac{dy}{dx}(10y + \cos y) = 2x$

$\frac{dy}{dx} = \frac{2x}{10y + \cos y}$

③ $2\sqrt{x} + \sqrt{y} = 3$

Sol. $2 \cdot \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$

$\frac{1}{2\sqrt{y}} \frac{dy}{dx} = \frac{-1}{\sqrt{x}}$ (نضرب الطرفين $\div 2\sqrt{y}$)

$\frac{dy}{dx} = \frac{-2\sqrt{y}}{\sqrt{x}}$

$= -2\sqrt{\frac{y}{x}}$

④ $\tan y = 3x^2$

Sol $\sec^2 y \frac{dy}{dx} = 6x$

$\frac{dy}{dx} = \frac{6x}{\sec^2 y}$

$= 6x \cos^2 y$

⑤ $x^2 + y^2 = 6xy$ ← ضرب

Sol. $2x + 2y\frac{dy}{dx} = 6x\cdot\frac{dy}{dx} + y\cdot 6$

$2y\frac{dy}{dx} - 6x\frac{dy}{dx} = 6y - 2x$

$\frac{dy}{dx}(2y - 6x) = 6y - 2x$

$\frac{dy}{dx} = \frac{6y - 2x}{2y - 6x}$

⑥ $\sin(x+y) = y^2\cos x$

Sol. $\cos(x+y)\cdot\left(1 + \frac{dy}{dx}\right) = -y^2\sin x + \cos x\, 2y\frac{dy}{dx}$

$\cos(x+y) + \cos(x+y)\frac{dy}{dx} = -y^2\sin x + 2y\cos x\frac{dy}{dx}$

$\cos(x+y)\frac{dy}{dx} - 2y\cos x\frac{dy}{dx} = -y^2\sin x - \cos(x+y)$

$\frac{dy}{dx}(\cos(x+y) - 2y\cos x) = -y^2\sin x - \cos(x+y)$

$\frac{dy}{dx} = \frac{-y^2\sin x - \cos(x+y)}{\cos(x+y) - 2y\cos x}$

H.W

⑦ $\sin x + \cos y = \sin x\cos y$

H.W

⑧ $x^2 - 2xy + y^3 = 4$

Q) Find equation of tangent for $x^2 + 2xy - y^2 + x = 2$ at $(1,2)$ ?

Sol "m" نشتق ضمنياً ونعوض مباشرة $(1,2)$

$2x + 2x\frac{dy}{dx} + y\cdot 2 - 2y\frac{dy}{dx} + 1 = 0$

$2 + 2\frac{dy}{dx} + 4 - 4\frac{dy}{dx} + 1 = 0$

$-2\frac{dy}{dx} = -7$

$\frac{dy}{dx} = \frac{-7}{2} = m$

equ. of tangent $(y-2) = \frac{-7}{2}(x-1)$

Ex) Find an equation of the tangent line of the curve $y^4 - 2x^2y^3 + 1 = 0$ at $(1,1)$ ?

Sol.

$4y^3\frac{dy}{dx} - (2x^2\cdot 3y^2\frac{dy}{dx} + y^3\cdot 4x) = 0$

$4\frac{dy}{dx} - (6\frac{dy}{dx} + 4) = 0$

$-2\frac{dy}{dx} - 4 = 0 \Rightarrow \frac{dy}{dx} = \frac{4}{-2} = -2$

equ. of tangent

$(y-1) = -2(x-1)$

(2) $y = y^2 + \sin(\sin x)$ , then $\frac{dy}{dx} =$

Sol. $\frac{dy}{dx} = 2y\frac{dy}{dx} + \cos(\sin x).\cos x$

$\frac{dy}{dx} - 2y\frac{dy}{dx} = \cos(\sin x)\cos x$

$\frac{dy}{dx}(1-2y) = \cos(\sin x)\cos x$

$\frac{dy}{dx} = \frac{\cos(\sin x)\cos x}{1-2y}$

(3.2) Derivative of Logarithmic fn.

(3.3) Derivative of exponantial and inverse Trigonometric function.

Thm:

(1) $\frac{d}{dx} e^{f(x)} = e^{f(x)} \cdot f'(x)$

(2) $\frac{d}{dx} a^{f(x)} = a^{f(x)} \cdot f'(x) \cdot \ln a$

a: any number

(3) $\frac{d}{dx} \ln(f(x)) = \frac{f'(x)}{f(x)}$

مشتقة ما داخل ln

على ما داخل ln

(4) $\frac{d}{dx} \log_a(f(x)) = \frac{f'(x)}{f(x) \cdot \ln a}$

(5) $\frac{d}{dx} \sin^{-1}(f(x)) = \frac{f'(x)}{\sqrt{1-(f(x))^2}}$

(6) $\frac{d}{dx} \cos^{-1}(f(x)) = \frac{-f'(x)}{\sqrt{1-(f(x))^2}}$

(7) $\frac{d}{dx} \tan^{-1}(f(x)) = \frac{f'(x)}{1+(f(x))^2}$

(8) $\frac{d}{dx} \cot^{-1}(f(x)) = \frac{-f'(x)}{1+(f(x))^2}$

(9) $\frac{d}{dx} \sec^{-1}(f(x)) = \frac{f'(x)}{|f(x)|\sqrt{(f(x))^2-1}}$

(10) $\frac{d}{dx} \csc^{-1}(f(x)) = \frac{-f'(x)}{|f(x)|\sqrt{(f(x))^2-1}}$

ex) Find $\frac{dy}{dx}$ :

1) $y = \tan^{-1} x$

مشتقة x

$\frac{dy}{dx} = \frac{1}{1+x^2}$

2) $y = \sin^{-1}(2x)$

Sol. $\frac{dy}{dx} = \frac{2}{\sqrt{1-(2x)^2}}$

$= \frac{2}{\sqrt{1-4x^2}}$

3) $y = \cot^{-1}(\sqrt{x})$

Sol. $\frac{dy}{dx} = \frac{-1}{1-(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}}$

$= \frac{-1}{2\sqrt{x}(1-x)}$

4) $f(x) = e^{x^2}$

Sol. $f'(x) = e^{x^2} \cdot 2x$

5) $f(x) = 7^{x^3}$

Sol $f'(x) = 7^{x^3} \cdot 3x^2 \cdot \ln 7$

6) $g(x) = \ln(\sin x)$

Sol. $g'(x) = \frac{\cos x}{\sin x} = \cot x$

7) $f(x) = \log_5 x^2$

Sol. $f'(x) = \frac{2x}{x^2 \cdot \ln 5} = \frac{2}{x \ln 5}$

8) $f(x) = \ln(x^3 + 2x)$

Sol. $f'(x) = \frac{3x^2+2}{x^3+2x}$

9) $f(x) = \log_3(1-x^4)$

Sol $f'(x) = \frac{-4x^3}{(1-x^4)\cdot \ln 3}$

10) $f(x) = \pi^{2x^2+7}$

Sol. $f'(x) = \pi^{2x^2+7} \cdot 4x \cdot \ln \pi$

11) $f(x) = e^x$ $f'(x) = e^x \cdot 1 = e^x$

12) $f(x) = \sqrt{e^x + 5x^8}$

Sol. $f'(x) = \frac{e^x + 40x^7}{2\sqrt{e^x+5x^8}}$

13) $f(x) = x^4 \cdot \ln x$

Sol. $f'(x) = x^4 \cdot \frac{1}{x} + \ln x \cdot 4x^3$

$= x^3 + 4x^3 \ln x$

14) $y = \cos^{-1}(e^x)$

Sol. $\frac{dy}{dx} = \frac{-e^x}{\sqrt{1-(e^x)^2}}$

$= \frac{-e^x}{\sqrt{1-e^{2x}}}$

15) $y = 2\cos(\sin^{-1}x)$

Sol $y' = -2\sin(\sin^{-1}x) \cdot \frac{1}{\sqrt{1-x^2}}$

16) $y = (\sin^{-1}5x)^4$

Sol. $y' = 4(\sin^{-1}5x)^3 \cdot \frac{5}{\sqrt{1-(5x)^2}}$

$= \frac{20(\sin^{-1}5x)^3}{\sqrt{1-25x^2}}$

H.W

17) $y = \tan^{-1}(2x)$

Sol $y' = \frac{2}{1+4x^2}$

18) $f(x) = \sin^{-1}(\log_5 x)$

Sol $f'(x) = \frac{1}{\sqrt{1-(\log_5 x)^2}} \cdot \frac{1}{x \ln 5}$

19) $f(x) = \sqrt{\tan^{-1}x} = (\tan^{-1}x)^{1/2}$

Sol. $f'(x) = \frac{1}{2}(\tan^{-1}x)^{-1/2} \cdot \frac{1}{1+x^2}$

$= \frac{1}{2\sqrt{\tan^{-1}x}\,(1+x^2)}$

20) $y = \sin^{-1}(\sqrt{\sin x})$

Sol. $y' = \frac{1}{\sqrt{1-(\sqrt{\sin x})^2}} \cdot \frac{\cos x}{2\sqrt{\sin x}}$

$= \frac{\cos x}{2\sqrt{\sin x}\sqrt{1-\sin x}}$

21) $y = \sqrt{x^3-1}\ \cot^{-1}x$

Sol. $\frac{dy}{dx} = \sqrt{x^3-1} \cdot \frac{-1}{1+x^2} + \cot^{-1}x \cdot \frac{3x^2}{2\sqrt{x^3-1}}$

$= \frac{-\sqrt{x^3-1}}{1+x^2} + \frac{3x^2\cot^{-1}x}{2\sqrt{x^3-1}}$

22) $y = \log_3(\sec x)$

Sol. $\frac{dy}{dx} = \frac{\sec x \cdot \tan x}{\sec x \cdot \ln 3}$

$= \frac{\tan x}{\ln 3}$

23) $y = \ln(8^x + 7x)$

Sol. $y' = \frac{8^x \cdot \ln 8 + 7}{8^x + 7x}$

24) $y = \frac{\tan x}{\pi}$

Sol. $y' = \frac{\tan x}{\pi} \cdot \sec^2 x \cdot \ln \pi$

25) $y = \ln(3x+5)^2$

Sol. $y = 2\ln(3x+5)$

$y' = 2 \cdot \frac{3}{3x+5} = \frac{6}{3x+5}$

26) $y = [\ln(1+e^x)]^2$

لا تنزل القوة لأن القوة ليست على ما داخل ln

Sol.

$y' = 2[\ln(1+e^x)]' \cdot \frac{e^x}{1+e^x}$

27) $y = \ln\left(\frac{(2x+1)^3}{(3x-1)^4}\right)$

Sol. يجب تبسيطها قبل الاشتقاق

$y = \ln(2x+1)^3 - \ln(3x-1)^4$

$y = 3\ln(2x+1) - 4\ln(3x-1)$

$\frac{dy}{dx} = 3 \cdot \frac{2}{2x+1} - 4 \cdot \frac{3}{3x-1}$

$= \frac{6}{2x+1} - \frac{12}{3x-1}$

H.W 28) $y = \ln(\csc x)$

H.W 29) $y = 3^x + x^3 + \ln(\cot x)$

H.W 30) $y = 7^{(x^2-2x)}$

H.W 31) $f(x) = \frac{e^{2x}}{\tan x}$

H.W 32) $h(x) = e^{3x} \cdot \cot x$

H.W 33) $g(x) = \sin^{-1}(\cos x)$

H.W 34) $R(x) = \sqrt{e^{x^2} - 3^{3x}}$

* مشتقة (اقتران) : مم

اقتران

مثل : $x^{x}$ , $x^{\cos x}$ , $x^{\tan^{-1}x}$ , $x^{\sqrt{x}}$ , $x^{\ln x}$

لا نستطيع الإشتقاق حتى ننزل الإقتران من القوة.

Ex) find $y'$ ?

1) $y = x^{\sqrt{x}}$

Sol. نأخذ ln على الطرفين حتى تنزل القوة

$\ln y = \sqrt{x} \ln x$

نشتق الطرفين

$\frac{y'}{y} = \sqrt{x} \cdot \frac{1}{x} + \ln x \cdot \frac{1}{2\sqrt{x}}$

$y' = y\left(\frac{\sqrt{x}}{x} + \frac{\ln x}{2\sqrt{x}}\right)$

$y' = x^{\sqrt{x}}\left(\frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}}\right)$

2) $y = x^{\sin x}$

Sol. $\ln y = \sin x \ln x$

$\ln y = \sin x \ln x$

$\frac{y'}{y} = \sin x \cdot \frac{1}{x} + \ln x \cos x$

$y' = y\left(\frac{\sin x}{x} + \ln x \cos x\right)$

$y' = x^{\sin x}\left(\frac{\sin x}{x} + \ln x \cos x\right)$

3) $f(x) = x^{x}$

Sol. $\ln f(x) = \ln x^{x}$

$\ln f(x) = x \ln x$

$\frac{f'(x)}{f(x)} = x \cdot \frac{1}{x} + \ln x \cdot 1$

$f'(x) = f(x)(1 + \ln x)$

$f'(x) = x^{x}(1 + \ln x)$

4) $y = (\cos x)^{x}$

Sol. $\ln y = \ln(\cos x)^{x}$

$\ln y = x \ln(\cos x)$

$\frac{y'}{y} = x \cdot \frac{-\sin x}{\cos x} + \ln(\cos x) \cdot 1$

$y' = y\left(\frac{-x \sin x}{\cos x} + \ln(\cos x)\right)$

$y' = (\cos x)^{x}\left(-x \tan x + \ln(\cos x)\right)$

5) $x^y = y^x$

Sol. $\ln x^y = \ln y^x$

$y \ln x = x \ln y$

$y \cdot \frac{1}{x} + \ln x \frac{dy}{dx} = x \cdot \frac{1}{y}\frac{dy}{dx} + \ln y \cdot 1$

$\ln x \frac{dy}{dx} - \frac{x}{y}\frac{dy}{dx} = \ln y - \frac{y}{x}$

$\frac{dy}{dx}\left(\ln x - \frac{x}{y}\right) = \ln y - \frac{y}{x}$

$\frac{dy}{dx} = \frac{\ln y - \frac{y}{x}}{\ln x - \frac{x}{y}}$

6) $y = \frac{x^2 e^x \sin x}{2^x (x+1)^4}$

Sol. نأخذ ln لتبسيط الاقتران

$\ln y = \ln\left(\frac{x^2 e^x \sin x}{2^x (x+1)^4}\right)$

$\ln y = \ln(x^2 e^x \sin x) - \ln(2^x (x+1)^4)$

$\ln y = \ln(x^2) + \ln(e^x) + \ln(\sin x) - \ln(2^x) - \ln(x+1)^4$

$\ln y = \ln x^2 + x + \ln(\sin x) - \ln(2^x) - 4\ln(x+1)$

$\frac{y'}{y} = \frac{2x}{x^2} + 1 + \frac{\cos x}{\sin x} - \frac{2^x \ln 2}{2^x} - 4 \cdot \frac{1}{x+1}$

$y' = y\left(\frac{2}{x} + 1 + \cot x - \ln 2 - \frac{4}{x+1}\right)$

$y' = \left(\frac{x^2 e^x \sin x}{2^x (x+1)^4}\right)\left(\frac{2}{x} + 1 + \cot x - \ln 2 - \frac{4}{x+1}\right)$

H.W 6) $y = (2x)^{\sin^{-1} x}$

Sol. $y' = (2x)^{\sin^{-1} x}\left(\frac{\sin^{-1} x}{x} + \frac{\ln(2x)}{\sqrt{1-x^2}}\right)$

H.W 7) $y = (\tan x)^{1/x}$

Sol. $y' = (\tan x)^{1/x}\left(\frac{\sec^2 x}{x \tan x} - \frac{\ln(\tan x)}{x^2}\right)$

H.W 8) $y = (\ln x)^{\tan x}$

Sol. $y' = (\ln x)^{\tan x}\left(\frac{\tan x}{x \ln x} + \ln(\ln x) \cdot \sec^2 x\right)$

H.W 9) $y = \ln\left(\frac{\sqrt{x}\sin x}{x+1}\right)$

* أسئلة السنوات على شابتر ٣

Q1) If $y = \ln|\sec x + \tan 3x|$, $x \in (0, \frac{\pi}{6})$ then $\frac{dy}{dx} =$

Sol.
$$\frac{dy}{dx} = \frac{\sec x \cdot \tan x + \sec^2(3x) \cdot 3}{\sec x + \tan(3x)}$$

Q2) If $f(x) = x \csc^{-1}(x)$ then $f'(2) =$

Sol.
$$f'(x) = x \cdot \frac{-1}{x\sqrt{x^2-1}} + \csc^{-1} x$$

$$f'(2) = \frac{-1}{\sqrt{3}} + \csc^{-1} 2$$

$$= \frac{-1}{\sqrt{3}} + \frac{\pi}{6}$$

Q3) $y = (\tan^{-1} x)^{-1}$, then $\frac{dy}{dx} =$

Sol.
$$\frac{dy}{dx} = -1(\tan^{-1} x)^{-2} \cdot \frac{1}{1+x^2}$$

$$= \frac{-1}{(\tan^{-1} x)^2 (1+x^2)}$$

Q4) $\frac{d^{97}}{dx^{97}}(xe^x) =$

Sol.
$$\frac{d}{dx} xe^x = xe^x + e^x \cdot 1$$

$$\frac{d^2}{dx^2} = xe^x + e^x + e^x = xe^x + 2e^x$$

$$\frac{d^3}{dx^3} = xe^x + e^x + 2e^x = xe^x + 3e^x$$

$$\vdots$$

$$\frac{d^{97}}{dx^{97}} = xe^x + 97e^x$$

Q5) $f(x) = \tan^{-1}(x^2)$

Sol. $f'(x) = \frac{1}{1+(x^2)^2} \cdot 2x$

$$f'(x) = \frac{2x}{1+x^4}$$

Q6) $\frac{d}{dx} \pi^{\sin(x^2)}$

Sol. $= \pi^{\sin(x^2)} \cdot \cos(x^2) \cdot 2x$

Q7) If $f(x)+e^{2x}(f(x))^3=-2$ and $f(0)=-1$, then $f'(0)=$

Sol. نشتق المعادلة

$$f'(x)+e^{2x}\cdot 3(f(x))^2\cdot f'(x)+(f(x))^3 e^{2x}\cdot 2=0$$

$$f'(0)+e^{0}\cdot 3(f(0))^2\cdot f'(0)+(f(0))^3\cdot e^{0}\cdot 2=0$$

$$f'(0)+1\cdot 3(-1)^2 f'(0)+(-1)^3\cdot 1\cdot 2=0$$

$$4f'(0)-2=0$$

$$4f'(0)=2 \Rightarrow f'(0)=\frac{1}{2}$$

Q8) The slope of tangent line to $y=(\cos^{-1}x)^2$ at $x=\frac{-1}{\sqrt{2}}$

Sol. slope = المشتقة

$$\text{slope}=\left.\frac{dy}{dx}\right|_{x=\frac{-1}{\sqrt{2}}}$$

$$\frac{dy}{dx}=2(\cos^{-1}x)\cdot\frac{-1}{\sqrt{1-x^2}}$$

$$\left.\frac{dy}{dx}\right|_{x=\frac{-1}{\sqrt{2}}}=2\left(\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)\right)\cdot\frac{-1}{\sqrt{1-\left(\frac{-1}{\sqrt{2}}\right)^2}}$$

$$=2\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)\cdot -\sqrt{2}$$

$$=2\cdot\frac{3\pi}{4}\cdot -\sqrt{2}$$

$$=\frac{3\pi}{2}\cdot -\sqrt{2}=\frac{3\pi}{\sqrt{2}\cdot\sqrt{2}}\cdot -\sqrt{2}$$

$$=\frac{-3\pi}{\sqrt{2}}$$

Q9) $\frac{d}{dx}\left(x\tan^{-1}(x^2)\right)$

Sol. $x\cdot\frac{1}{1+(x^2)^2}\cdot 2x+\tan^{-1}(x^2)\cdot 1$

$$=\frac{2x^2}{1+x^4}+\tan^{-1}(x^2)$$

Q10) $y=\ln(x^2+y^2)$, then $\frac{dy}{dx}=$

Sol.

$$\frac{dy}{dx}=\frac{2x+2y\frac{dy}{dx}}{x^2+y^2}$$

$$(x^2+y^2)\frac{dy}{dx}=2x+2y\frac{dy}{dx}$$

$$(x^2+y^2-2y)\frac{dy}{dx}=2x$$

$$\frac{dy}{dx}=\frac{2x}{x^2+y^2-2y}$$

Q11) If $f(x)=\sin^{-1}(\sqrt{x+2})$ then $\frac{dy}{dx}=$ (8-1-2017)

Sol. $f'(x)=\frac{1}{\sqrt{1-(\sqrt{x+2})^2}}\cdot\frac{1}{2\sqrt{x+2}}$

$=\frac{1}{\sqrt{1-x-2}}\cdot\frac{1}{2\sqrt{x+2}}$

$=\frac{1}{\sqrt{-1-x}}\cdot\frac{1}{2\sqrt{x+2}}$

Q12) If $f(x)=2^x$, find $\frac{d^{100}f}{dx^{100}}$.

Sol. $f'(x)=2^x\cdot\ln 2=2^x\cdot\ln 2$

$f''(x)=2^x\cdot 1\cdot\ln 2\cdot\ln 2=2^x(\ln 2)^2$

$f'''(x)=2^x\cdot\ln 2(\ln 2)^2=2^x(\ln 2)^3$

$\vdots$

$f^{(100)}(x)=2^x(\ln 2)^{100}$

Q13) Let $y=ax+b$ ($a$, $b$ are constants) be the equation of the tangent line of the function $f(x)=2xe^{x^2}$ at $x=0$. Then $a+b$ equals to

Sol. $x=0\longrightarrow y=f(0)=2\cdot 0\cdot e^0=0$

$(x_0,y_0)=(0,0)$

$m=f'(0)$

$f'(x)=2xe^{x^2}\cdot 2x+e^{x^2}\cdot 2$

$f'(0)=e^0\cdot 2=2$

$m=2$

eq. of tangent $y-0=2(x-0)$

$y=2x$

$y=ax+b$

$y=2x+0$

$\therefore a=2$, $b=0$

$a+b=2$

Q14) $y=\left(\frac{4^x}{2^{x+1}}\right)^2$, $y^{(15)}=$

a) $\frac{4(\ln 2)^{15}}{2^{x-15}}$ b) $2^{2x+25}(\ln 2)^{15}$ c) $2^{2x+13}(\ln 2)^{15}$

Sol. $y=\left(\frac{2^x\cdot 2^x}{2^x\cdot 2}\right)^2=\frac{2^{2x}}{4}$

$y^{(15)}=\frac{1}{4}\left(2^{2x}\cdot 2^{15}\cdot(\ln 2)^{15}\right)$

$=\frac{1}{2^2}\left(2^{2x}\cdot 2^{15}(\ln 2)^{15}\right)$

$=2^{2x}\cdot 2^{13}(\ln 2)^{15}=2^{2x+13}(\ln 2)^{15}$

(c)

Q15) If $f(x) = \frac{1}{\pi^2} x^3 + \cos x$, then $(f(2x))'$ at $x = \pi$ equals

Sol.
$(f(2x))' = f'(2x) \cdot 2 \Big|_{x=\pi}$
$= f'(2\pi) \cdot 2$
$= 12 \cdot 2 = 24$

$f'(x) = \frac{3}{\pi^2} x^2 - \sin x$

$f'(2\pi) = \frac{3}{\pi^2} (2\pi)^2 - \sin(2\pi)$

$f'(2\pi) = \frac{3}{\pi^2} \cdot 4\pi^2 - 0 = 12$

H.W
Q16) If $x e^y + 5 \sec x = y - 1$, then $\frac{dy}{dx} =$

H.W
Q17) Let $y = (\sin x)^{x^3}$, Find $\frac{dy}{dx} =$

H.W
Q18) $\frac{d}{dx} (\tan^{-1}(\ln 2x)) =$

H.W
Q19) $y = \ln \left( \sqrt[3]{\frac{x-1}{x+1}} \right)$, $\frac{dy}{dx} =$

H.W
Q20) If $x = \cos y$, then $\frac{dy}{dx} =$

Sol.
$\frac{dy}{dx} = \frac{-1}{\sin y}$

H.W
Q21) $y = (\sin(3x))^{5x}$, $\frac{dy}{dx} =$

H.W
Q22) If $f(x) = \log_9 (x^4 + 3)$, then $f'(x) =$

H.W
Q23) Let $f(x) = \ln \sqrt[9]{x^2+1}$. Then $f'(x) =$

a) $\frac{2x}{9(x^2+1)}$ b) $\frac{2x}{\sqrt[9]{x^2+1}}$

c) $\frac{2x}{9\sqrt[9]{x^2+1}}$ d) $\frac{2x}{x^2+1}$

b

Q24) Let $f(x) = 6^x$. Then $f''(0) =$ $(\ln 6)^2$

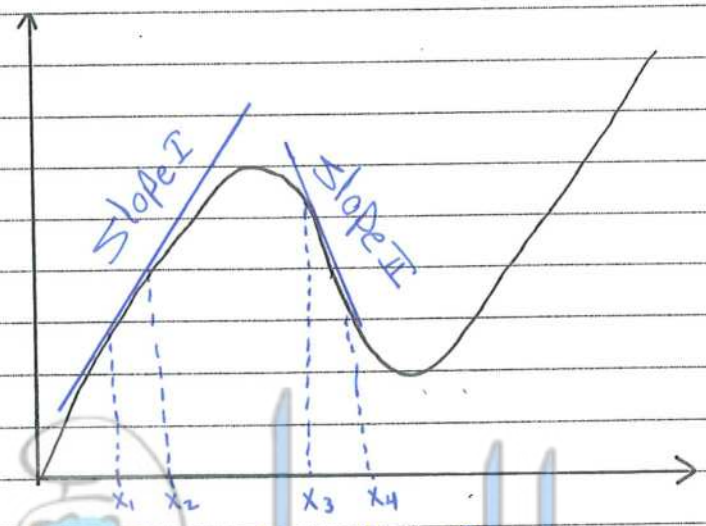
Q25) Let $f(x) = e^{6x}$. Then $f''(0) =$ 36

Q26) Let $f(x) = \log_x 5$. Then $f'(e) =$ $\frac{-\ln 5}{e}$

Q27) The slope of the tangent line to the curve of $x^3 + x\sqrt{y^2+3} = 3y^2$ at $(1,-1)$ is $\frac{-10}{11}$

# CH-4 The Derivative in Graphing Application

## 4.1 Analysis function I : Increasing, Decreasing and Concavity




- If $x_2 > x_1$ and $f(x_2) > f(x_1)$ then $f(x)$ increasing. (متزايد)
- If $x_3 < x_4$ and $f(x_3) > f(x_4)$ then $f(x)$ decreasing (متناقص)

- If $f'(x) > 0$, then $f(x)$ increasing ($\nearrow$) on interval I.
- If $f'(x) < 0$, then $f(x)$ decreasing ($\searrow$) on interval I.
- If $f'(x) = 0$, then $f(x)$ is Constant.

**Def:** (Critical points) النقاط الحرجة

If $x_0 \in Dom(f)$, then $(x_0, f(x_0))$ is called a Critical point of $f$ if $f'(x_0) = 0$ OR $f'(x_0)$ (d.n.e)

* النقطة الحرجة : هي النقطة التي يكون الاقتران عندها متصل و نخرجها إذا كان الاقتران كثير حدود بأصفار المشتقة الأولى / أما الكسور فبأصفار المشتقة الأولى والقيم التي تجعل المشتقة (d.n.e)

(Ex) Find a Critical numbers

1) $f(x) = 5x^2 + 4x$

Sol. $f'(x) = 10x + 4$

$f'(x) = 0$

$10x + 4 = 0 \Rightarrow x = \frac{-4}{10}$ is Critical number

2) $f(x) = 4x - x^2$, $x \in [0,5]$

من أطراف الفترة تعتبر نقاطاً حرجة

Sol. $f'(x) = 4 - 2x$

$f'(x) = 0$

$4 - 2x = 0 \Rightarrow x = 2$

Criticals numbers: $\{0, 2, 5\}$

3) $f(x) = x + \frac{1}{x}$

عشان كسر ⇐ حالتان

$f'(x) = 0$ او $f'(x)$ d.n.e

Sol. $f'(x) = 1 - \frac{1}{x^2}$

$f'(x) = \frac{x^2 - 1}{x^2}$

$f'(x) = 0 \rightarrow x^2 - 1 = 0$

$x = \pm 1$

عندما البسط = صفر

d.n.e $\rightarrow x^2 = 0$

$x = 0$

عندما المقام = صفر

$0 \notin D(f)$ لأن

Criticals numbers: $\{-1, 1\}$

4) $f(x)=\sin x$ , $x\in[0,2\pi]$

Sol. $f'(x)=\cos x$

$f'(x)=0$

$\cos x=0$ , $x=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}$ خارج الفترة

Criticals numbers:

$\{0,\frac{\pi}{2},\frac{3\pi}{2},2\pi\}$

5) $f(x)=\sin x+\cos x$ , $x\in[0,\frac{\pi}{2}]$

$f'(x)=\cos x-\sin x$

$f'(x)=0$

$\cos x-\sin x=0$

$\cos x=\sin x$

$\Rightarrow x=\frac{\pi}{4}$

Criticals numbers: $\{0,\frac{\pi}{4},\frac{\pi}{2}\}$

6) $f(x)=\sqrt{3}\sin x+\cos x$ , $x\in[0,\pi]$

Sol. $f'(x)=\sqrt{3}\cos x-\sin x$

$f'(x)=0$

$\sqrt{3}\cos x-\sin x=0$

$\sqrt{3}\cos x=\sin x$

$\sqrt{3}=\tan x$

$\Rightarrow x=\frac{\pi}{3}$

Criticals numbers: $\{0,\frac{\pi}{3},\pi\}$

7) $f(x)=\sin^2 x$ , $x\in[0,2\pi]$

Sol. $f'(x)=2\sin x.\cos x$

$=\sin(2x)$

$f'(x)=0$

$\sin(2x)=0$ , $2x=0,\pi,2\pi,3\pi,4\pi$

$x=0,\frac{\pi}{2},\pi,\frac{3\pi}{2},2\pi$

Criticals numbers:

$\{0,\frac{\pi}{2},\pi,\frac{3\pi}{2},2\pi\}$

8) $f(x)=x^{2/3}(6-x)^{1/3}$

Sol. $f'(x)=\frac{4-x}{x^{1/3}(6-x)^{2/3}}$

$f'(x)=0\Rightarrow 4-x=0$

$x=4$

$f'(x)$ d.n.e $\Rightarrow x^{1/3}(6-x)^{2/3}=0$

$x=0$ , $x=6$

Criticals numbers:

$\{0,4,6\}$

H.W

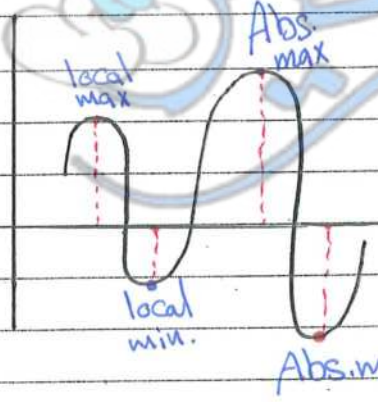
9) $f(x)=x^{3/5}(4-x)$

Sol. $\{0,\frac{12}{8}\}$

# Extrema Value
## القيم العظمى

- محلية (local/Relative)
  - Max
  - Min
  - القيم المحلية لا تكون إلا داخل الفترات .
- مطلقة Absolute
  - Max
  - Min
  - القيم المطلقة ممكن أن تكون عند الأطراف أو داخل الفترة .

ex)




Abs. : أكبر/أصغر من جميع القيم على الفترة

local. : أكبر/أصغر من جيرانها فقط

**Thm:** Every cts function on closed interval has both Abs. Max and Abs. Min.

كل اقتران متصل على فترة مغلقة سيكون له قيمة عظمى مطلقة و صغرى مطلقة .

⇒ Method to find Abs. max and Abs. Min For cts function $f(x)$ on closed interval $[a,b]$ :

① Find critical number. ١- نوجد النقاط الحرجة

② Find values of critical numbers and $f(a)$, $f(b)$ ٢- نجد صور النقاط الحرجة و أطراف الفترة $a, b$

③ The largest value is Abs. max ٣- أكبر قيمة تكون عظمى مطلقة

④ The Smallest value is Abs. Min. ٤- أصغر قيمة تكون صغرى مطلقة

(Ex) Find Abs. Max and Abs. Min for :

i) $f(x) = x^3 - 3x^2 + 1$ on $[-1, 4]$

Sol. Criticals

$f'(x) = 0$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$x = 0 \;,\; x = 2$

$f(-1) = -3$ نعوض

$f(0) = 1$

$f(2) = -3$

$f(4) = 17$

Abs. max = 17 at $x = 4$

Abs. min = 3 at $x = -1, 2$

ii) $f(x) = x^2 - 4x + 1$ on $[1, 3]$

Sol. Criticals

$f'(x) = 0$

$2x - 4 = 0$

$\Rightarrow x = 2$

$f(1) = -2$

$f(2) = -3$

$f(3) = -2$

Abs. max = $-2$ at $x = 1, 3$

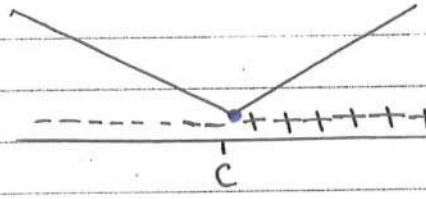
Abs. min = $-3$ at $x = 2$

H.W iii) $f(x) = \frac{1}{3}x^3 - 9x \;,\; [-6, 6]$

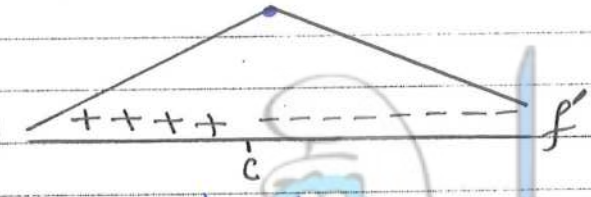
## ⇒ First Derivative Test

نستخدم لإيجاد local max/min

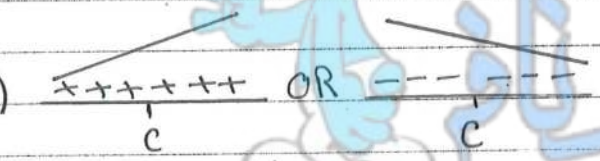
C : Critical number




at c local min




at c local max




No Max
No Min.

• القيم المحلية (local) لا تكون إلا داخل الفترات ولا تكون عند الأطراف.

• أطراف الفترة إن لم تكن Abs. فإنها تحتمل.

• في حال لم نعطى فترة في السؤال فلا يوجد Abs. أبداً.

(Ex) Find

i) interval of increasing and decreasing.
ii) Find local min and max values (If exists)
iii) Find Abs. max and Abs. min (If exists)

a) $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$

Sol. $f'(x) = 0$

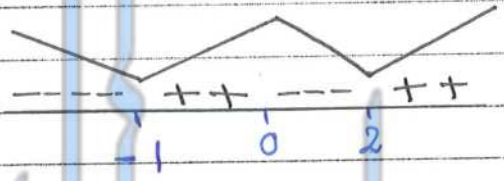
$12x^3 - 12x^2 - 24x = 0$

$12x(x^2 - x - 2) = 0$

$12x(x-2)(x+1) = 0$

$x = 0, \quad x = 2, \quad x = -1$

نعوض القيم هنا للسؤال




f is increasing on $[-1,0]$ , $[2,\infty)$

f is decreasing on $(-\infty,-1]$ , $[0,2]$

At -1 local min with value $f(-1) = 0$

At 0 local max with value $f(0) = 5$

At 2 local min with value $f(2) = -27$

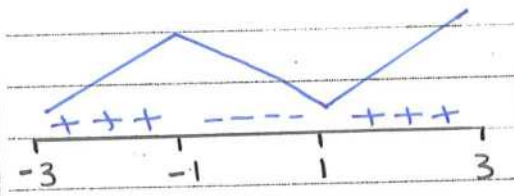
No Abs. max } لأنه
No Abs. min } يوجد عندنا فترة مغلقة

b) $f(x) = x^3 - 3x + 5$ , $x \in [-3, 3]$

Sol: $f'(x) = 0$

$3x^2 - 3 = 0 \Rightarrow x^2 = 1$

$x = \pm 1$




$f(x)$ is increasing on $[-3,-1]$ , $[1,3]$

$f(x)$ is decreasing on $[-1,1]$

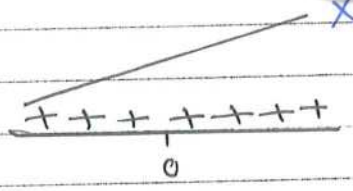
ملحوظة: الفترات مغلقة لأن الإقتران معرف عند أطراف الفترات

c) $f(x) = x^3$

Sol: $f'(x) = 3x^2$

$f'(x) = 0 \Rightarrow 3x^2 = 0$

$x = 0$




$f(x)$ is increasing on $\mathbb{R}$

No Max , No Min

H.W

d) $f(x) = x^3 - 6x^2 + 9x$ , $x \in [0,4]$

H.W

e) $f(x) = x^3 - 3x + 10$

Sol: $f \nearrow$ on $(-\infty,-1]$ , $[1,\infty)$

$f \searrow$ on $[-1,1]$

at $x=-1$ local max with value $f(-1)=12$

$x=1$ local min " " $f(1)=8$

f) $f(x) = \sin(2x)$ , $x \in [0, \pi]$

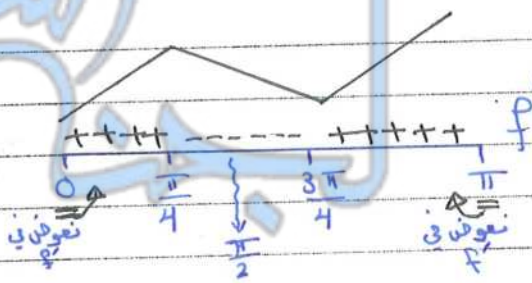
Sol: $f'(x) = 2\cos(2x)$

$f'(x) = 0$

$2\cos(2x) = 0 \Rightarrow \cos(2x) = 0$

$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$ خارج الفترة




$f \nearrow$ on $[0, \frac{\pi}{4}]$ , $[\frac{3\pi}{4}, \pi]$

$f \searrow$ on $[\frac{\pi}{4}, \frac{3\pi}{4}]$

$f(0) = 0$ نهاية

$f(\frac{\pi}{4}) = 1$ Abs.max / local max

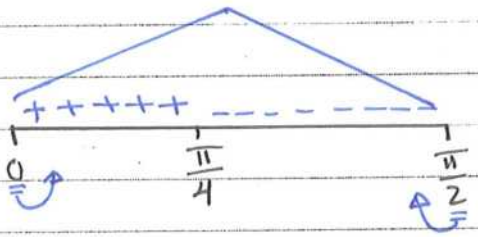
$f(\frac{3\pi}{4}) = -1$ Abs.min / local min

$f(\pi) = 0$ نهاية

g) $f(x) = \sin x + \cos x$, $x \in [0, \frac{\pi}{2}]$

Sol. Critical $x = \frac{\pi}{4}$ (نقطة 3)

$f'(x) = \cos x - \sin x$




$f(x)$ is increasing on $[0, \frac{\pi}{4}]$

$f(x)$ is decreasing on $[\frac{\pi}{4}, \frac{\pi}{2}]$

$f(0) = 1$ Abs. min

$f(\frac{\pi}{4}) = \frac{2}{\sqrt{2}}$ Abs. max / local max

$f(\frac{\pi}{2}) = 1$ Abs. min

H.W

h) $f(x) = \sqrt{3}\sin x + \cos x$, $x \in [0, \pi]$

Sol. $f \nearrow [0, \frac{\pi}{3}]$

$f \searrow$ on $[\frac{\pi}{3}, \pi]$

at $x = \pi$ Abs. min with value $f(\pi) = -1$

$x = \frac{\pi}{3}$ Abs. and local max $f(\frac{\pi}{3}) = 2$

H.W

k) $f(x) = (x^2 - 1)^3$

Sol. $f \nearrow [0, \infty)$ $f \searrow (-\infty, 0]$

$f(0) = -1$ local min.

l) $f(x) = \sin^2 x$, $x \in [0, 2\pi]$

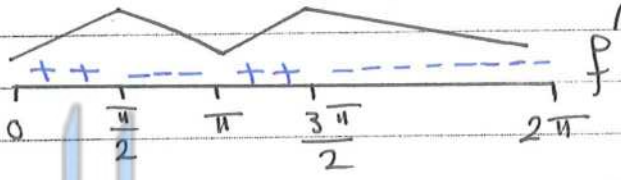
Sol. $f'(x) = 0$

$2\sin x \cos x = 0$

$\sin(2x) = 0$

$2x = 0, \pi, 2\pi, 3\pi, 4\pi$

$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$




$f \nearrow [0, \frac{\pi}{2}], [\pi, \frac{3\pi}{2}]$

$f \searrow [\frac{\pi}{2}, \pi], [\frac{3\pi}{2}, 2\pi]$

$f(0) = 0$ Abs. min

$f(\frac{\pi}{2}) = 1$ Abs. max / local max

$f(\pi) = 0$ Abs min / local min

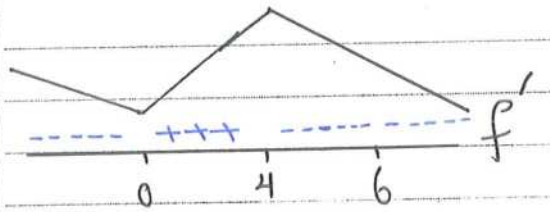
$f(\frac{3\pi}{2}) = 1$ Abs max / local max.

$f(2\pi) = 0$ Abs. min

m) $f(x) = x^{2/3}(6 - x)^{1/3}$

Sol. $f'(x) = \frac{4 - x}{x^{1/3}(6 - x)^{2/3}}$

Criticals numbers: $\{0, 4, 6\}$ (3 نقاط)




$f \nearrow [0,4]$ $\quad f \searrow (-\infty,0], [4,6), (6,\infty)$

$f(0) = 0$ local min

$f(4) = 2^{5/3}$ local max

at $x=6$ No Max, No Min

---

n) $f(x) = x + \frac{9}{x+2}$, $x \in [-1,4]$

Sol. $f'(x) = 1 - \frac{9}{(x+2)^2}$

$= \frac{(x+2)^2 - 9}{(x+2)^2}$ كسر حالتان

$f'(x) = 0 \Rightarrow (x+2)^2 - 9 = 0$

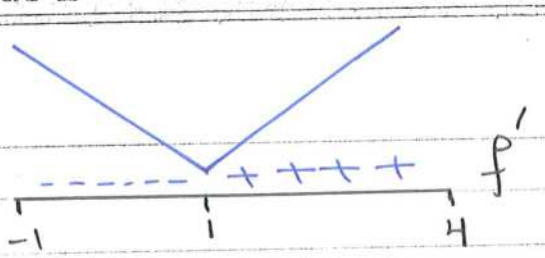
$(x+2)^2 = 9$

$|x+2| = 3$

$\Rightarrow x = 1$ , $x = -5$ (نهمل خارج)

$f'(x)$ d.n.e $\Rightarrow (x+2)^2 = 0$

$x+2 = 0 \Rightarrow x = -2$ نهمل خارج




$f \nearrow$ on $[-1,1]$ $\quad f \searrow$ on $[1,4]$

$f(-1) = 8$ Abs. max

$f(1) = 4$ Abs. min / local min

$f(4) = \frac{11}{2}$ تهمل لأنها ليست الأكبر

---

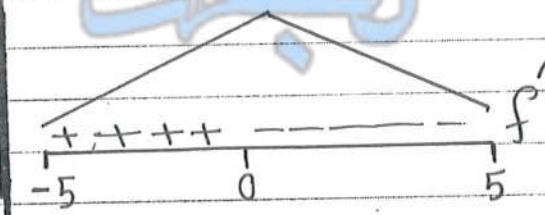
o) $f(x) = \frac{1}{x^2+1}$, $-5 \le x \le 5$

Sol. $f'(x) = \frac{-2x}{(x^2+1)^2}$ حالتان

$f'(x) = 0 \Rightarrow -2x = 0 \Rightarrow x = 0$

$f'(x) = 0 \Rightarrow (x^2+1)^2 = 0$

$x^2 + 1 = 0$ ليس للعدم أصفار




$f \nearrow$ on $[-5,0]$ $\quad f \searrow [0,5]$

$f(-5) = \frac{1}{26}$ Abs. min

$f(0) = 1$ Abs. max / local max

$f(5) = \frac{1}{26}$ Abs. min

p) $f(x)=\sqrt{4-x^2}$ , $x\in[-2,2]$

∴ إذا لم تعطى قوة (الجال) للجذور الزوجية فعليك أن تجدها.

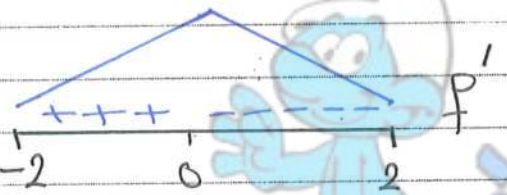
$$f'(x)=\frac{-2x}{2\sqrt{4-x^2}}=\frac{-x}{\sqrt{4-x^2}}$$

$f'(x)=0 \Rightarrow -x=0 \Rightarrow x=0$

$f'(x)$ d.n.e $\Rightarrow \sqrt{4-x^2}=0$

$4-x^2=0$

$x^2=4 \Rightarrow x=\pm 2$




$f\nearrow [-2,0]$ $f\searrow [0,2]$

$f(-2)=0$ Abs. min

$f(0)=2$ Abs. max / local max

$f(2)=0$ Abs. min.

---

g) $f(x)=\sqrt[3]{9-x^2}$ جذر فردي مجاله كل $\mathbb{R}$

Sol. $f(x)=(9-x^2)^{1/3}$

$f'(x)=(9-x^2)^{-2/3}\cdot(-2x)$

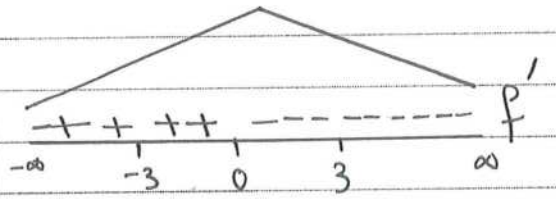
$$f'(x)=\frac{-2x}{(9-x^2)^{2/3}}$$

$f'(x)=0 \Rightarrow -2x=0$

$x=0$

$f'(x)$ d.n.e $\Rightarrow (9-x^2)^{2/3}=0$

$x=\pm 3$




$f\nearrow$ on $(-\infty,0]$

$f\searrow$ on $[0,\infty)$

$f(0)=\sqrt[3]{9}$ local max.

---

r) $f(x)=\sqrt{\sin x}$ , $x\in[0,\pi]$

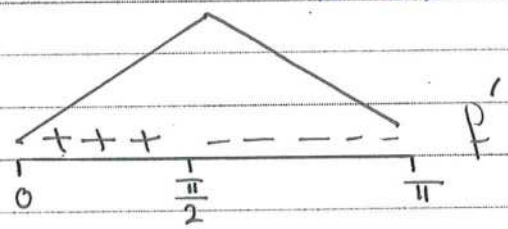
Sol. $f'(x)=\frac{\cos x}{2\sqrt{\sin x}}$

$f'(x)=0 \Rightarrow \cos x=0$ , $x=\frac{\pi}{2}$

$f'(x)$ d.n.e $\Rightarrow 2\sqrt{\sin x}=0$

$\sin x=0$

$x=0,\pi$




$f\nearrow [0,\frac{\pi}{2}]$ $f\searrow [\frac{\pi}{2},\pi]$

$f(0)=0$ Abs. min

$f(\frac{\pi}{2})=1$ Abs. max, local max

$f(\pi)=0$ Abs. min

Q) final 2017 $f(x) = 6^{x^2-2x}$ has

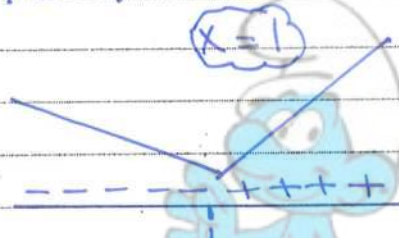
a) abs.min at 1 b) abs.max at 1
c) relative min at 1 d) relative max at 1

Sol. $f'(x) = 6^{x^2-2x}(2x-2)\ln 6$

$f'(x) = 0$

$6^{x^2-2x}(2x-2)\ln 6 = 0$

$6^{x^2-2x} \neq 0$ , $2x-2=0$



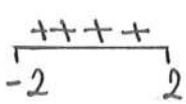

at $x=1$ relative min.

Q) final 2017 $f(x) = \sin^{-1}\left(\frac{x}{2}\right)$ find the increasing interval

Sol. $f'(x) = \frac{1}{\sqrt{1-(\frac{x}{2})^2}} \cdot \frac{1}{2}$

$f'(x) = \frac{\frac{1}{2}}{\sqrt{1-\frac{x^2}{4}}}$

$f'(x)$ d.n.e $\Rightarrow x = \pm 2$




$f \nearrow$ on $[-2,2]$

Q) Find Criticals number and interval of increasing and decreasing and local (max/min) and Abs.(max/min)

for i) $f(x) = \begin{cases} x^2+4x, & -3 \le x < 0 \\ x^2-4x, & 0 \le x \le 4 \end{cases}$

at $x \in [-3,4]$

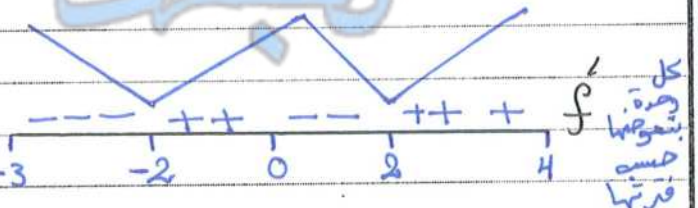
Sol. f is Cts.

$f'(x) = \begin{cases} 2x+4, & -3 < x < 0 \\ 2x-4, & 0 < x < 4 \\ \text{d.n.e}, & x = 0 \end{cases}$

$f'_+(0) = -4$
$f'_-(0) = 4$
$f'_+(0) \neq f'_-(0)$

Criticals $\xrightarrow{\text{أطراف الفترة}} -3, 4$
$\xrightarrow{\text{d.n.e}} 0$
$\xrightarrow{f'(x)=0} 2x+4=0 \Rightarrow x=-2$
$2x-4=0 \Rightarrow x=2$

Criticals $\{-3, 4, 0, 2, -2\}$




كل واحدة بتعوضها في f' قريب

$f \nearrow [-2,0], [2,4]$
$f \searrow [-3,-2], [0,2]$

$f(-3) = -3$ Abs.min / local min
$f(0) = 0$ Abs.max / local max.
$f(-2) = -4$ Abs.min / local min
$f(2) = -4$ Abs.min / local min
$f(4) = 0$ Abs.max.

ii) $f(x)=\begin{cases} x^2+1 & , x>1 \\ 2x & , x\le 1 \end{cases}$

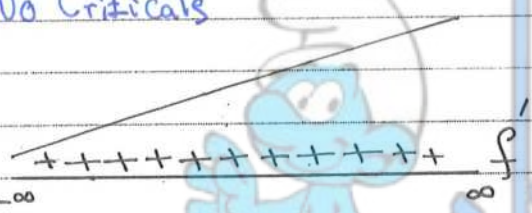
Sol. f is cts at $x=1$

$f'(x)=\begin{cases} 2x & , x>1 \\ 2 & , x<1 \\ 2 & , x=1 \end{cases}$ $\quad f'_+(1)=2$, $f'_-(1)=2$, $f'(1)=2$

$f'(x)=0 \to 2x=0 \Rightarrow x=0$ مهمل لأن $x>1$

$\to 2\neq 0$

No Criticals




f(x) increasing on R

No Max, No Min.

* Concavity Tests: اختبار التقعر

① If $f''(x)>0 \to f$ Concaved upward $(\cup)$ on I.

② If $f''(x)<0 \to f$ Concaved down $(\cap)$ on I.

الفترات لـ Concavity دائماً مفتوحة.

Def: Inflection point نقطة الإنعطاف

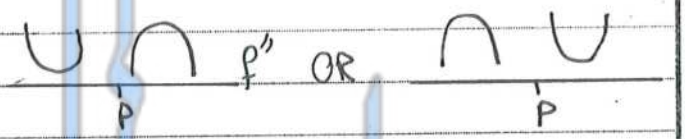
A point "p" is called inflection point if

① f is cts at D

② f(x) changes its Concavity at p

نقطة الإنعطاف : هي النقطة التي يتغير عندها الإقتران من متقعر لأعلى إلى متقعر لأسفل و العكس.

لتكون النقطة inflection يجب أن يكون الإقتران متصل عندها و يغير من تقعره حولها.




Q) ① Find Inflection point (If exists)

② Find interval of Concavity

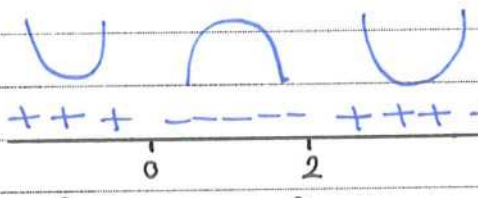
i) $f(x)=x^4-4x^3$

Sol. $f'(x)=4x^3-12x^2$

$f''(x)=12x^2-24x$

$f''(x)=0$

$12x(x-2)=0 \Rightarrow x=0, x=2$




$(0,f(0))=(0,0)$ inflection

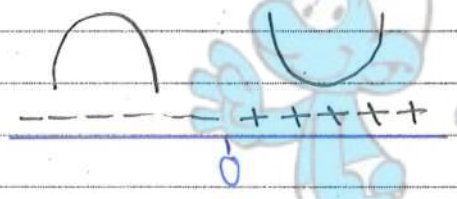
$(2,f(2))=(2,-16)$ inflection

$f(x)$ Concave up $(-\infty,0),(2,\infty)$
$f(x)$ Concave down $(0,2)$

ii) $f(x)=x^3-12x+1$

Sol. $f'(x)=3x^2-12$
$f''(x)=6x$
$f''(x)=0$
$6x=0 \Rightarrow x=0$

At 0: Cont. at 0 and changes it Concavity.




$f$ is Concaved up on $(0,\infty)$
$f$ is Concave down on $(-\infty,0)$

iii) The function $f(x)=x^4-4x^3+1$ on $[-1,1]$ has inflection point at $x=$

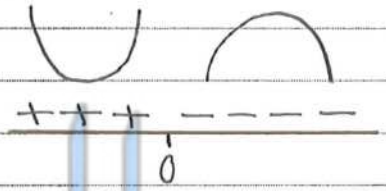
Sol. $f'(x)=4x^3-12x^2$
$f''(x)=12x^2-24x$
$f''(x)=0$
$12x(x-2)=0$
$\boxed{x=0}$ , $x \neq 2$ خارج العشرة

iv) $f(x)=x^{1/3}$

Sol. $f'(x)=\frac{1}{3}x^{-2/3}$ $f''(x)=\frac{-2}{9}x^{-5/3}$

$f''(x)=\frac{-2}{9x^{5/3}}$

$f''(x)$ $=0 \rightarrow$ البسط لا يساوي
d.n.e $\rightarrow 9x^{5/3}=0 \Rightarrow x=0$




At 0 $f$ is cts $\Rightarrow (0,f(0))=(0,0)$ is inflection point

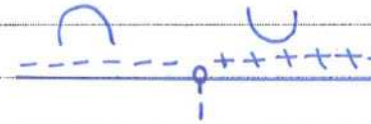
$f$ is Concave up $(-\infty,0)$
$f$ is Concave down $(0,\infty)$

v) $f(x)=\frac{1}{x-1}$ , $x\neq 1$

Sol. $f'(x)=\frac{-1}{(x-1)^2}$ $f''(x)=\frac{2}{(x-1)^3}$

$f''(x)$ d.n.e $\Rightarrow x=1$ تمثل

At $x=1$ No inflection since $1 \notin Df$




$f$ Concave up $(1,\infty)$
$f$ Concave down $(-\infty,1)$

• اختبار المشتقة الأولى والثانية يستخدمان لإيجاد local (max/min)

• يفضل استخدام المشتقة الأولى لإيجاد local max/min كلما إذا فرض علينا استخدام اختبار المشتقة الثانية

H.W

vi) $f(x) = \sin(2x)$ , $x \in [0, 2\pi]$

f Concave up $(\frac{\pi}{2}, \pi)$, $(\frac{3\pi}{2}, 2\pi)$

f Concave down $(0, \frac{\pi}{2})$, $(\pi, \frac{3\pi}{2})$

ex) If $f(x) = (x+1)e^x$, then f is Concave upward on

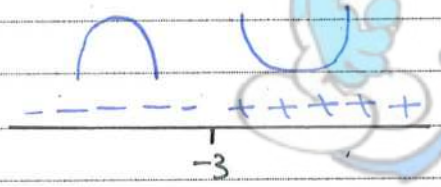
Sol. $f'(x) = (x+1)e^x + e^x = xe^x + 2e^x$

$f''(x) = xe^x + e^x + 2e^x = xe^x + 3e^x$

$f''(x) = 0$

$xe^x + 3e^x = 0 \Rightarrow e^x(x+3) = 0$

$x = -3$




f Concave upward on $(-3, \infty)$

Ex) If $f(x) = x^3 - 3x$ Find local max and local min by second derivative test?

Sol. $f'(x) = 3x^2 - 3$

$f'(x) = 0$

$3x^2 - 3 = 0 \Rightarrow x^2 = 1$

$x = \pm 1$

$f'(\pm 1) = 0$

$f''(x) = 6x$

$f''(1) = 6 > 0$ at $x = 1$ local min with value $f(1) = -2$

$f''(-1) = -6 < 0$ at $x = -1$ local max with value $f(-1) = 2$

→ Second Derivative Test:
اختبار المشتقة الثانية

Suppose $f''$ is cts at c and $f'(c) = 0$ c : Critical

① If $f''(c) < 0$, then f has a local max at c (local max)

② If $f''(c) > 0$, then f has a local min at c (local min)

③ If $f''(c) = 0$ || Back to First Derivative Test

* If the point (a, b) Critical or extreme point (local max/min)

$\Rightarrow f(a) = b$ and $f'(a) = 0$

* If the point (a, b) inflection point

$f(a) = b$ and $f''(a) = 0$

Ex) If $f(x)=ax^3-3x^2$ has local min at $x=1$, Find a?

Sol. local min at $x=1 \Rightarrow f'(1)=0$

$f'(x)=3ax^2-6x$

$f'(1)=0$

$3a-6=0 \Rightarrow a=2$

---

Ex) If $f(x)=4x-ax^2$ has a critical at $x=2$, Find a?

Sol. Critical at $x=2 \Rightarrow f'(2)=0$

$f'(x)=4-2ax$

$f'(2)=0$

$4-4a=0 \Rightarrow a=1$

---

بواسطة

Ex) The function $f(x)=px^2+qx+2$ has $(1,4)$ as an extreme point, then $p=$ ____ and $q=$ ____

Sol. extreme point $(1,4)$

$\Rightarrow f(1)=4$ and $f'(1)=0$

* $f(1)=4 \Rightarrow p+q+2=4$

$p+q=2$ ----- (I)

* $f'(x)=2px+q$

$f'(1)=0 \Rightarrow 2p+q=0$ --- (II)

$p+q=2$

$2p+q=0$ نضرب المعادلة الأولى $\times -1$

$-p-q=-2$

$2p+q=0$

$p=-2$

نعوض p بأي معادلة لنحصل على q

$-2+q=2 \Rightarrow q=4$

$p=-2, \ q=4$

---

H.W

Ex) If $f(x)=ax^3-bx^2$ has inflection point at $(1,-4)$, Find a, b?

Sol. $a=2, \ b=6$

---

Q) 2017 $f(x)=x|x|$, find inflection point?

Sol. $f(x)=x.\begin{cases} x, & x\geq 0 \\ -x, & x<0 \end{cases}$

$=\begin{cases} x^2, & x\geq 0 \\ -x^2, & x<0 \end{cases}$

f cts at 0 $\Rightarrow f'(x)=\begin{cases} 2x, & x>0 \\ -2x, & x<0 \\ 0, & x=0 \end{cases}$

f' cts at 0 $\Rightarrow f''(x)=\begin{cases} 2, & x>0 \\ -2, & x<0 \\ d.n.e, & x=0 \end{cases}$

$f''(0^+)\neq f''(0^-)$

$f''$ (d.n.e) at $x=0$ ∩ ---- ∪ ++++ (0)

inflection

بوكتاي

Q) $f(x) = x + 4x^{-1}$ has Critical points at $x =$ _____

Sol. $f'(x) = 1 - 4x^{-2}$

$= 1 - \frac{4}{x^2}$

$f'(x) = \frac{x^2 - 4}{x^2}$

$f'(x) = 0 \Rightarrow x^2 - 4 = 0 \Rightarrow x = \pm 2$

$f'(x)$ d.n.e $\Rightarrow x^2 = 0 \Rightarrow x = 0 \notin D_f$ تهمل

at $x = \pm 2$

بوكتاي

Q) If $f(x) = x^2 - \frac{8}{x-1}$, then the interval of increasing is

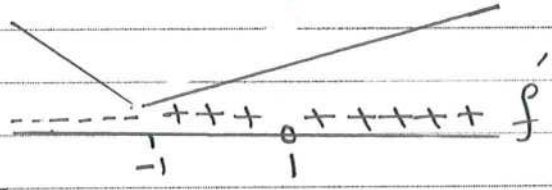
Sol. $f'(x) = 2x + \frac{8}{(x-1)^2}$

$f'(x) = 0 \Rightarrow 2x = \frac{-8}{(x-1)^2}$

$\Rightarrow 2x(x-1)^2 = -8$

$x = -1$ بالتجريب

$f'(x)$ d.n.e $\Rightarrow x \neq 1 \notin D_f$




$f \nearrow$ on $[-1,1), (1,\infty)$

بوكتاي

Q) Consider the function

8-12-2015

$$f(x) = \frac{1}{2}\left(\frac{(x-1)^4}{2} + (x-1)^2 + 3\right)$$

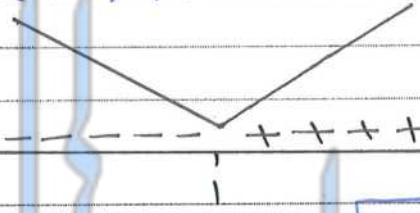
Then the function $f$ is decreasing on

Sol. $f(x) = \frac{(x-1)^4}{4} + \frac{(x-1)^2}{2} + \frac{3}{2}$

$f'(x) = (x-1)^3 \cdot 1 + (x-1)^1 \cdot 1 + 0$

$f'(x) = 0$

$(x-1)^3 + (x-1) = 0 \Rightarrow x = 1$




a) $[-1,\infty)$ b) $[1,\infty)$ c) $(-\infty,1]$ d) $(-\infty,-1]$

2017

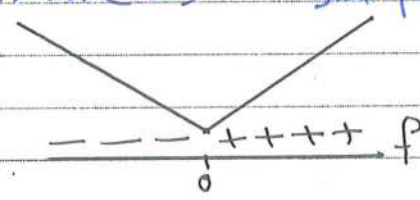
Q) The increasing interval of $f(x) = \tan^{-1}(x^2+1)$ is

Sol. $f'(x) = \frac{1}{1+(x^2+1)^2} \cdot 2x$

$= \frac{2x}{1+(x^2+1)^2}$

$f'(x) = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$

$f'(x)$ d.n.e $\Rightarrow$ ليس للمقام أصفار




increasing $[0,\infty)$

H.w
Q) A function $f(x) = 5^{x^2-x}$ has

Sol. (local / Relative) min at $x = \frac{1}{2}$

final
Q) The function $f(x) = x + \frac{4}{x}$ has critical points at $x =$

Sol. $D_f = \mathbb{R} - \{0\}$

$f'(x) = 1 - \frac{4}{x^2}$

$f'(x) = \frac{x^2-4}{x^2}$

$f'(x) = 0 \Rightarrow x^2 - 4 = 0 \rightarrow x = \pm 2$

$f'(x)$ d.n.e $\Rightarrow x^2 = 0 \Rightarrow x = 0$

$0 \notin D_f$ تهمل

at $x = \pm 2$

أسئلة تزويدية على شابتر ٤

H.w
Q1) Find all inflection points of the function

$f(x) = 2x^3 + 3x^2 - 36x$

a) $x = 2, -3$ b) $x = \frac{1}{2}$

c) $x = 0, \frac{1}{2}$ d) $x = -3, 2$

b

Q2) Find the absolute max. and absolute min. values of

$f(x) = \frac{x^2-4}{x^2+4}$ on the interval $[-4, 4]$

Sol. نخرج النقاط الحرجة

$f'(x) = \frac{16x}{(x^2+4)^2}$

$f'(x) = 0 \Rightarrow 16x = 0 \Rightarrow x = 0$

$f'(x)$ d.n.e $\Rightarrow$ ليس للمقام أصفار

$f(-4) = \frac{3}{5}$ $f(0) = -1$

$f(4) = \frac{3}{5}$

at $x = -4, 4$ Abs. Max with value $\frac{3}{5}$

$x = 0$ Abs. Min " " $-1$

H.w
Q3) On which interval is $f(x) = 4x^3 - x^4$ increasing?

a) Only on $[-3, 0]$ b) $[-3, 0], [0, \infty)$

c) $(-\infty, 0], [0, 3]$ d) $[0, 3]$

c

H.w
Q4) On which intervals is $f(x) = (x^2 - 4x + 2)e^x$ Concave down?

a) $(-1, 1)$ b) $(-2, 2)$

c) $(-\infty, -2), (2, \infty)$ d) $(-\sqrt{2}, \sqrt{2})$

b

H.W

Q5) What are the x-values of the critical points of

$$f(x) = \sqrt{4-(x-1)^2}$$

a) $x=-1,1,3$ b) $x=-1, x=3$
c) $x=1$ d) no criticals

[a]

H.W

Q6) Determine the intervals over which

$$f(x) = (x-4)^3$$

is Concaved down

a) $(-\infty,4)$ b) $(-\infty,-4)$
c) $(4,\infty)$ d) $(-4,\infty)$

[a]

Q7) The function

$$f(x) = x^2+18x^2+2$$

is decreasing.

a) $[-12,0]$ b) $(-\infty,-12],[0,\infty)$
c) $(-\infty,-12],[-12,0]$ d) $[0,12],[12,\infty)$

[a]

H.W

Q8) Determine the intervals where

$$f(x) = \frac{x^2}{x-3}$$

is decreasing

a) $[0,3),(3,6]$ b) $[0,6]$
c) $(-\infty,0],[6,\infty)$ d) $[0,3),(6,\infty)$

[a]

Q9) Use the first derivative test to determine the local extrema, if any, for the function

$$f(x) = x^3+3x^2-24x+6$$

a) local max at $x=-4$ and local min $x=2$
b) local max at $x=-4$
c) local min at $x=2$
d) local max at $x=2$ and local min at $x=-4$

[a]

Q10) Find the inflection points for

$$f(x) = x^3-6x-1$$

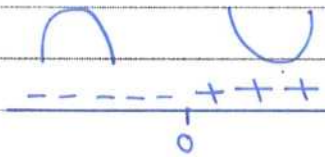
a) $(0,-1)$ b) $(0,-6)$ c) $(1,-1)$
d) $(-1,6)$

Sol: $f'(x) = 3x^2-6$

$f''(x) = 6x$

$f''(x) = 0$

$6x = 0 \Rightarrow x = 0$




0: cts at 0 and changes it Concavity

$(0, f(0)) = (0,-1)$ is inflection point

(a)

H.W

Q11) Find the inflection points for $f(x)=\sqrt{x+7}$

a) $(-7,0)$ b) $(-3,2)$
c) $(-6,1)$ d) There are no points of inflection

[d]

H.W

Q12) Find the inflection points for

$$f(x)=\frac{1}{4}x^4-x^3+6$$

a) $(0,6)$ and $(2,2)$ b) $(0,0)$ and $(2,2)$
c) $(0,0)$ d) $(0,6)$ and $(2,4)$

[a]

H.W

Q13) $f(x)=x^2e^{-x^2}$, f is increasing on

Sol. $(-\infty,-1]$ and $[0,1]$

H.W

Q14) $f(x)=\frac{x}{x^2+1}$, $x\in[-5,5]$, then f has abs. max at $x=$

Sol. Abs. max of $f(x)$ is $\frac{1}{2}$ at $x=1$

Q15) List all inflection points of

$$y=\frac{1}{3}x^6+\frac{1}{2}x^5-5x^4+4x+8$$

a) $-3,0,2$ b) $-3,2$
c) $0,2$ d) $-6,0,1$

Sol. $y'=2x^5+\frac{5}{2}x^4-20x^3+4$

$y''=10x^4+10x^3-60x^2$

$y''=0$

$10x^4+10x^3-60x^2=0$

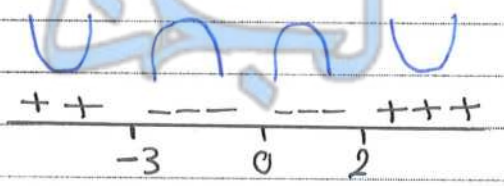
$10x^2(x^2+x-6)=0$

$10x^2(x+3)(x-2)=0$

$x=0$ , $x=-3$ , $x=2$

لتبحث متى تكون النقطة نقطة انعطاف (inflection) يجب أن تحقق شرطان :

1) متصل عندها f و يتغير من تقعر (∪) لتقعر (∩)




$x=-3$ , 2 inflection points

(b)

$x=0$ Not inflection point
لأن عند 0 لم يتغير التقعر

# Chapter 5: Integration

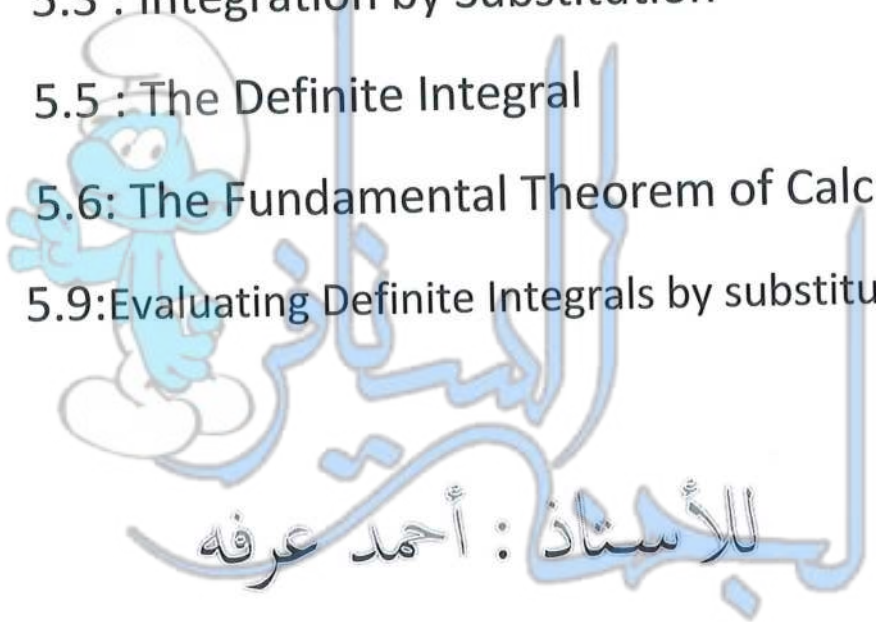
5.2 : The Indefinite Integral

5.3 : Integration by Substitution

5.5 : The Definite Integral

5.6: The Fundamental Theorem of Calculus

5.9:Evaluating Definite Integrals by substitution

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## Ch5 : Integration $\int$

**Def : Antiderivative function**

F(x) called antiderivative of f(x) on I where $F'(x) = f(x)$

$F \xrightarrow[\text{بمعطى}]{\text{اشتق}} F$

**Ex :** Let $F(x) = \frac{x^3}{3}$, Find $f(x) = ?$

**Sol :** $F'(x) = f(x)$

$x^2 = f(x)$

**Def : Integration**

It's a process of find all antiderivatives of F

$\int f(x)\,dx = F(x) + c$

$\int f'(x)\,dx = f(x) + c$

## 5.2 The Indefinite Int.

**Rule :**

1) $\int k\,dx = kx + c$
2) $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad ; n \neq -1$
3) $\int x^{-1}\,dx = \int \frac{1}{x}\,dx = \ln x + c$
4) $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{(n+1)\,a} + c$
5) $\int e^x\,dx = e^x + c$
6) $\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + c$
7) $\int a^x\,dx = \frac{a^x}{\ln a} + c$
8) $\int \sin x\,dx = -\cos x + c$
9) $\int \cos x\,dx = \sin x + c$
10) $\int \sec x \tan x\,dx = \sec x + c$
11) $\int \sec^2 x\,dx = \tan x + c$
12) $\int \csc^2 x\,dx = -\cot x + c$
13) $\int \csc x \cdot \cot x\,dx = -\csc x + c$
14) $\int \frac{dx}{1+x^2} = \tan^{-1} x + c$
15) $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c$
16) $\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1} x + c$
17) $\int cf(x)\,dx = c\int f(x)\,dx$
18) $\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

ملاحظة : في تكامل الاقترانات المثلثية نوضح السالب الناتج الذي يسبق أ ب ج

Ex :- Evaluate :

1) $\int (3x^4 + \frac{1}{x^2} - \sqrt[3]{x} + 5)\, dx$

Sol $= \frac{3}{5}x^5 + \frac{x^{-1}}{-1} - \frac{3}{4}x^{4/3} + 5x + c$

2) $\int \frac{x^2 + x^2\sqrt{x} + 1}{x^2}\, dx$

Sol : نوزع البسط على جميع حدود المقام

$= \int (\frac{x^2}{x^2} + \frac{x^2\sqrt{x}}{x^2} + \frac{1}{x^2})\, dx$

$= \int (1 + \sqrt{x} + x^{-2})\, dx$

$= x + \frac{2}{3}x^{3/2} + \frac{x^{-1}}{-1} + c$

3) $\int (x+1)(2x+1)\, dx$

Sol : $= \int (2x^2 + x + 2x + 1)\, dx$

$= \int (2x^2 + 3x + 1)\, dx$

$= \frac{2}{3}x^3 + \frac{3}{2}x^2 + x + c$

4) $\int x(x^2+1)^2$

Sol $= \int x(x^4 + 2x^2 + 1)\, dx$

$= \int (x^5 + 2x^3 + x)\, dx$

$= \frac{x^6}{6} + \frac{2}{4}x^4 + \frac{x^2}{2} + c$

5) $\int e^{3x+1}\, dx$

Sol $= \frac{e^{3x+1}}{3} + c$

6) $\int 7^x\, dx = \frac{7^x}{\ln 7} + c$

7) $\int 10^x + x^{10} + 10\, dx$

$= \frac{10^x}{\ln 10} + \frac{x^{11}}{11} + 10x + c$

8) $\int \frac{\cos x}{\sin^2 x}\, dx$

Sol $= \int \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x}\, dx$

$= \int \cot x \cdot \csc x\, dx$

$= -\csc x + c$

9) $\int \frac{\sin 2x}{\sin x}\, dx$

Sol $= \int \frac{2 \sin x \cos x}{\sin x}\, dx$

$= \int 2\cos x\, dx = 2\sin x + c$

متطابقتان :

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \csc^2 x$

10) $\int (1 + \tan^2 x)\, dx$

Sol : هنا لا يوجد تكامل لـ $\tan^2 x$ لذا نستخدم متطابقة

$= \int \sec^2 x\, dx = \tan x + c$

11) $\int \frac{1+\cos^2 x}{\cos^2 x}\, dx$

Sol $= \int \left( \frac{1}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \right) dx$

$= \int (\sec^2 x + 1)\, dx$

$= \tan x + x + c$

12) $\int \frac{1-x^2}{1-x^4}\, dx$

Sol $= \int \frac{(1-x^2)}{(1-x^2)(1+x^2)}\, dx$

$= \int \frac{1}{1+x^2}\, dx = \tan^{-1} x + c$

13) $\int \sec^2(7x+5)\, dx$

Sol $= \frac{\tan(7x+5)}{7} + c$

14) $\int 2^x e^x\, dx$

Sol : $= \int (2e)^x\, dx$

$= \frac{(2e)^x}{\ln 2e} + c$

15) $\int (\sin x + \cos x)^2\, dx$

Sol : $= \int (\sin^2 x + 2\sin x \cos x + \cos^2 x)\, dx$

$= \int (1 + \sin 2x)\, dx$

$= x + -\frac{\cos 2x}{2} + c$

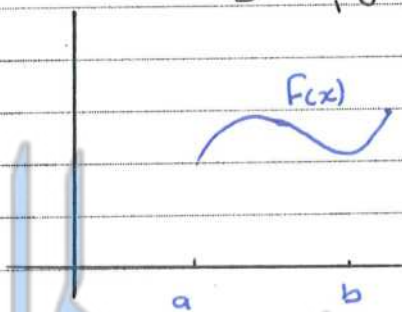
H.W : Evaluate

1) $\int \left(3x^2 - 2x + \frac{3}{x^2+1}\right) dx$

2) $\int \left(\sqrt{2x+1} + e^{5x+1}\right) dx$

3) $\int \left(2^x + \frac{3}{x}\right) dx$

## 5.3 Definite Integeral التكامل المحدود




$\int_a^b F(x)\, dx$ = area under the curve of $F(x)$ on the interval $[a,b]$.

Properties:

1) $\int_a^a f(x)\, dx = 0$ لأننا لم نتحرك فلم تنتج مساحة

2) $\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx$

3) $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$ "تجزئة التكامل"

4) $\int_a^b c f(x) = c \int_a^b f(x)\, dx$

5) $\int_a^b [f(x) \pm g(x)]\, dx = \int_a^b F(x)\, dx \pm \int_a^b g(x)\, dx$

* التكامل لا يتوزع على الضرب والقسمة

Ex : If $\int_0^{10} f(x)\,dx = 17$, $\int_0^{8} f(x)\,dx = 12$

Find $\int_8^{10} f(x)\,dx$ ?

Sol نأخذ أكبر فترة بينهم ثم نجزئ التكامل

$$\int_0^{10} f(x)\,dx = \int_0^{8} f(x)\,dx + \int_8^{10} f(x)\,dx$$

$$17 = 12 + \int_8^{10} f(x)\,dx$$

$$\Rightarrow \int_8^{10} f(x)\,dx = 5$$

Ex: $\int_0^{5} f(x)\,dx = 7$, $\int_3^{0} f(x)\,dx = 5$,

Find $\int_3^{5} f(x)\,dx$ ?

Sol $$\int_0^{5} f(x)\,dx = \int_0^{3} f(x)\,dx + \int_3^{5} f(x)\,dx$$

$$7 = -5 + \int_3^{5} f(x)\,dx$$

$$\Rightarrow \int_3^{5} f(x) = 12$$

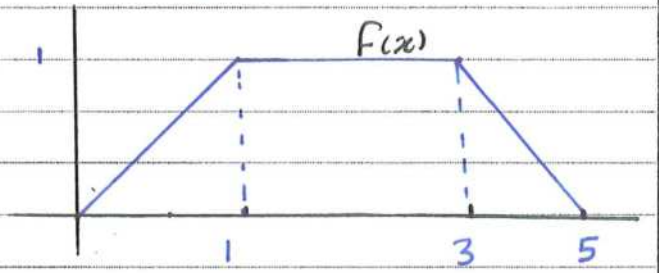
Ex H.W) If $\int_0^{9} f(x)\,dx = 37$ and $\int_9^{0} g(x) = 16$, find

$$\int_0^{9} [2f(x) + 3g(x)]\,dx$$

Sol : 26

Q: let $g(x) = \int_0^{x} f(t)\,dt$.

where $f(x)$ as shown at this Figure :




Find g

① $g(0)$ , ② $g(1)$ ③ $g(3)$

④ H.W $g(5)$

Sol

1) $g(0) = \int_0^{0} f(t)\,dt = 0$

2) $g(1) = \int_0^{1} f(t)\,dt$

= The area of Triangle

$= \frac{1}{2} * \text{base} * \text{hight}$

$= \frac{1}{2} * 1 * 1 = \frac{1}{2}$

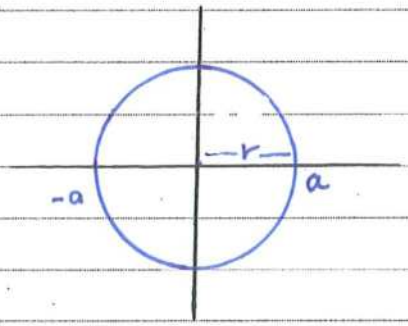
3) $g(3) = \int_0^{3} f(t)\,dt$

$= \int_0^{1} f(t)\,dt + \int_1^{3} f(t)\,dt$

$= \frac{1}{2} + (2*1) = \frac{1}{2} + 2 = \frac{5}{2}$

4) $g(5) = 7/2$

* Equation of circle :




$x^2 + y^2 = a^2$

$\Rightarrow y^2 = a^2 - x^2$

$y = \pm\sqrt{a^2 - x^2}$

$y = \sqrt{a^2 - x^2}$ القسم العلوي للدائرة

$y = -\sqrt{a^2 - x^2}$ القسم السفلي للدائرة

• Circle equation :

$(x-a)^2 + (y-b)^2 = r^2$

where : * circle center : $(a,b)$

* radius : $r$

• Area of circle $= \pi r^2$

Ex: find :

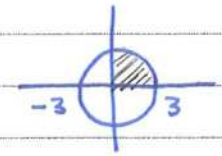
i) $\int_0^3 \sqrt{9-x^2}\,dx$

$y = \sqrt{9-x^2}$

$\Rightarrow y^2 = 9 - x^2 \Rightarrow x^2 + y^2 = 9$

* circle center $(0,0)$

* $r = 3$




$\int_0^3 \sqrt{9-x^2}\,dx$

$= \frac{1}{4} *$ area of circle

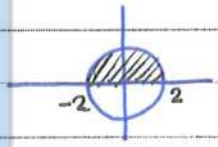
$= \frac{1}{4}(\pi \cdot 3^2) = \frac{9\pi}{4}$

(ii) $\int_{-2}^{2} \sqrt{4-x^2}\,dx$

Sol : $y = \sqrt{4-x^2}$

$\Rightarrow x^2 + y^2 = 4$

القسم العلوي للدائرة




$\int_{-2}^{2} \sqrt{4-x^2}\,dx = \frac{1}{2} * \pi r^2$

$= \frac{1}{2}\pi (2)^2 = 2\pi$

(iii) HW $\int_0^1 \sqrt{1-x^2}\,dx$

Sol $\frac{\pi}{4}$

(iv) $\int_0^3 \sqrt{6x-x^2}\,dx$

Sol $y = \sqrt{6x-x^2}$

$y^2 = 6x - x^2$

$y^2 = -(x^2 - 6x + 9 - 9)$

إكمال مربع

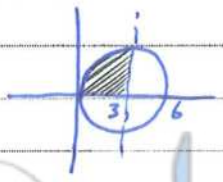
$y^2 = -((x-3)^2 - 9)$

$= 9 - (x-3)^2$

$\Rightarrow (x-3)^2 + y^2 = 9$

Circle center $(3,0)$

$r = 3$




$$\int_0^3 \sqrt{6x - x^2}\, dx$$

$$= \frac{1}{4}\pi r^2 = \frac{\pi}{4}(9) = \frac{9\pi}{4}$$

** عند إجراء التكامل المحدود لا نضيف في الناتج $(+C)$ وإنما نعوض القيمة العليا - القيمة السفلى .

$$\int_a^b f'(x)\, dx = f(x)\Big|_a^b = f(b) - f(a)$$

Ex: Evaluate

1) $$\int_0^1 x\, dx = \frac{x^2}{2}\Big|_0^1 = \frac{1}{2} - \frac{0}{2} = \frac{1}{2}$$

2) $$\int_1^2 \frac{1}{x}\, dx = \ln x\Big|_1^2$$

$$= \ln 2 - \ln 1 = \ln 2$$

3) $$\int_0^1 \frac{1}{1+x^2}\, dx = \tan^{-1} x\Big|_0^1$$

$$= \tan^{-1} 1 - \tan^{-1} 0$$

$$= \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

4) $$\int_0^1 10^x\, dx = \frac{10^x}{\ln 10}\Big|_0^1$$

$$= \frac{10}{\ln 10} - \frac{1}{\ln 10} = \frac{9}{\ln 10}$$

5) $$\int_0^{\pi/4} \sec^2 x\, dx = \tan x\Big|_0^{\pi/4}$$

$$= \tan\frac{\pi}{4} - \tan 0$$

$$= 1 - 0 = 1$$

6) $$\int_{-2}^{5} 6\, dx = 6(5 - (-2))$$

$$= 6(7) = 42 .$$

7) $$\int_0^{\pi/4} \sec x \tan x\, dx$$

$$= \sec x\Big|_0^{\pi/4}$$

$$= \sec\frac{\pi}{4} - \sec 0 = \sqrt{2} - 1$$

8) $$\int_0^{\pi} f(x)\, dx$$ where

$$f(x) = \begin{cases} \sin x, & 0 \le x \le \pi/2 \\ \cos x, & \pi/2 \le x \le \pi \end{cases}$$

Sol: يجب تجزئة التكامل .

$$\int_0^{\pi} f(x)\, dx = \int_0^{\pi/2} \sin x\, dx + \int_{\pi/2}^{\pi} \cos x\, dx$$

$$= -\cos x\Big|_0^{\pi/2} + \sin x\Big|_{\pi/2}^{\pi}$$

$$= (0 - 1) + (0 - 1)$$

$$= 1 - 1 = 0 .$$

9) H.W $\int_{-2}^{2} f(x)\,dx$ where

$$f(x)=\begin{cases}2 & , -2\le x\le 0\\ 4-x^2 & , 0<x\le 2\end{cases}$$

10) $\int_{-1}^{2}|x|\,dx$ $\qquad |x|=\begin{cases}x & , x\ge 0\\ -x & , x<0\end{cases}$

Sol

$$\int_{-1}^{2}|x|\,dx=\int_{-1}^{0}-x\,dx+\int_{0}^{2}x\,dx$$

$$=\frac{-x^2}{2}\Big|_{-1}^{0}+\frac{x^2}{2}\Big|_{0}^{2}$$

$$=\frac{-1}{2}(0-1)+\frac{1}{2}(4-0)$$

$$=\frac{5}{2}$$

11) H.W $\int_{0}^{10}|x-5|\,dx$

Remark :

a) If $f$ is even $[f(-x)=f(x)]$, then $\int_{-a}^{a}f(x)\,dx=2\int_{0}^{a}f(x)\,dx$

b) If $f$ is odd $[f(-x)=-f(x)]$, then $\int_{-a}^{a}f(x)\,dx=0$

Ex: Evaluate :

i) $\int_{-2}^{2}1+\frac{\tan x}{x^2+\cos x}\,dx$

$$=\int_{-2}^{2}1\,dx+\int_{-2}^{2}\frac{\tan x}{x^2+\cos x}\,dx$$

نلاحظ هنا أن التكامل من $(-2,2)$ و

الاقترانان : $\frac{odd}{even}=odd$

$$=1\,(2+2)+0=4.$$

ii) $\int_{-1}^{1}\tan^3 x\,dx=0$ (odd)

iii) $\int_{-5}^{5}\frac{\cot x}{|x|+x^4}\,dx=0$ (odd)

iv) $\int_{-2}^{2}|x|\,dx=$ (even)

$$2\int_{0}^{2}x\,dx=2\cdot\frac{x^2}{2}\Big|_{0}^{2}$$

$$=4$$

v) H.w $\int_{-1}^{1}\frac{\tan x}{1+x^2+x^4}\,dx$

vi H.W $\int_{3}^{3}\frac{\cos x+10^x}{\tan x-\ln x}\,dx$

vii) $\int_{-\pi}^{\pi}\sin x\,dx=0$ (odd)

viii) $\int_{-\pi}^{\pi}\cos x\,dx=$

$$2\int_{0}^{\pi}\cos x\,dx=2\sin x\Big|_{0}^{\pi}$$

$$=2(0-0)=2(0)=0$$

## 5.4 Fundamental Thm of calculus ..

IF $F(x)$ is cont and $g(x) = \int_{u(x)}^{v(x)} f(t)\,dt$ , then

$g'(x) = F(v(x)).v'(x) - F(u(x)).u'(x)$

**Ex :** Use the Fundamental theorem of Calculus to find the derivative of the function

1) $g(x) = \int_{1}^{x} \frac{1}{t^3+1}\,dt$

**Sol :** نعوض القيمة العليا في المقدار ده نضربه بمشتقتك ثم نعوض القيمة السفلى في المقدار ثم نشتري

$\Rightarrow g'(x) = \frac{1}{x^3+1}.1 - \frac{1}{1^3+1}.0$

$= \frac{1}{x^3+1}$

2) $g(x) = \int_{1}^{x^2} \ln t\,dt$

**Sol :** $g'(x) = \ln x^2 . 2x - \ln 1 . 0$

$= 2x \ln x^2$

3) $g(x) = \int_{0}^{x} t^2 \sin t\,dt$

**Sol :** $g'(x) = x^2 . \sin x . 1 - 0^2 \sin 0 . 0$

$= x^2 \sin x$

4) $f(x) = \int_{x}^{\pi} \sqrt{1+\sec t}\,dt$

**Sol :** $f'(x) = \sqrt{1+\sec\pi}\cdot 0 - \sqrt{1+\sec x}\cdot 1$

$= -\sqrt{1+\sec x}$

5) H.W $h(x) = \int_{0}^{1/x} \tan^{-1} t\,dt$

6) $g(x) = \int_{2x}^{3x} \frac{u^2-1}{u^2+1}\,du$

**Sol :** $g'(x) = \frac{9x^2-1}{9x^2+1}.3 - \frac{4x^2-1}{4x^2+1}.2$

7) $g(x) = \int_{\tan x}^{x^2} \frac{1}{\sqrt{2+t^4}}\,dt$

**Sol :** $g'(x) = \frac{1}{\sqrt{2+(x)^8}}.2x - \frac{1}{\sqrt{2+\tan^4 x}}.\sec^2 x$

8) H.W $y = \int_{\sqrt{x}}^{x^3} \sqrt{t}\sin t\,dt$

**ملاحظة :** ١) مشتقة التكامل غير المحدود هي ما داخل التكامل نفسه

٢) مشتقة التكامل المحدود إذا كانت حدوده أرقام (ثوابت) هي صفر

Ex : Find $\frac{dy}{dx}$ :

1) $y = \int \tan x + x^3 + 5 \; dx$

$\frac{dy}{dx} = \tan x + x^3 + 5$

2) $y = \int_{-1}^{1} \cos x + x^3 \; dx$

$\frac{dy}{dx} = 0$

Ex: If $F(x) = \int_0^x (1-t^2) e^{t^2} dt$.

On what interval is $f(x)$ increasing?

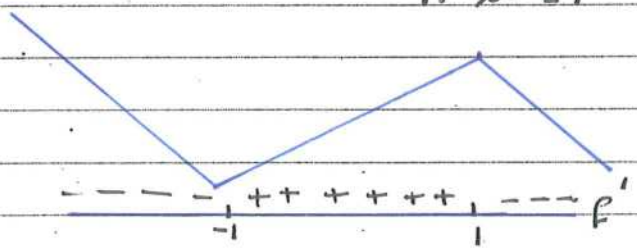
Sol: $F'(x) = (1-x^2)e^{x^2} \cdot 1 - (1-0)e^0 \cdot 0$

$F'(x) = (1-x^2)e^{x^2}$

$F'(x) = 0 \Rightarrow (1-x^2)e^{x^2} = 0$

$e^{x^2} \neq 0 \Rightarrow (1-x^2) = 0$

$\therefore x = \pm 1$




$F \nearrow$ on $[-1, 1]$.

## 5.5 Subsitution Integral التكامل بالتعويض

عادة" ما يكون في السؤال الاقتران ومشتقته أو جزء من المشتقة

Ex: Evaluate :

1) $\int e^{\sin x} \cos x \; dx$

Sol: نفرض: الاقتران $y = \sin x$

$dy = \cos x \; dx$

$\Rightarrow \int e^y \, dy = e^y + c$

$= e^{\sin x} + c$

2) $\int \frac{\tan^{-1} x}{1+x^2} \; dx$

Sol: $y = \tan^{-1} x$

$dy = \frac{1}{1+x^2} dx$

$\int y \, dy = \frac{y^2}{2} + c$

$= \frac{(\tan^{-1} x)^2}{2} + c$

3) $\int \frac{\ln x}{x} \; dx$

Sol: $y = \ln x$

$dy = \frac{1}{x} dx$

$\int \frac{\ln x}{x} dx = \int y \, dy$

$= \frac{y^2}{2} + c = \frac{(\ln x)^2}{2} + c$

$= \frac{2}{3}(e^{y} + 2y) + c$

$= \frac{2}{3}(e^{\sqrt{x}} + 2\sqrt{x}) + c$

4) $\int \frac{1}{\sin^{-1} x \sqrt{1-x^2}}\, dx$

Sol $y = \sin^{-1} x$

$dy = \frac{1}{\sqrt{1-x^2}}\, dx$

$\int \frac{1}{\sin^{-1} x \sqrt{1-x^2}}\, dx = \int \frac{1}{y}\, dy$

$= \ln y + c$

$= \ln(\sin^{-1} x) + c$

5) $\int x^3 \cos(x^4+2)\, dx$

Sol $y = x^4 + 2$

$dy = 4x^3\, dx$

$\Rightarrow x^3\, dx = \frac{1}{4}\, dy$

$\int x^3 \cos(x^4+2)\, dx =$

$\int \frac{1}{4} \cos y\, dy = \frac{1}{4} \sin y + c$

$= \frac{1}{4} \sin(x^4+2) + c$

6) $\int \frac{e^{\sqrt{x}} + 2}{3\sqrt{x}}\, dx$

Sol $y = \sqrt{x}$

$dy = \frac{1}{2\sqrt{x}}\, dx$

$\Rightarrow 2\sqrt{x}\, dy = dx$

$\int \frac{e^{\sqrt{x}} + 2}{3\sqrt{x}}\, dx = \int \frac{e^{y} + 2}{3\sqrt{x}} \cdot 2\sqrt{x}\, dy$

$= \frac{2}{3} \int e^{y} + 2\, dy$

7) $\int \tan x\, dx = \int \frac{\sin x}{\cos x}\, dx$

Sol $y = \cos x$

$dy = -\sin x\, dx$

$= -\int \frac{1}{y}\, dy = -\ln y + c$

$= -\ln(\cos x) + c$

$= \ln(\cos x)^{-1} + c$

$= \ln(\sec x) + c$

Ex

1) $\int_1^e \frac{\ln x}{x}\, dx$

Sol $y = \ln x$

$dy = \frac{1}{x}\, dx$

يجب هنا تغيير حدود لأن حدود 1 & e هي حدود x ، ويجب أن تغير لتصبح حدوداً لـ y .

IF $x = 1 \Rightarrow y = \ln 1 = 0$

$x = e \Rightarrow y = \ln e = 1$

$\int_1^e \frac{\ln x}{x}\, dx = \int_0^1 y\, dy = \frac{y^2}{2}\Big|_0^1 = \frac{1}{2}$

نعوض مباشرة ولا نغير y .

2) $\int_0^1 \frac{e^x+1}{e^x+x}\,dx$

Sol ⇐ $y = e^x + x$

$dy = e^x + 1\,dx$

If $x=0 \Rightarrow y = e^0 + 0 = 1$

$x=1 \Rightarrow y = e+1$

$$\int_0^1 \frac{e^x+1}{e^x+x}\,dx = \int_1^{e+1} \frac{1}{y}\,dy$$

$$= \ln y \Big|_1^{e+1}$$

$$= \ln(e+1) - \ln 1$$

$$= \ln(e+1)$$

* يمكننا حلها سريعاً باستخدام :

$$\int \frac{f'(x)}{f(x)}\,dx = \ln(f(x)) + c$$

(القاعدة من المعلوم بمعنى $\int \frac{1}{x}\,dx = \ln x + c$)

3) $\int_1^2 x\sqrt{x-1}\,dx$

Sol ⇐ $y = x-1 \Rightarrow x = y+1$

$dy = dx$

If $x=1 \rightarrow y=0$

$x=2 \rightarrow y=1$

$$\int_1^2 x\sqrt{x-1}\,dx = \int_0^1 (y+1)\sqrt{y}\,dy$$

$$= \int_0^1 y^{3/2} + y^{1/2}\,dy$$

$$= \frac{2}{5}y^{5/2} + \frac{2}{3}y^{3/2}\Big|_0^1$$

$$= \frac{2}{5} + \frac{2}{3} = \frac{16}{15}$$

4) $\int_0^{\ln x} \frac{e^{2x}}{1+e^{4x}}\,dx$

Sol $\int_0^{\ln x} \frac{e^{2x}}{1+(e^{2x})^2}\,dx$

let $y = e^{2x}$

$dy = 2e^{2x}\,dx$

$\Rightarrow \frac{1}{2}dy = e^{2x}\,dx$

If $x=0 \rightarrow y=1$

$x=\ln x \rightarrow y = e^{2\ln x} = e^{\ln x^2} = x^2$.

$$\int_0^{\ln x} \frac{e^{2x}}{1+e^{4x}}\,dx = \frac{1}{2}\int_1^{x^2} \frac{dy}{1+y^2}$$

$$= \frac{1}{2}\left[\tan^{-1} y\right]_1^{x^2}$$

$$= \frac{1}{2}\left(\tan^{-1} x^2 - \tan^{-1} 1\right)$$

$$= \frac{1}{2}\left(\tan^{-1} x^2 - \pi/4\right).$$

5) $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$

Sol ⇐ $y = \ln x$

$dy = \frac{1}{x}\,dx$

If $x=e \rightarrow y = \ln e = 1$

$x=e^4 \rightarrow y = \ln e^4 = 4$.

$$\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}} = \int_1^4 \frac{dy}{\sqrt{y}} = \int_1^4 y^{-1/2}\,dy$$

$$= 2y^{1/2}\Big|_1^4$$

$$= 2\sqrt{4} - 2\sqrt{1} = 4 - 2$$

$$= 2.$$

6) H.W $\int_0^1 x\, e^{-x^2}\,dx$

$y = x^2$

7) H.W $\int_1^2 \frac{e^{1/x}}{x^2}\,dx$

Sol: $e - \sqrt{e}$

H.W

1) $y = \int_{\cos x}^{5x} \cos(u^2)\,du$, Find $y'$

2) $\int_{-\pi/2}^{\pi/2} \frac{x^2 \sin x}{1+x^6}\,dx$

3) $\int e^{\tan x} \sec^2 x\,dx$

4) $\int e^x \sqrt{1+e^x}\,dx$

5) $\int \frac{(\ln x)^2}{x}\,dx$

6) $\int_{\pi/2} e^x \sin(e^x)\,dx$

7) $\int_0^{\sqrt{\pi}} \cos x \sin(\sin x)\,dx$

8) $\int_0 x \cos(x^2)\,dx$

Rule 8

1) $\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{(n+1)\,a} + C$

2) $\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + C$

3) $\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + C$

4) $\int \sin(ax+b)\,dx = \frac{-\cos(ax+b)}{a} + C$

وهذه القاعدة تعمم لكل الاقترانات المثلثية التي تكون الزاوية بها خطية

5) $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln(ax+b) + C$

ex: $\int \frac{1}{2x+3}\,dx = \frac{1}{2}\ln(2x+3) + C$

6) $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

ex: $\int \frac{1}{9+x^2}\,dx = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C$

7) $\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right) + C$

ex: $\int \frac{1}{\sqrt{4-x^2}}\,dx = \sin^{-1}\left(\frac{x}{2}\right) + C$

8) $\int \frac{F'(x)}{F(x)}\,dx = \ln|F(x)| + C$

* ex: $\int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx$ ($F'$ over $F$)

$= \ln|\sin x| + C$

* ex: $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$

$$= -\int \frac{-\sin x}{\cos x}\,dx \quad (+)$$

$$= -\ln|\cos x| + c$$

$$= \ln(\cos x)^{-1} + c$$

$$= \ln\left(\frac{1}{\cos x}\right) + c$$

$$= \ln(\sec x) + c$$

* ex: $\int \frac{e^x}{e^x+1}\,dx$ ($F'$ over $e^x$, $F$ over $e^x+1$)

$$= \ln|e^x+1| + c$$

* ex: $\int \frac{x}{x^2+7}\,dx =$

$$\frac{1}{2}\int \frac{2x}{x^2+7}\,dx =$$ ($F'$ over $2x$, $F$ over $x^2+7$)

$$\frac{1}{2}\ln|x^2+7| + c$$

---

Ex: $\int \frac{x}{x+1}\,dx$

Sol: هنا لأن درجة البسط ≥ درجة المقام : نضيف ونطرح للبسط 1 ثم نوزع المقام على البسط

$$\Rightarrow \int \frac{x+1-1}{x+1}\,dx$$

$$= \int \frac{x+1}{x+1} - \frac{1}{x+1}\,dx$$

$$= \int 1 - \frac{1}{x+1}\,dx$$

$$= x - \ln|x+1| + c$$

---

Ex: $\int \frac{x^2}{x^2+4}\,dx$

Sol:

$$\int \frac{x^2+4-4}{x^2+4}\,dx$$

$$= \int 1 - \frac{4}{x^2+4}\,dx$$

$$= \int 1\,dx - 4\int \frac{1}{x^2+4}\,dx$$

$$= x - 4\cdot\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + c$$

$$= x - 2\tan^{-1}\left(\frac{x}{2}\right) + c.$$

---

Ex: $\int \sec x\,dx$

Sol $\int \sec\cdot\frac{\sec x+\tan x}{\sec x+\tan x}\,dx$

$$= \int \frac{\sec^2 x + \sec x\tan x}{\sec x+\tan x}\,dx$$ ($F'$ over numerator, $F$ over denominator)

$$= \ln|\sec x+\tan x| + c.$$

---

Ex: $\int \csc x\,dx$

Sol: $\int \csc x\cdot\frac{\csc x+\cot x}{\csc x+\cot x}\,dx$

$$= -\int \frac{-(\csc^2 x + \csc x\cot x)}{\csc x+\cot x}\,dx$$

$$= -\ln|\csc x+\cot x| + c$$

## Ch.5 أسئلة سنوات على

Q: The Integral $\int_0^{\ln(\ln 3)} 2\, e^{x+e^x}\, dx$ equals to

Sol $\int_0^{\ln(\ln 3)} 2\, e^x \cdot e^{e^x}\, dx$

let $y = e^x$

$dy = e^x\, dx$

$x = 0 \Rightarrow y = 1$

$x = \ln(\ln 3) \Rightarrow y = e^{\ln(\ln 3)} = \ln 3$

$\Rightarrow \int_1^{\ln 3} 2\, e^y\, dy = 2e^y \Big|_1^{\ln 3}$

$= 2[e^{\ln 3} - e^1]$

$= 2[3 - e^1]$ (D)

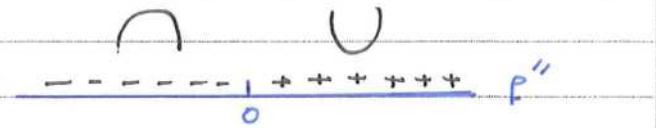
Q: The concave up interval of $f(x) = \int_0^{2x} e^{t^2}\, dt$ is : (قابل 2017)

Sol $F'(x) = e^{4x^2} \cdot 2 = 2e^{4x^2}$

$F''(x) = 2e^{4x^2} \cdot 8x = 16x\, e^{x^4}$

$F''(x) = 0$

$16x\, e^{x^4} = 0 \Rightarrow x = 0$




Concave up $(0, \infty)$.

Q (قابل 2017) $\int_0^{\pi/2} (\sin x + \cos x)^2\, dx$

Sol $\int_0^{\pi/2} (\sin^2 x + 2\sin x \cos x + \cos^2 x)\, dx$

$= \int_0^{\pi/2} (1 + 2\sin x \cos x)\, dx$

$= \int_0^{\pi/2} [1 + \sin(2x)]\, dx$

$= x + \frac{-\cos(2x)}{2} \Big|_0^{\pi/2}$

$= (\frac{\pi}{2} - 0) - (0 - \frac{1}{2}) = \frac{\pi}{2} + 1$

Q (قابل 2017) $\int_1^2 \frac{x+1}{x}\, dx$

Sol $\int_1^2 1 + \frac{1}{x}\, dx = x + \ln x \Big|_1^2$

$= (2 + \ln 2) - (1 + \ln 1)$

$= 1 + \ln 2$.

Q (قابل 2017) $\int \frac{8\tan^{-1} x}{1+x^2}\, dx$

Sol $y = \tan^{-1} x$

$dy = \frac{1}{1+x^2}\, dx$

$$\Rightarrow 8\int y\,dy =$$

$$8\cdot\frac{y^2}{2}+c = 4(\tan^{-1}x)^2+C$$

Q) فاينل 2017

$$\int_{-2}^{2} \sin^5 x + \sqrt{4-x^2}\,dx$$

Sol

$$\int_{-2}^{2} \underbrace{\sin^5 x}_{\text{odd}}\,dx + \int_{-2}^{2}\sqrt{4-x^2}\,dx$$

$$= 0 + \left(\frac{1}{2}\cdot\pi\cdot 4\right)$$

$$= 2\pi$$

Q) فاينل 2017

Complete the following answer :

If $2+\int_a^x f(t)\,dt = e^{4x}$ then

$a = ?$

Sol

$$\int_a^x f(t)\,dt = e^{4x} - 2$$

نشتق الطرفين بالنسبة لـ $x$

$$f(x)\cdot 1 - f(a)\cdot 0 = 4e^{4x}$$

$$\therefore f(x) = 4e^{4x}$$

نعوض قيمة $f(x)$ في التكامل.

$$\Rightarrow 2+\int_a^x 4e^{4t}\,dt = e^{4x}$$

$$\Rightarrow 2+\frac{4e^{4t}}{4}\Big|_a^x = e^{4x}$$

$$\Rightarrow 2+(e^{4x}-e^{4a}) = e^{4x}$$

$$\Rightarrow e^{4a} = 2$$

$$\Rightarrow 4a = \ln 2$$

$$\Rightarrow a = \frac{\ln 2}{4}$$

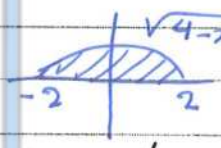
Q) $\int_{-2}^{0} f(x)\,dx = 5$ , $\int_{0}^{2} f(x)\,dx = -2$.

Find $\int_{-2}^{2}(f(x)+2\sqrt{4-x^2})\,dx$.

Sol

$$\int_{-2}^{2}(f(x)+2\sqrt{4-x^2})\,dx$$

$$= \int_{-2}^{2} f(x)\,dx + 2\int_{-2}^{2}\sqrt{4-x^2}\,dx$$




$$= \int_{-2}^{0} f(x)\,dx + \int_{0}^{2} f(x)\,dx + 2\left(\frac{1}{2}\cdot\pi\cdot 4\right)$$

$$= 5+(-2)+4\pi = 3+4\pi$$

Q) If $\int_1^3 f(x)\,dx = 4$ , then

$\int_1^2 f(2x-1)\,dx = ?$

Sol let $y = 2x-1$

$$dy = 2dx \Rightarrow \frac{1}{2}dy = dx$$

$x=1 \Rightarrow y=1$

$x=2 \Rightarrow y=3$

$$\int_1^2 f(2x-1)\,dx = \int_1^3 \frac{1}{2} f(y)\,dy$$

$$= \frac{1}{2} \cdot 4 = 2.$$

Q: $\frac{d^2}{dx^2}\left(\int_1^{x^3} \frac{1}{t}\,dt\right) = ?$

Sol: $\frac{d}{dx}\left(\int_1^{x^3} \frac{1}{t}\,dt\right) =$

$$\frac{1}{x^3} \cdot 3x^2 - \frac{1}{1} \cdot 0 =$$

$$\frac{3x^2}{x^3} - 0 = \frac{3}{x}$$

$$\frac{d^2}{dx^2}\left(\int_1^{x^3} \frac{1}{t}\,dt\right) = \frac{d}{dx}\left(\frac{3}{x}\right)$$

$$= \frac{-3}{x^2}$$

Q: $\int \frac{\sin 2x}{1+\cos^2 x}\,dx$

Sol $\int \frac{2\sin x \cos x}{1+\cos^2 x}\,dx$

let $y = \cos x$

$dy = -\sin x\,dx$

$$\rightarrow \int \frac{-2y}{1+y^2}\,dy =$$

$$-\ln|1+y^2| + c =$$

$$-\ln|1+\cos^2 x| + c =$$

$$\ln(1+\cos^2 x)^{-1} + c =$$

$$\ln\left(\frac{1}{1+\cos^2 x}\right) + c$$

Q) $\int_{\frac{-1}{\sqrt{2}}}^{1/\sqrt{2}} \frac{\sin^{-1} x}{1+x^4}\,dx$

Sol: $\frac{\sin^{-1} x}{1+x^4}$ odd / even = odd

$$\int_{\frac{-1}{\sqrt{2}}}^{1/\sqrt{2}} \text{odd}\,dx = 0$$

Q: Evaluate $\int_0^1 x\sqrt{1-x^4}\,dx$

Sol: $\int_0^1 x\sqrt{1-(x^2)^2}\,dx$

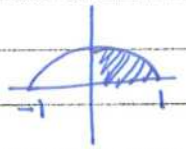
let $y = x^2$

$dy = 2x\,dx$

$x=0 \rightarrow y=0$

$x=1 \rightarrow y=1$

$$\rightarrow \frac{1}{2}\int_0^1 \sqrt{1-y^2}\,dy$$




$$= \frac{1}{2}\left(\frac{1}{4}\pi \cdot 1^2\right) = \frac{\pi}{8}$$

Q: $\int \frac{-2}{1+2x^2}\,dx$

Sol: $-2\int \frac{1}{2\left(\frac{1}{2}+x^2\right)}\,dx =$

$$-\left(\frac{1}{\frac{1}{\sqrt{2}}}\tan^{-1}\left(\frac{x}{\frac{1}{\sqrt{2}}}\right)\right) + c =$$

$$-\sqrt{2}\tan^{-1}(\sqrt{2}\,x) + c$$

Q: The Integral: How $\int_0^{\sqrt{\frac{\pi}{4}}} -x\tan(x^2)\,dx = ?$

Sol $y = x^2$